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Understanding and Improving Novice Kinesthetic Teaching for Robot Learning from Demonstration

by

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degree of Doctor of Philosophy

in the
School of Computing and Information Systems
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Abstract

Robotic systems hold tremendous potential to assist humans not only in industrial environments but also in everyday tasks. However, effectively deploying these systems typically requires programming expertise, creating a significant barrier for novice users. Learning from Demonstration (LfD) offers a promising alternative by allowing users to teach robots through demonstrations rather than writing sophisticated code, thereby promoting the democratization of robotics. To enhance the efficacy of LfD, it is essential to understand not only the underlying learning algorithms but also the quality and information content embedded in human-provided demonstrations.

First, we quantified the quality of human demonstrations using two motion-related features: manipulability and joint-space jerk. We then investigated how these quality measures influence the performance of LfD models. Our results show that demonstration quality directly transfers to the learned models, highlighting the critical role of human-provided data in shaping robot performance. While most existing LfD approaches focus primarily on successful demonstrations—emphasizing what to imitate—failed and sub-optimal demonstrations can provide valuable information about what not to imitate. To address this gap, we proposed a novel learning algorithm that not only learns from successful demonstrations but also explicitly repels the model from failed ones. This approach enables more robust and efficient learning even when demonstrations are imperfect, as commonly occurs with novice users.

Next, we examined the role of kinematic redundancy in LfD through a user study involving 24 novice participants. Specifically, we analyzed how the robot’s degree of redundancy, reflected in the number of available joint-space solutions, affects both human teaching behavior and robot performance. Our findings indicate that reduced redundancy makes it more difficult for novice users to identify desirable joint configurations, negatively impacting both the quality of demonstrations and the resulting learned models. Finally, we developed an augmented reality (AR)-based real-time feedback system to assist users in providing higher-quality demonstrations. By offering intuitive visual guidance during teaching, the system supports more effective human-robot interaction and improves overall learning outcomes.

Overall, this research advances LfD from a purely algorithmic perspective toward a more human-centered framework. By quantifying demonstration quality, leveraging failed attempts, analyzing the impact of redundancy, and introducing interactive feedback mechanisms, this work contributes to the development of intuitive and accessible human-robot

interaction systems in which everyday users can teach robots naturally and effectively, expanding the possibilities for real-world, user-centered robotic applications.

Declaration of Authorship

I, Muhammad Bilal, declare that this thesis titled, ‘Understanding and Improving Novice Kinesthetic Teaching for Robot Learning from Demonstration’ and the work presented in it are my own. I confirm that:

- The thesis comprises only my original work towards the Doctor of Philosophy except where indicated in the preface;
- Due acknowledgment has been made in the text to all other material used; and
- The thesis is fewer than the maximum word limit (100,000 words) in length, exclusive of tables, bibliographies and appendices.

Signed:

Muhammad Bilal

Date:

June 2026

Preface

This thesis is submitted in total fulfilment of the requirements for the degree of Doctor of Philosophy at the University of Melbourne. The research presented here was primarily conducted at the School of Computing and Information Systems, Faculty of Engineering and Information Technology, The University of Melbourne, under the supervision of A/Prof Wafa Johal, A/Prof Nir Lipovetzky, and Prof Denny Oetomo.

Below is the list of publications and manuscripts arising from this thesis. I am the principal author of all papers and I am responsible for designing the algorithm architectures, processing data, implementation, running experiments, and analyzing the experimental results. My co-authors contributed with feedback to all these processes and the revision of the papers.

- Part of the contents of [chapter 3](#) has been published: **Muhammad Bilal**, Nir Lipovetzky, Denny Oetomo, and Wafa Johal. *Beyond success: Quantifying demonstration quality in learning from demonstration. In 2024 IEEE/RSJ International conference on Intelligent Robots and Systems (IROS). IEEE, 2024.*
DOI: [10.1109/IROS58592.2024.10802187](https://doi.org/10.1109/IROS58592.2024.10802187).
- Part of the contents of [chapter 5](#) has been published: **Muhammad Bilal**, D. Antony Chacon, Nir Lipovetzky, Denny Oetomo, and Wafa Johal. *Investigating the Impact of Robot Degree of Redundancy on Learning from Demonstration. In the 21st ACM/IEEE International Conference on Human-Robot Interaction, 2026.*
DOI: [10.1145/3757279.3785606](https://doi.org/10.1145/3757279.3785606).
- Part of the contents of [chapter 6](#) has been published: **Muhammad Bilal**, Tharaka S. Ratnayake, D. Antony Chacon, Nir Lipovetzky, Denny Oetomo, and Wafa Johal. *Design and Evaluation of AR-Based Real-Time Feedback System for Kinesthetic Robot Teaching. In the ACM Designing Interactive Systems (DIS) conference 2026.* DOI: [10.1145/3800645.3812951](https://doi.org/10.1145/3800645.3812951).

Honourable Mention Award

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I would also like to sincerely thank my co-supervisors, A/Professor Nir Lipovetzky and Professor Denny Oetomo, for their valuable advice, thoughtful feedback, and continued support throughout my candidature. Your expertise and encouragement have greatly contributed to the quality of this research and my academic growth.

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I am deeply grateful to my parents for their unconditional love, endless prayers, and countless sacrifices. Their unwavering belief in me has been my greatest source of strength and motivation. I would also like to thank my siblings for their constant encouragement, patience, and support from afar.

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This thesis is dedicated to my son,
Muhammad Hasan Bilal

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Chapter 1

Motivation and Introduction

1.1 Motivation

Robots are becoming increasingly capable, with advances in sensing, control, and computation enabling them to perform a growing range of tasks while simultaneously decreasing in size, cost, and improving robustness [1, 2]. Despite these advances, the adoption of robots in many workplaces such as Small & Medium Enterprises (SMEs) remains limited [3]. In addition, we observe persistent concerns that automation may displace human workers [4–6] and impact negatively the quality of work [3, 7]. Such concerns are often exacerbated by the limited accessibility of robotic systems for novice users, as most robots still require specialized technical expertise to be programmed and operated [3, 8, 9].

Traditional robot programming exemplifies this barrier. It typically demands coding expertise and considerable development time, requiring users to explicitly specify action sequences and low-level control logic [10]. Although motion planning techniques reduce the need for manual trajectory specification, they still rely on precise high-level inputs such as goal poses and waypoints. Such programs often lack flexibility and must be reconfigured when the environment changes [11, 12]. Even within the active *Robot Operating System (ROS)* community¹, the number of users worldwide with the skills necessary to program robotic systems remains below 150,000, underscoring the limited pool of technically proficient operators relative to the global workforce.

Alternative paradigms such as reinforcement learning (RL) aim to reduce manual programming by enabling robots to learn through trial and error. However, RL-based approaches frequently require domain-specific expertise to design appropriate reward

¹ROS metrics — <https://metrics.ros.org/index.html>

functions, tune hyperparameters, and ensure safe exploration—factors that complicate their deployment in real-world environments [13–15]. Moreover, tasks that are difficult to express as optimization objectives or that involve nuanced human preferences remain challenging to encode through reward engineering alone.

On the other side, LLM-based approaches face several fundamental limitations in embodied settings [16–20]. First, regarding robot hardware, these models typically do not account for platform-specific dynamics during training, making generalization to previously unseen manipulators or mobile bases difficult [21–23]. Second, concerning the robot’s environment, LLMs lack grounded knowledge of physical properties such as object weight, material composition, friction, and texture—factors that are essential for safe and effective manipulation. Third, in the context of human–robot interaction, models trained predominantly on textual and visual datasets do not inherently encode human behavior models, physical safety constraints, or real-time risk assessment, which are critical during collaborative tasks [8, 16]. As a result, relying solely on LLM-driven reasoning for low-level control or safety-critical decisions may lead to unstable performance or unsafe behavior. These limitations highlight the gap between high-level semantic reasoning and physically grounded skill acquisition, reinforcing the importance of learning paradigms that directly incorporate embodied interaction and task-specific human expertise.

Recent advances in neural network–based learning algorithms, particularly deep learning methods, have significantly expanded the capabilities of robotic systems. Deep neural networks can model high-dimensional sensory inputs, learn complex nonlinear policies, and generalize across varied task conditions when trained on sufficiently large datasets [24, 25]. In robotics, such approaches have enabled progress in visuomotor control, policy learning, and end-to-end skill acquisition [26, 27]. However, these methods typically require vast amounts of labeled or interaction data, extensive computational resources, and careful architecture design and tuning. Collecting large-scale robotic datasets in real-world environments is often expensive, time-consuming, and constrained by safety considerations. Moreover, purely data-driven models may lack interpretability and can struggle to incorporate structured prior knowledge or human intent.

In this context, Learning from Demonstration (LfD) has emerged as a compelling alternative. LfD allows robots to acquire new skills by observing and imitating human demonstrations, thereby shifting the burden from programming explicit instructions to providing examples of desired behavior [10, 28, 29]. This paradigm is particularly attractive when tasks are not easily scripted using conventional programming or cannot be naturally formulated as reward optimization problems. By leveraging human expertise directly through demonstrations, LfD has the potential to make robot skill acquisition

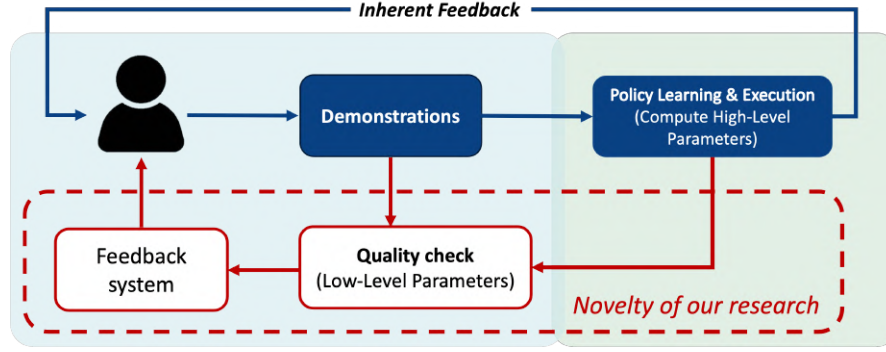


FIGURE 1.1: Block diagram of revised LfD framework

more intuitive, accessible, and adaptable—especially for users without specialized technical backgrounds. In conventional LfD, as illustrated in Figure 1.1, a user provides a *set of demonstrations* of the desired task, followed by *policy learning*, and finally execution on the robot using the learned model.

What is robot learning from demonstration?

Robot learning from demonstration allows users to teach robot new skills through demonstrations rather than writing complex code.

It is well-established in the machine learning literature that the overall performance of a learning process is largely driven by two key factors: the *quality of the input data* and the *capability of the learning algorithm*. In practice, high-quality data often enables even moderately complex algorithms to achieve strong performance, whereas high-capacity algorithms cannot fully compensate for poor or noisy data. This insight has recently been formalized in the emerging *data-centric* AI paradigm, which emphasizes that systematic improvement of dataset quality can be more effective for enhancing model performance than focusing solely on algorithmic complexity [30, 31]. Similarly, the effectiveness of an LfD process depends on improving both (1) the robustness of the *robotic learning algorithms* [28, 32, 33] and (2) the *information content* in the human-provided demonstrations [34, 35]. In the literature, researchers have devoted considerable attention to the algorithmic approaches, with comparatively less emphasis on the human-centric aspects for enhancing LfD [32]. In this thesis, we primarily investigate the information content (quality of demonstration) that enable a robot to achieve higher performance, and then explore ways to enhance the teaching ability of novice users to provide desired quality of demonstrations. The resulting revised LfD framework is shown in Figure 1.1.

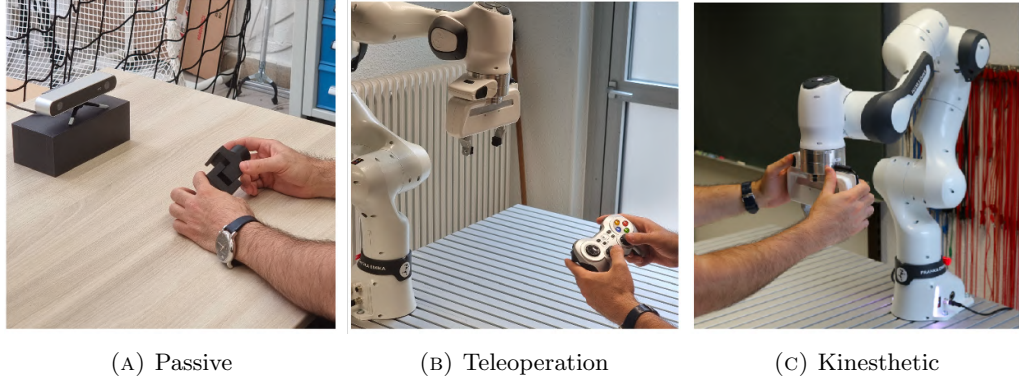


FIGURE 1.2: Modalities of Demonstration [38]

1.2 Introduction

Learning from demonstration is a supervised learning approach, allowing novice users to teach robots new tasks through demonstrations—typically represented as sequences of state-action pairs that capture the intended behavior [9, 36, 37]. There are multiple modalities of demonstration, as summarized in the following subsection, through which novice users teach desired skills to the robotic systems.

1.2.1 Demonstration Modalities

In literature [9, 10, 36], several modalities have been explored for teaching robots, including: (1) passive observation, (2) teleoperation, and (3) kinesthetic teaching. Approaches based on *passive observation* involve tracking the demonstrator using external sensors, such as motion capture systems or biomedical sensors [29, 39], as illustrated in Figure 1.2a. Although these methods are non-intrusive for the robot, they suffer from issues like occlusions, fast human movements, and the complexity of mapping human motions to robot kinematics. In contrast, *teleoperation* based approaches, as shown in Figure 1.2b, allows novice users to control the robot through interfaces such as joysticks, haptic devices or virtual/augmented reality systems [27, 40, 41]. This approach is well-suited for teaching multi-DoF robots; however, users typically control only the robot’s end-effector while the joint configurations are computed through inverse kinematics. The derivation of inverse kinematics for higher degrees of freedom robots is itself a challenging task [42]. Finally, *kinesthetic teaching* offers an intuitive method that allows novice users to physically guide the robot through the desired motion, eliminating reliance on additional sensors, external inputs, or specialized interfaces [43, 44], as depicted in Figure 1.2c. This approach enables to capture the preferences of users encoded not only at Cartesian space (as done in teleoperation) but also at joint space level. This allows for more comprehensive demonstrations of constrained tasks—such

TABLE 1.1: Comparison of demonstration modalities in robot learning [9, 10, 36, 37]

Modality	Advantages	Limitations	Example
Kinesthetic Teaching	Intuitive for non-experts; Accurate data in robot’s workspace.	Requires a compliant robot; Physically demanding; Limited to close proximity.	Human physically moves the robot arm to demonstrate a task.
Teleoperation	No physical contact with robot; Remote demonstration possible; High control fidelity.	Requires specialized interface (e.g., haptic device); Needs skilled operator; Latency can affect performance.	Using a 6-DoF space mouse or VR controller to teach pick-and-place.
Passive Observation	Natural and unintrusive demonstrations; Scalable data collection; No robot setup required.	Requires complex perception and mapping; Ambiguity in human intent; Lower precision in motion capture.	Robot observes a human pouring water into a cup.

as those involving joint limits, conceptual constraints, or physical joint constraint—by permitting users to manipulate all the robot joints in order to meet the task requirements [45–47]. Based on these insights, we used kinesthetic teaching approach to collect demonstrations throughout this thesis. A summary of these demonstration modalities is provided in Table 1.1.

1.2.2 Types of Demonstrations

Most existing LfD approaches suppose that users are skilled and capable of providing optimal demonstrations [10]. However, this assumption overlooks the heterogeneous nature of human demonstrators, often resulting in suboptimal demonstrations [28, 48] or even failed attempts [49]. Previous research explored different types of demonstrations, categorized based on users teaching capabilities to provide effective demonstrations [34, 50]. An illustration of different types of demonstrations is shown in Figure 1.3. The outermost circle represents all possible configurations of robotic system. The term R represents the task space under learned policy while the term B depicts the demonstration space which user would like to provide. Furthermore, the terms r and b indicate the task realization through rollout the policy and actual demonstrations provided by the user, respectively.

1.3 Thesis Objectives

This thesis addresses human- and data-centric challenges that limit the effectiveness and broader adoption of LfD systems. Since LfD also relies on human users—often without specialized technical expertise—to teach robots through physical demonstrations, it is

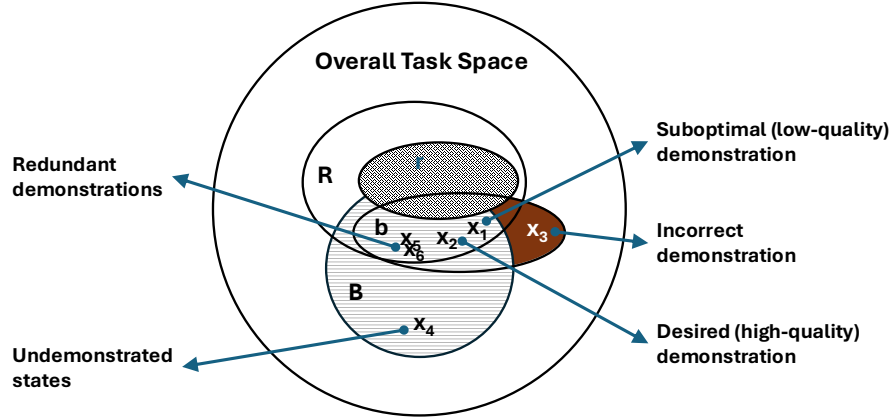


FIGURE 1.3: The figure is adapted from [50] and illustrates the relationship between the user’s intended demonstration space (B), the robot’s learned model (R), and realized executions (r). The overlap represents the shared region where the learned model aligns with user intent, while deviations indicate mismatch. Each type of demonstration is summarized in Table 1.2.

TABLE 1.2: Types of demonstrations with their descriptions. The terms $x_1 - x_6$ represent each type of demonstration as shown in Figure 1.3.

Demonstration Type	Description
Successful demonstration	A demonstration that completes the intended task.
Suboptimal (Low-quality) (x_1)	A successful demonstration with poor execution quality (e.g., high jerk and/or low manipulability).
Desired (High-quality) (x_2)	A successful demonstration with smooth and efficient execution, representing desired behavior.
Failed demonstration	An attempt that does not complete the task due to self-collision, excessive force/torque, and joint limits violations.
Incorrect demonstration (x_3)	An unintended or irrelevant behavior that does not align with the task objective.
Undemonstrated states (x_4)	Regions of the state space that are not covered by any demonstration.
Redundant demonstration (x_5, x_6)	Repetitive demonstrations that do not provide additional learning benefit to the model.

essential to first understand how the quality of these demonstrations affects robot performance. Building on this understanding, we can then develop methods to help novice users provide more effective and reliable demonstrations. The objectives of this thesis are summarized in chronological order as follows:

Regarding the quality of demonstrations, existing evaluation mainly focuses on appraising the robot’s task performance, classifying demonstrations as either high-quality or low-quality based on the success or failure of the task [35, 51]. However, the underlying reason for the given demonstrations being of high or low quality remains unclear. Therefore, it becomes crucial to systematically measure and assess each demonstration. We

introduce a way to quantify demonstration’s quality using motion-related features, such as smoothness and manipulability. In later studies, we also incorporated other metrics, including task completion rate, demonstration completion time, distance to joint limits, distance to self-collision, etc. This allows us to reason more precisely about why some demonstrations lead to better learning outcomes than others (details in [chapter 3](#)).

Second, real-world teaching sessions are inherently imperfect [48, 52]. Human demonstrators naturally make mistakes, explore alternative strategies, and occasionally fail while demonstrating a task. Inspired by the theory of productive failure [53, 54] which shows failed attempts also contain important information, we leverage both failed and successful demonstrations to train an LfD model. While a limited body of prior work has explored learning from mixed sets of successful and failed demonstrations, these efforts have largely framed the problem in terms of task completion using a binary notion of success or failure. As a result, the quality of task execution—how effectively and robustly the robot performs the task—has received comparatively little attention. This thesis addresses this gap by developing a learning framework that explicitly incorporates failed demonstrations as informative negative evidence, enabling the learned model not only to imitate successful behaviors but also to actively avoid suboptimal or undesirable ones (details in [chapter 4](#)).

Third, most commercial robots, including Franka Emika, KUKA, Sawyer, have 7-DoF joint configurations, making them redundant for majority of daily life tasks. Kinematic redundancy in robots (i.e., more joint DoF than required for a given task) enables multiple joint configurations for performing a task, offering flexibility during demonstrations. On one hand, higher redundancy offers flexibility during kinesthetic demonstrations, since users have larger set of feasible configurations to provide demonstrations. On the other hand, reducing the number of joints that need to be manipulated could impact the user’s demonstration effort, by reducing the number of solutions that need to be considered in demonstrations. This trade-off highlights the importance of understanding how robot redundancy affects kinesthetic demonstrations, as it can impact user experience, demonstration efficiency, and ultimately, the robot’s ability to perform the task using the learned model (details in [chapter 5](#)).

Finally, building on the insights from the preceding studies regarding demonstration quality, failure, and novice teaching behavior, this thesis investigates an interactive feedback system that supports users during the teaching process. Teaching a robot involves more than simply showing it how to perform a task; novice users often provide demonstrations that are incomplete, inconsistent, or poorly aligned with the robot’s learning mechanisms. One key reason for this difficulty is the opacity of the robot’s internal state—such as what it has learned, where it is uncertain, and why it fails—which

deprives users of actionable feedback comparable to the cues naturally exchanged in human–human instruction. As a result, users may form inaccurate mental models of the robot’s capabilities, leading to frustration and suboptimal teaching strategies [49, 55]. To address this challenge, we explore an interactive feedback framework that communicates relevant information about the robot’s state and the quality of ongoing demonstrations, enabling users to adapt their teaching behavior in real time. By guiding users toward higher-information demonstrations and making the learning process more transparent, this approach aims to transform robot teaching into a shared and responsive interaction rather than a trial-and-error process. Such communication-driven interfaces are critical for improving the robustness, usability, and real-world deployment of LfD systems [56, 57] (details in [chapter 6](#)).

Overall, our research aims to achieve a higher performance for the robot along with improving the teaching abilities of novice users with the help of an interactive feedback system. The overview of this thesis can be illustrated as shown in [Figure 1.4](#) where each chapter addresses one of the following *research questions*:

Research Questions

- RQ1** *How does the quality of demonstration impact both robot task completion and the quality of task execution? (Chapter 3)*
- RQ2** *How does an iterative refinement method improve robot task execution quality when training data includes suboptimal or failed demonstrations? (Chapter 4)*
- RQ3** *How does a robot’s degree of redundancy influence cognitive workload, demonstration time, number of failed attempts, and users’ hand placement and adjustment strategies during task demonstrations? (Chapter 5)*
- RQ4** *How can robots communicate internal constraints to users in human–robot Learning from Demonstration interactions? (Chapter 6)*

1.4 Thesis Organization

The summary of each chapter is briefly explained in this section as follows:

Chapter 3 represents the first step to quantify the quality of individual demonstration at each step by incorporating motion-related quality features such as manipulability and joint-space jerk. To assess the efficacy of these features, we initially evaluated the given demonstrations—taking into account each quality feature—to rank them from high- to

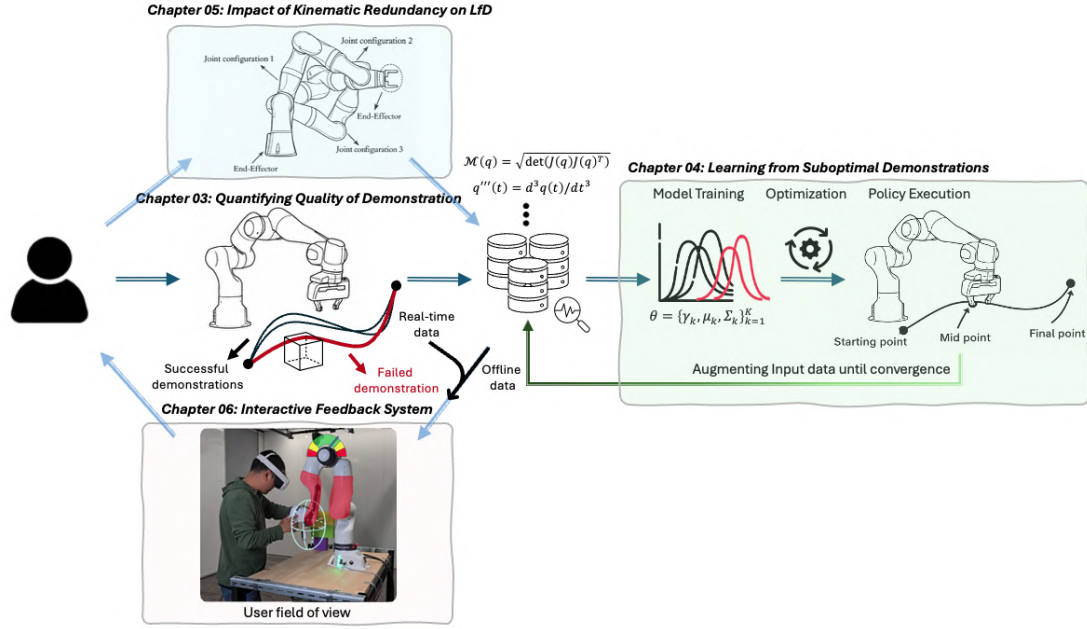


FIGURE 1.4: **Thesis Overview**: This research initiates with quantifying the information content in human demonstrations (Ch 03), enables robot learning even from failed and suboptimal demonstrations (Ch 04), examines the role of kinematic redundancy in LfD (Ch 05), and then develops interactive feedback systems to support more efficient human teaching (Ch 06).

low-quality. Subsequently, we investigated the impact of demonstration’s quality on task performance and the quality of task execution. In this pursuit, we trained a series of LfD models for two distinct manipulation tasks: cube lifting and pick-and-place of soda can. Our results illustrate a strong correlation between ranked demonstrations and the quality of task execution. Interestingly, we observed that the quality features have a significant impact on task performance, particularly when the provided demonstrations exhibit diversity in terms of quality. The results signify that not only the system can learn to perform the task (i.e. typically assessed through success rate) but also how to perform the task (through quality features). This work can lead to develop new ways to assess LfD models optimizing quality features rather than only looking at success rates.

Chapter 4 presents a novel quality-aware LfD framework that leverages a revised weighted Gaussian Mixture Model (GMM) to learn from both failed and suboptimal demonstrations. The proposed algorithm adjusts the GMM components by repelling its away from failed demonstrations while attracting them toward successful demonstrations. Importantly, the algorithm not only learns from successful demonstrations but also weighs more the high-quality ones during learning process. The quality of demonstration is assessed using motion-related quality features—such as manipulability and joint-space jerk—which are then used to assign importance weights to the given successful demonstrations. We employed a non-linear cost function (with some quadratic components) to optimize the reproduced trajectory and then iteratively augment it to

the input dataset until convergence. The proposed framework not only ensures task completion but also guides the system toward high-quality task execution. We validated our approach through extensive 2D simulations with a 3R planar redundant robot and then 3D real-world experiments using a Franka 7-DoF Panda robot.

Chapter 5 investigates how the number of degrees of redundancy impact human performance during kinesthetic demonstrations, and then how these demonstrations influence robot performance. We simulated high and low degree of redundancy by locking one of the robot joint on a 7-DoF Panda robotic arm. We conducted a within-subject user study ($N=24$) with two conditions: unconstrained condition (high degree of redundancy) and constrained condition (low degree of redundancy). We used a motion capture system to capture participants physical interaction with the robot when demonstrating two tasks: button pressing and cuboid block insertion. The results show that the robot's degree of redundancy significantly affects mental workload, demonstration time, number of failed attempts, and physical interaction with the robot. Likewise, joint constraints significantly influence robot performance, measured by task completion using the learned model. These findings highlight the importance of considering robot degree of redundancy during demonstrating constrained tasks, allowing novice users to provide effective demonstrations.

Chapter 6 investigates the design of a feedback system that enables effective communication between the human and the robot during demonstrations. We first conducted a focus group study ($N = 9$) to identify effective ways of visualizing key robot information, including joint limits, self-collisions, and manipulability. Guided by these insights, we designed an AR-based real-time feedback system and evaluated it in a between-subjects user study ($N = 36$) on a Franka Emika 7-DoF robot. Participants performed two manipulation tasks—block insertion and drink pouring, with the second task enabling assessment of participant's learning across tasks. Results show that real-time AR feedback reduced demonstration time, increased task completion rate, lowered perceived mental workload, and improved adherence to robot kinematic constraints, leading to better robot task execution. These findings demonstrate the value of real-time AR feedback for intuitive and effective robot teaching.

Chapter 2

Related Work

This chapter presents a structured review of prior research on LfD [58], with particular attention to how learning algorithms, demonstration data, and human–robot interaction mechanisms have been designed and evaluated. The reviewed works are organized thematically to clarify their assumptions and limitations and to contextualize the human- and data-centric contributions of this thesis.

2.1 Robot Learning Algorithms

This section reviews major families of robot learning methods commonly used in, or alongside, LfD. To keep the discussion accessible, we organize these methods according to the type of information available to the robot: (i) demonstrated actions to imitate, (ii) reward functions to optimize, and (iii) interactive guidance provided during execution.

2.1.1 Imitation Learning

Imitation learning (IL) is a relatively simple LfD approach, treating demonstrations as behaviors to be directly mimicked [29]. In *behavioral cloning*, the robot learns a mapping from observations—such as camera images and robot states—to actions using standard supervised learning techniques [33, 59–61]. This method is appealing due to its simplicity and effectiveness when demonstrations sufficiently cover the states the robot will encounter at deployment [60, 62]. However, this method requires large number of demonstrations to train an LfD model to achieve satisfactory policy outcomes.

More recent imitation learning methods aim to move beyond pure copying by encouraging the robot to capture higher-level task structure. For instance, *adversarial* approaches

such as GAIL learn a discriminator that distinguishes expert behavior from the robot’s behavior, and then train the policy to produce expert-like trajectories [61, 63]. While these methods can be more robust in complex or long-horizon tasks, they typically require more careful training and tuning than standard behavioral cloning.

2.1.2 (Inverse) Reinforcement Learning

Reinforcement learning (RL) enables robots to learn through trial and error by optimizing a reward function [64]. However, in many real-world tasks it is difficult to specify a reward that accurately reflects human intent (e.g., concepts such as “safe driving”), and different users may value different trade-offs. *Inverse reinforcement learning* (IRL) addresses this challenge by inferring a reward function that explains demonstrated behavior, and then learning a policy that optimizes the inferred reward [61, 65]. Bayesian variants, such as BIRL, further model uncertainty over the reward, which can be particularly useful when demonstrations are sparse or noisy [66, 67].

A closely related line of work retains the RL framework but relies on humans to provide reward feedback directly, often referred to as human-in-the-loop RL or reinforcement learning from human feedback (RLHF) [68, 69]. In these approaches, users may score robot behaviors or, more commonly, express preferences between pairs of behaviors [59, 70–73]. Because repeatedly collecting human feedback can be costly, many methods aim to minimize queries by soliciting input only when the robot is uncertain or when comparisons between candidate behaviors are expected to be most informative [72, 74].

2.1.3 Active Learning

Active learning methods improve the efficiency of human feedback by carefully selecting which questions to ask a teacher. In preference-based reward learning, the robot presents a small set of alternatives (e.g., two candidate trajectories) and asks the user to choose the better one; this feedback is then used to update the reward model and policy [75–78]. The design of these queries is critical: options should be both informative for learning and easy for users to compare, in order to reduce cognitive burden and improve feedback quality [77, 79–81]. Similar approaches have been applied in assistive settings, for example to adapt gait parameters in lower-limb exoskeletons based on user preference feedback [82].

2.1.4 Combined Imitation and Reinforcement Learning

Demonstrations can also be used to bootstrap reinforcement learning. Combined IL+RL approaches initialize a policy from demonstrations and then refine it through trial-and-error, often by combining imitation losses with RL objectives or by incorporating demonstrations as additional experience in off-policy learning [83, 84]. In practice, these methods can substantially improve sample efficiency and enable learning in tasks where pure behavioral cloning suffers from distribution shift and pure reinforcement learning is too slow or unsafe [26, 85].

2.1.5 Interactive Teaching during Execution

Many robot learning systems acquire data while the robot is acting, rather than relying solely on a fixed dataset. In *interactive imitation learning*, the robot executes its current policy and a human intervenes when errors occur, providing corrective action labels that are used for further learning [59, 86]. Approaches vary in how interventions are triggered: in some cases the human decides when to step in, while in others the robot actively requests assistance when it detects uncertainty [87–89].

A closely related paradigm is *physical corrections*, where the user kinesthetically guides the robot to adjust a trajectory during execution [90–92]. Rather than treating these corrections as full demonstrations, many methods interpret them as incremental improvements and update the policy online or in small batches [93–97]. From a human-centered perspective, update timing plays an important role: faster updates can make the impact of feedback more transparent to users, whereas slower batch updates may reduce computational load but provide less immediate insight into how the robot is learning. Despite these great efforts, the problem still needs attention due to multiple reasons, for instance, whether the teacher (human demonstrator) will be able to effectively test the learner’s (robot’s) knowledge, whether the teacher will be able to correctly interpret the attempts, and whether the teacher will be able to provide an effective demonstration. This necessitates the step to provide feedback and guidance to the teacher to help them become informed about the learning process, resulting in improved teaching performance [98].

2.1.6 Learning with Imperfect Demonstrations

Several LfD methods have been proposed to improve robot skill reproduction by refining or weighting demonstrations [49, 99–102]. Prior work includes iterative refinement schemes that down-weight less effective demonstrations over time [101], kinesthetic

correction methods that update skill models using physical corrections and geometric constraints [102], and probabilistic frameworks that weigh demonstrations based on task-relevant factors such as obstacle proximity [103]. Other approaches combine LfD with reinforcement learning to refine initially suboptimal demonstrations, but rely on manually designed reward functions and extensive interaction, which limits practicality for real-world use [104, 105]. While some methods explicitly model failed demonstrations—either by searching for successful trajectories near failures [106] or by inferring rewards from success and failure examples [105]—they often assume strong similarity between failures and successes or depend on hand-crafted features. In contrast, our proposed approach weights demonstrations directly based on their quality and learns from both successful and failed examples without reward engineering or user-defined features, enabling efficient learning from limited, labeled demonstrations.

2.2 Quantifying Quality of Demonstration

Assessing the quality of demonstrations is a central challenge in LfD, yet there is no single, generalize accepted metric [10, 11, 107]. Existing approaches evaluate quality from different perspectives, including task performance, motion characteristics, information content, and their effectiveness for training learning models.

2.2.1 Task Performance-Based Metrics

The most direct way to assess demonstration quality is through task performance and outcome-based measures. Prior work consistently shows a strong correlation between demonstration quality and learning performance: higher-quality demonstrations tend to produce policies with better task execution [35, 108]. This relationship has been observed across a range of learning settings, where demonstrations that score highly under performance metrics lead to improved success rates in trained models.

Several methods evaluate demonstration quality by measuring how effectively a learning algorithm can reproduce the demonstrated task [109, 110]. In this view, high-quality demonstrations are those that enable a learned policy to achieve success rates above a predefined threshold while generalizing to nearby task conditions [35, 51]. Performance-based evaluation can be applied both during training—by identifying demonstrations that contribute most to learning progress—and after training, for example by using success classifiers that estimate the probability of task completion associated with each demonstration.

More recent work seeks to estimate demonstration quality without requiring full model training. By analyzing spatial and temporal consistency across demonstrations, these methods can predict a large fraction of task success and generalization performance [111, 112]. Such approaches enable quality assessment prior to expensive training, making them well suited for large-scale demonstration filtering and dataset curation.

2.2.2 Motion Related Metrics

Trajectory- and motion-based approaches assess demonstration quality through the physical properties of the demonstrated movements, rather than task outcomes alone. A common observation is that expert demonstrations tend to produce smooth and consistent trajectories, whereas lower-quality demonstrations are often noisier and less stable [10, 113, 114]. This insight has motivated the use of trajectory smoothness and related kinematic measures as indicators of demonstration quality. The underlying reason for the given demonstrations being of high or low quality remains unclear to the user. In response to these limitations, it becomes crucial to systematically measure and assess each demonstration, taking into consideration various quality features. Feature-based methods [115, 116] present a promising approach to address the mentioned limitations, specifically the issues of being time-intensive and lacking clarity on why a demonstration is classified as high or low quality.

Path efficiency is another widely used trajectory-level metric. Some methods quantify quality based on trajectory length, with shorter paths typically indicating more efficient and purposeful demonstrations [51, 111]. In more advanced formulations, predictive models estimate expected path lengths for a task and weight demonstrations according to deviations from these expectations.

Recent work has also incorporated information-content measures into trajectory-based quality assessment [111, 117, 118]. By analyzing action diversity and predictability through the mutual information between states and actions over entire trajectories, these methods capture both structure and variability in demonstrations. Empirical evaluations show that such metrics can reliably separate high- and low-quality demonstrations, as confirmed by expert human annotations [117].

2.3 Information Content and Communication Interfaces

Learning from demonstrations under constraints reflects the realities of physical robots operating in complex environments. Robots with high degrees of freedom often exploit

redundancy to accomplish tasks efficiently while satisfying physical, environmental, or task-specific constraints [119, 120]. Capturing these constraints from demonstrations is therefore essential, yet challenging, because many constraints are implicit and not directly observable from motion alone. To address this, the notion of *conceptual constraints* was introduced to augment low-level trajectory data with higher-level, abstract task information [46]. Subsequent work automated the extraction of such constraints and integrated them into human-aware motion planning frameworks [47].

Several learning approaches explicitly infer constraints from demonstrations. For example, combining positive and negative demonstrations enables robots to distinguish feasible from infeasible behaviors and thereby learn task constraints more effectively [118]. Other methods adopt multi-stage frameworks that infer continuous constraint functions from demonstrations [121]. While these algorithmic advances improve constraint-aware learning, they largely assume that relevant constraint information is implicitly encoded in demonstrations. In practice, however, how constraints are perceived and acted upon by human teachers depends strongly on what information the robot communicates. In particular, robot-side constraints—such as differing levels of kinematic redundancy—shape human teaching strategies and influence the quality of demonstrations provided. This highlights the need for communication interfaces that make constraints and constraint violations explicit, linking information content directly to more effective human teaching and improved learning outcomes.

2.3.1 Visual Interfaces Beyond Screens

Traditional interfaces often rely on 2D screens to display planned trajectories or learned policies [122, 123]. While flexible, screens can divert attention away from the physical interaction and obscure spatial cues such as depth, contact, or proximity. To address this, many systems move feedback closer to the task space using in-situ visual interfaces, including augmented reality overlays, projections, LEDs, or intentionally designed robot motions [123–126]. Embedding feedback directly into the environment allows users to interpret learning signals while remaining focused on the task, resulting in more natural and robust interaction [127, 128].

2.3.2 Implicit Visual Communication

In addition to external visual aids, robots can use their own motion to communicate intent and internal state [128]. Techniques such as legible motion exaggerate or modulate trajectories to clarify goals, limitations, or uncertainty [129]. Gestures and gaze cues, particularly in humanoid robots, further support intuitive interpretation of robot

behavior [130–132]. While these forms of implicit communication can be powerful and socially meaningful, they are often limited in the complexity of information they can convey and may not generalize well to non-humanoid platforms.

2.3.3 Auditory Feedback

Auditory interfaces provide an effective non-visual channel for communicating robot learning, especially when users are visually occupied or mobile [133, 134]. Verbal feedback can convey explanations of robot decisions or policies, improving transparency and understanding when carefully designed [135]. Non-verbal sounds, such as tones or alerts, offer lightweight and direction-independent cues for states like uncertainty or the need for assistance [136, 137]. However, excessive or poorly timed auditory feedback can overload attention, making selective and context-aware use essential.

2.3.4 Haptic Feedback

Haptic interfaces leverage the sense of touch to communicate information that is difficult to convey visually or verbally, such as forces, contact, or urgency [138–140]. Wearable haptic devices can provide continuous, attention-free feedback, while touch-based interfaces on the robot can localize information to specific joints or interaction points [141, 142]. Haptics have proven particularly effective for conveying physical quantities and guiding corrections during kinesthetic teaching, enabling teachers to adapt demonstrations based on the robot’s sensed state [143, 144].

2.3.5 Multi-modal Communication

No single modality is sufficient to communicate all aspects of robot learning [145–147]. Multi-modal interfaces combine visual, auditory, and haptic feedback to distribute information according to modality strengths [148]. Common design strategies include separating feedback by timescale (continuous vs. intermittent), by data type (spatial vs. abstract), or by priority (primary vs. secondary channels) [143, 149, 150]. When designed carefully, multi-modal interfaces improve clarity, trust, and performance without increasing cognitive load [151, 152]. However, poorly integrated feedback can confuse users, highlighting the importance of deliberate allocation rather than simply adding more channels [153].

2.4 Open Challenges in LfD

One of the main challenges in LfD is demonstration quality. LfD often assumes expert demonstrations, but in practice users are not always experts and data can be inconsistent or suboptimal [49, 111, 154, 155]. This makes learning harder, especially when methods only evaluate whole trajectories instead of identifying good and bad parts [156, 157]. It is still unclear how to measure whether we have enough good-quality demonstrations and how to select better ones when needed.

Second challenge in LfD is the need for least number of good quality demonstrations, which are often expensive or difficult to collect [158, 159]. This becomes worse in multi-task settings, where each task may require new data [160]. Another issue is coverage: demonstrations are often limited and do not capture all situations a robot may face [161]. Data collection in real environments is also affected by noise, missing information, and sensing limits, which further reduces data quality [162, 163].

Third challenge in LfD is variability across demonstrators. Most studies assume expert or consistent teachers, but in practice users differ in experience, skill, and familiarity with robots, which affects both demonstration quality and efficiency [28, 161, 164]. This variability is not well handled by current methods, even though evidence shows that teacher expertise strongly influences the quantity and quality of usable demonstrations [165–167]. As a result, learning from heterogeneous users remains an open problem, particularly when the system must generalise across different teaching styles and skill levels.

Another challenge is human burden and interaction design. Collecting demonstrations is time-consuming and physically/mentally demanding, especially in multi-task or long-horizon settings, where fatigue can further degrade performance [160, 168, 169]. Poor interface design can also make teaching harder, as users struggle to coordinate motion, orientation, and gripper control, even when they are experienced [166, 170]. While active LfD can reduce data requirements, it may increase cognitive load due to frequent queries, creating a trade-off between data efficiency and usability [171]. These issues highlight the need for more interactive, adaptive LfD systems that reduce effort and better support non-expert users, for example through correction-based or feedback-driven teaching.

Chapter 3

Beyond Success: Quantifying Demonstration Quality in Learning from Demonstration¹

To overcome the obstacle of using robots for daily life tasks without programming skills, the ‘Learning from Demonstration’ (LfD) approach offers a promising solution. This strategy, inspired by human interactions, enables novice users to instruct robots through demonstrations instead of navigating the complexities of coding [11]. Through LfD, users can seamlessly impart tasks to robots, making the integration of robotic assistance more accessible and user-friendly for individuals without programming expertise.

One of the key goals in LfD is to enable a robotic system to efficiently learn and execute tasks with a minimal set of demonstrations. Achieving this goal is heavily contingent upon the quality of the provided demonstrations as well as modality of demonstration [10], as it directly shapes the overall competency of the robotic system. According to Laskey et al. [172], LfD algorithms fall into two broad categories: robot-centric and human-centric. In the former type, the robotic system initiates its learning process from a given set of demonstrations and then actively learns from its surroundings, incorporating feedback provided by the demonstrator [33]. On the other side, the human-centric approach involves learning an LfD model from a set of demonstrations provided by novice users; an example of this approach is imitation learning [28]. In the literature, researchers have devoted considerable attention to the robot-centric side, with comparatively less emphasis on the human-centric aspect for enhancing LfD [32].

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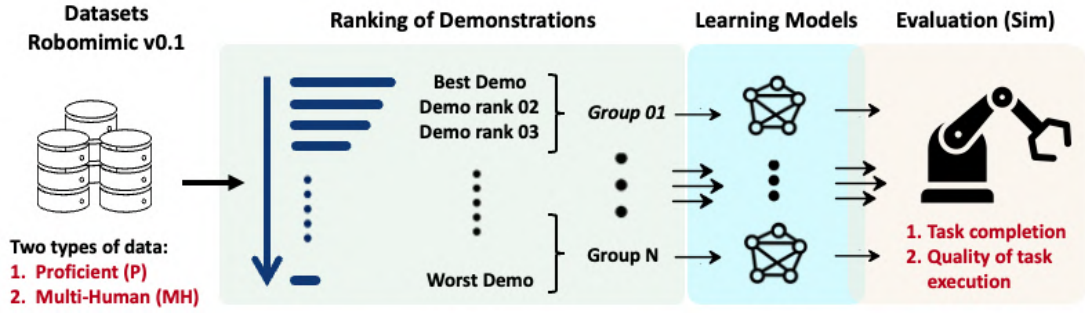


FIGURE 3.1: **Overview** The given datasets have been evaluated based on the quality features, including manipulability and joint-space jerk, and ranked them from high to low quality. At next step, we trained an LfD model for each group, where N_G denotes the total number of groups, followed by rollouts the learned LfD model in order to compute the task performance and assess the quality of task execution based on the quality features.

In the domain of LfD, the *quality of demonstration* represents the manipulability and smoothness exhibited by novice users [10]. Meanwhile, the *quality of task execution* measures how well the robot complete the learned task, taking into consideration features such as manipulability [173] and joint-space jerk [52]. This dual focus on both the quality of demonstration and the quality of task execution significantly contributes to the enhanced performance and adaptability of the robotic system within the LfD paradigm.

Existing evaluation of demonstration’s quality mainly focuses on appraising the robot’s task performance, classifying demonstrations as either high-quality or low-quality based on the success or failure of the task [35, 51]. While quality assessment of task performance is crucial for human-centric LfD, previous approaches do not allow for assessment of individual demonstration. Instead, the task performance’s evaluation is typically applied to a set of demonstrations utilized for training an LfD model—a machine learning model that reproduces actions observed during demonstrations—employing task performance as the primary metric. The inclusion of a learning phase to train the LfD model makes this process potentially time-intensive. Furthermore, the underlying reason for the given demonstrations being of high or low quality remains unclear to the user. In response to these limitations, it becomes crucial to systematically measure and assess each demonstration, taking into consideration various quality features. Feature-based methods [115, 116] present a promising approach to address the mentioned limitations, specifically the issues of being time-intensive and lacking clarity on why a demonstration is classified as high or low quality. Therefore, this study aimed to investigate a set of quality features by quantifying individual demonstrations at each step based on quality features and subsequently assessing the impact of the demonstration’s quality on task performance and the quality of task execution.

Research Question 01

How does the quality of demonstration impact both robot task completion and the quality of task execution?

This study makes three primary contributions: firstly, it advocates for the utilization of motion-based features to quantify the quality of individual demonstrations at an early stage and regardless of learning an LfD model. Secondly, it highlights the transferability of quality features from provided demonstrations to the corresponding LfD model generated motions. Finally, it posits that our quality features can serve as an indicator of task performance when used in a diverse set of demonstrations.

3.1 Background and Related Work

3.1.1 Assessing Novice Users Input in LfD

Several LfD algorithms assume that users are proficient and can provide optimal demonstrations [10]. However, this assumption often neglects the heterogeneous nature of human demonstrators, leading to sub-optimal demonstrations [48]. The quality of demonstrations can vary from user to user, influenced significantly by their understanding of the task and their capabilities to provide effective demonstrations [174].

Pais et al. [175] proposed three metrics to evaluate the user’s ability, including maneuvering the tool, consistency in teaching, and arm coordination. The metric such as consistency in teaching may not inherently depict the quality of a demonstration, as users could potentially exhibit similar errors across different demonstrations. Fischer et al. [176] compared teaching modalities in LfD, and used three common issues associated with novice operators: 1) self-collision, 2) singularity, and 3) excessive force on the end-effector. However, multiple ways of controlling the robot have been studied but did not evaluate the impact of these issues on the learned LfD model.

Recently, Sakr et al. [35] have quantified the quality of demonstrations based on: 1) task performance, model evaluation on the same task as seen during the demonstrations, and 2) generalization performance i.e. model evaluation on adjacent task to the learned task. The distinction between low- and high-quality demonstrations was established by training an LfD model on a set of demonstrations and then assessing whether the learned LfD model attained a success threshold. However, the findings are contingent on both the learning algorithm and task performance, which may loosely reflect the actual quality of demonstrations. To evaluate the diversity of demonstrations (i.e., how well demonstrations cover the relevant state and task space) across task space, Sena et

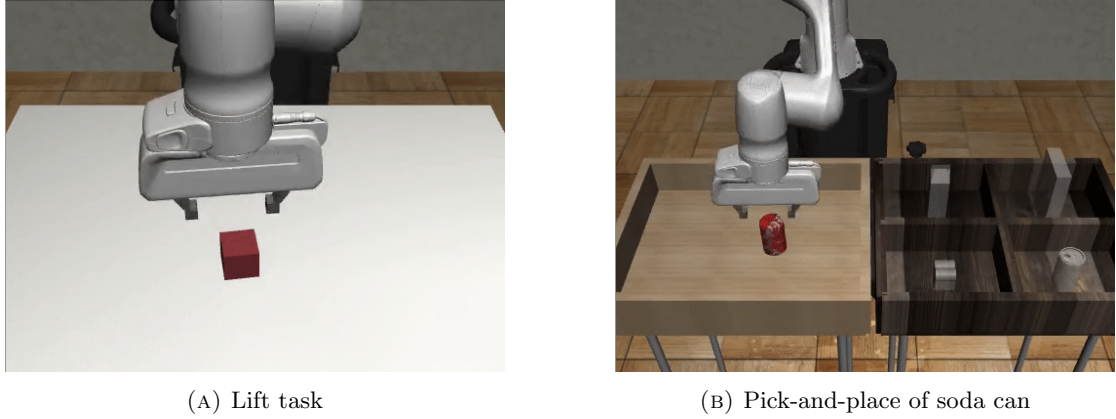


FIGURE 3.2: Manipulation tasks using Robosuite platform [177]

al. [50] proposed two metrics: 1) teaching efficacy and 2) teaching efficiency. Teaching efficacy gauges the extent to which a robot can generalize the learned task from given demonstrations while teaching efficiency illustrates the achievement of a task with the minimum number of demonstrations.

These metrics alone were insufficient for assessing the quality of individual demonstrations, as they primarily addressed the issue of data sparsity. Furthermore, the concept of entropy has been utilized to identify areas where additional demonstrations are required for effective robot learning [161]. The entropy approach is an interesting but focuses on diversity of the demonstrations instead of quantification of individual demonstrations at each step.

3.1.2 Assessing Quality of Task Execution

To quantify the quality of task execution, previous research collected six core quality features: *manipulability*, *jerk* at Cartesian and joint-space, *trajectory length* in Cartesian and joint-space, and *robot's joint limits* [51]. These quality metrics were used to quantify the quality of demonstrations [111]. Among these quality features, Chen et al. [52] investigated the importance of smoothness in robot learning, where they proposed an approach to identify and eliminate noises by providing alternative optimal control commands to the robot. Secondly, the manipulability is an important feature to consider because it directly impacts the maneuverability of the robotic system [173]. Furthermore, Jaquier et al. [178] proposed an interesting approach to learn and replicate manipulability ellipsoids from provided demonstrations. However, we aimed to investigate the impact of quality features irrespective of learning particular feature from given demonstrations.

Our work introduced innovation by explicitly assessing individual demonstration based on quality features and evaluating their influence on both task performance and the quality of task execution. We selected two quality features—manipulability and joint-space jerk—based on the relationships established by [10]. To clarify, high-quality demonstrations are those with the highest measures of these features, while low-quality demonstrations have the lowest measures. We hypothesized:

Hypotheses

H1 *The quality of demonstration correlates with task performance.*

H2 *The quality of demonstration correlates with the quality of task execution.*

3.2 Features Formulation

In this section, we present the mathematical formulation of motion-related quality features, including manipulability and joint-space jerk. Additionally, we provide a detailed description of the ranking criteria applied to the given demonstrations.

3.2.1 Manipulability

Manipulability measure quantifies how far the current configuration of a robotic system is from a singular configuration, where the robotic system cannot move in a specific direction [179]. In assessing the demonstrations, higher measures of manipulability correspond to high-quality demonstrations, while the opposite holds for lower measures of manipulability.

Consider a robot with n degrees of freedom whose joint variables are denoted by $\vec{q} = \langle q_0, q_1, \dots, q_n \rangle$, where $q_j \in \mathbb{R}$. A demonstration (D) can be defined as a joint-space trajectory $D = \langle \vec{q}_0, \dots, \vec{q}_T \rangle$ over the time period $0 \leq t \leq T$, where $\vec{q}_t$ and T represent the joint configuration of the robot at time step t and the total duration of the demonstration, respectively.

Let $\mathcal{D} = \langle D_1, D_2, \dots, D_K \rangle$ depict the set of all given demonstrations, where D_i indicates the i^{th} demonstration. The total number of demonstrations is represented by K , where $K = |\mathcal{D}|$. For a task with m dimensions, the Jacobian matrix $J(\vec{q}) : \mathbb{R}^n \rightarrow \mathbb{R}^{m \times n}$ is computed for the given joint configuration ($\vec{q}$) of the robot. The manipulability measure (ω) for demonstration D_i at time step t is then expressed as:

$$\omega(D_i, t) = \sqrt{\det J(\vec{q}_t) J'(\vec{q}_t)} \quad (3.1)$$

Finally, we computed the manipulability based metric $\alpha_i \in \mathbb{R}$ for demonstration D_i , by finding the average difference between the manipulability measure at each time step and the global maximum value $r = \{\max\langle\vec{\omega}(D_1), \vec{\omega}(D_2), \dots, \vec{\omega}(D_K)\rangle\} \in \mathbb{R}$, where $\vec{\omega}(D_i) = \langle\omega(D_i, 0), \dots, \omega(D_i, T)\rangle$.

$$\alpha_i = \frac{1}{|D_i|} \sum_{t=0}^T (r - \omega(D_i, t)) \quad (3.2)$$

A higher α value indicates a poorer manipulability measure, representing a low quality demonstration, and vice versa.

3.2.2 Joint-space Jerk

For a robot joint index j in a given demonstration D_i , the jerk $\zeta(D_i, j, t)$ is computed by taking the triple derivative of the robot's joint position for the given period $0 \leq t \leq T$.

Next, the mean jerk value $\bar{\zeta}(D_i, j) \in \mathbb{R}$ is calculated followed by computing a jerk based metric $\beta_i \in \mathbb{R}$ for demonstration D_i , indicating how much demonstration D_i contains jerk motion at joint-space level.

$$\bar{\zeta}(D_i, j) = \frac{1}{|D_i|} \sum_{t=0}^T \zeta(D_i, j, t) \quad (3.3)$$

$$\beta_i = \frac{1}{n} \sum_{j=1}^n \bar{\zeta}(D_i, j) \quad (3.4)$$

Based on the β value, ranking is determined with lower values indicating smoother demonstrations, while higher values correspond to a higher magnitude of jerks, indicating low quality demonstrations.

3.3 Analyses Setup

In this section, we discuss the dataset and the procedure employed for computing task performance. Additionally, we elaborate on the criteria for selecting an algorithm for training LfD models, focusing on task performance while utilizing a minimum number of demonstrations.

3.3.1 Dataset

In this study, we used the online available robomimic v0.1 dataset for the analyses [180, 181]. The details of the dataset are as follows: the demonstrations in the dataset were collected using the RoboTurk tool [182]; among multiple tasks, we considered the lift and pick and place tasks because they encapsulate fundamental manipulation skills such as grasping, lifting, and placement as shown in Figure 3.2. To investigate the influence of diverse-quality demonstrations, we considered both a “proficient” (P) and a “multi-human” (MH) dataset, with a total of 200 and 300 successful demonstrations for each manipulation task, respectively. In the multi-human dataset, the demonstrations were gathered by six operators with varying expertise, comprising two proficient, two normal, and two worst performers. Each individual contributed a total of 50 demonstrations, resulting in a dataset with mixed quality. On the other side, the proficient dataset consists of demonstrations provided by expert user.

3.3.2 Performance Criteria

To compute the success rates (or robot task performance), we utilized the learned LfD models—obtained using one of the LfD algorithms—to execute the same task as observed during the demonstrations. In both the demonstration and execution phases, the initial pose of the object was set randomly. For the lift task, the learned model requires to successfully lift a cube to a specific height of $0.06m$, while for the pick-and-place of soda can, the robot needs to pick up a soda can and accurately place it in its designated location (see Figure 3.2b).

3.3.3 Learning Algorithms

In real-world scenarios, it is often impractical for human users to provide a large number of demonstrations; therefore, one of the objectives in LfD is to train a model with a minimum number of demonstrations [24]. In this pursuit, we propose to identify which of the algorithm provided by Mandlkar et al. [180] exhibits the best performance with a minimal number of demonstrations. We test: Behavioral Cloning (BC) [183], Hierarchical Behavioural Cloning (HBC) [184], Batch-Constrained Q-Learning (BCQ) [185], and Implicit Reinforcement without Interaction at Scale (IRIS) [86].

The BC algorithm executes simple regression over a sequence of state-action pairs in the given dataset. Several variants of BC algorithm, including BC based on Gaussian mixture model (GMM), have been investigated for different types of manipulation tasks [180]. The HBC and IRIS algorithms are identical, except the latter includes

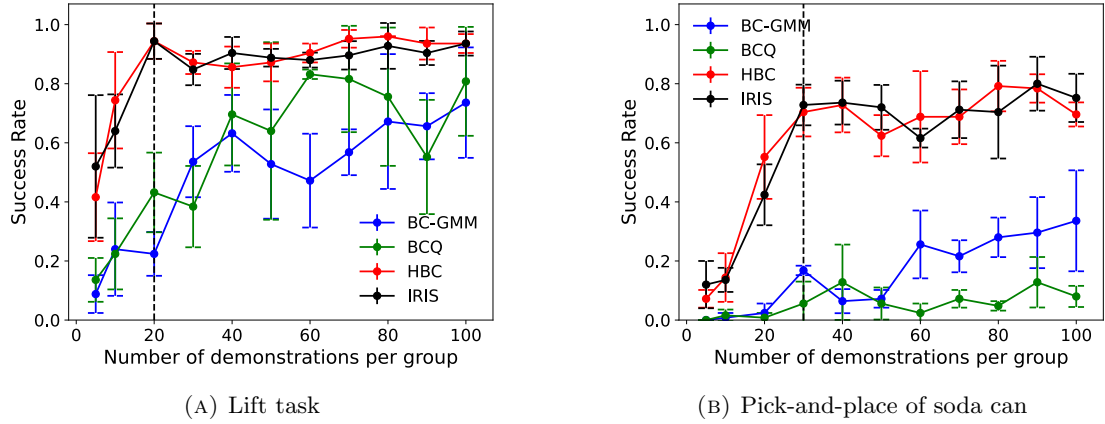


FIGURE 3.3: The performance of algorithms for the given different number of demonstrations per group, considering both the (a) lift task and (b) pick-and-place of soda can.

the addition of a value function in a high-level mechanism to locate a state with the maximum expectation. In this way, the IRIS algorithm is suitable for identifying near optimal state-action pair. Lastly, the BCQ algorithm is an offline batch reinforcement learning [185]. In this study, we considered an offline learning, where the algorithms are not permitted to consider additional samples.

For each algorithm, we computed the success rates — representing the number of successful task executions divided by the total number of rollouts [180] — for each set of demonstrations. The term “rollout” represents the deployment of the learned LfD model on a robotic system to evaluate its performance on the given task. To determine the minimal number of demonstrations to achieve an acceptable success rate, we devised 10 groups of varying sizes ranging from 10 to 100 demonstrations, with an incremental factor of 10 demonstrations. Additionally, we devised one group which consist of 5 demonstrations. Overall, this resulted in training 44 models per task, with 11 models for each algorithm. To validate the effectiveness of the models’ performance, we employed the k-fold cross-validation technique for each group.

3.3.4 Experiment Parameters

Observing the learning curves [180], a suitable number of epochs for both the lift and pick-and-place tasks is 250, utilizing a batch size of 100. Next, the learned model was subjected to 25 rollouts after every 50 epochs, allowing the computation of success rates. To ensure the adaptability of the learned model to an actual robotic system, the rollouts were performed on the same robotic system using the Robosuite platform [177]. Throughout the experiments, the demonstration data was partitioned into training and validation sets with a split ratio of 80% and 20%, respectively.

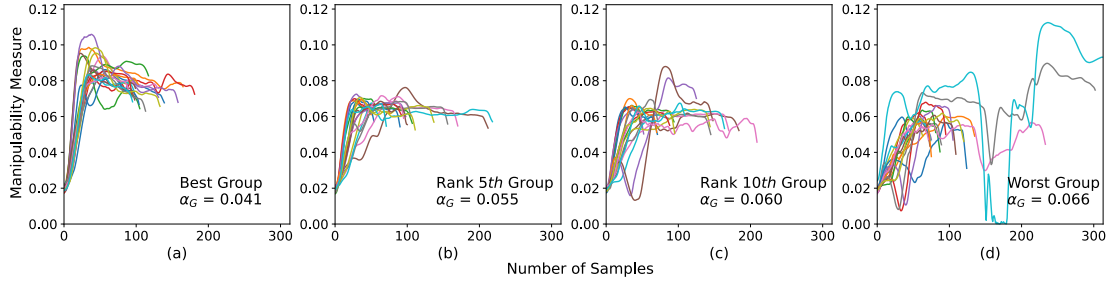


FIGURE 3.4: Each sub-figure illustrates ranked demonstrations based on the manipulability feature, where α_G represents the average of manipulability based metrics (α) of the corresponding group. In Figure 3.4d, the worst demonstration (cyan line) in the dataset falls to zero values for some interval, indicating a singular configuration.

3.4 Selection of Learning Algorithm

In our preliminary analyses, the algorithm with the minimum number of demonstrations required to achieve an acceptable success rate of 90% and 70%, values observed from the learning curves where the algorithm converged [184], for the lift and pick-and-place tasks, respectively. For each algorithm, the models were trained on the same set of demonstrations using 5-fold cross-validation with same performance criteria and experimental parameters. Additionally, the multi-human dataset was used for both manipulation tasks: lift task and pick-and-place of soda can.

Regarding BC-GMM algorithm, the success rates are significantly low for a small number of demonstrations; however, it improves with the increasing number of demonstrations as shown in Figure 3.3a. Each algorithm, as illustrated in Figure 3.3b, exhibited lower success rates as the task complexity increased from the simple lift task to the relatively complex pick-and-place of soda can. For example, the success rates of BC-GMM algorithm are almost halved for the pick-and-place task as compared to the lift task. In addition, the BCQ algorithm also exhibited poor task performance, less than 0.2 for all the given sets of demonstrations as depicted in Figure 3.3b. In the case of HBC and

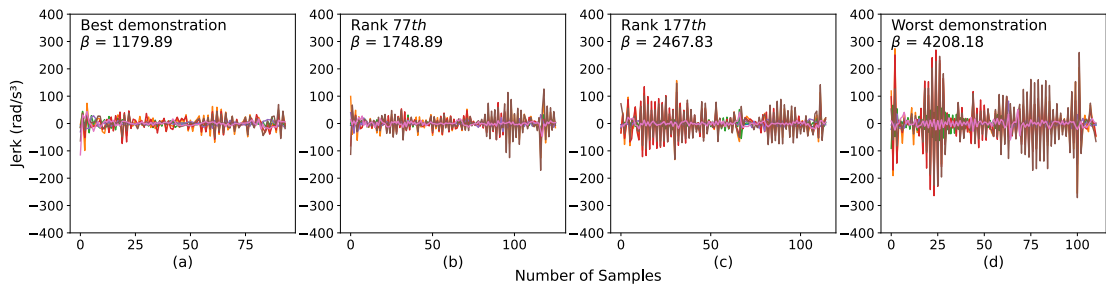


FIGURE 3.5: Each sub-figure depicts a ranked demonstration based on the jerk feature. The worst demonstration exhibits jerky motion throughout the entire length of the demonstration, while the best demonstration indicates smoother motion profiles.

TABLE 3.1: Statistics of dataset based on quality features along with correlation coefficients and the corresponding p-values.

Stats	Manip	mean $\pm$ std mean $\pm$ std	Lift task		Pick-and-Place Soda Can	
			P	MH	P	MH
	Jerk		0.032 ± 0.004 685 ± 212	0.057 ± 0.007 1401 ± 1256	0.035 ± 0.009 1940 ± 513	0.042 ± 0.011 3031 ± 1868
H1	Manip	corr	0.17	-0.11	-0.28	0.11
		p-value	0.635	0.680	0.590	0.750
	Jerk	corr	0.38	-0.60	0.31	-0.66
		p-value	0.270	0.019	0.55	0.036
H2	Manip	corr	0.68	0.78	0.77	0.60
		p-value	< 0.001	< 0.001	< 0.001	< 0.001
	Jerk	corr	0.75	0.84	0.86	0.73
		p-value	< 0.001	< 0.001	< 0.001	< 0.001

IRIS algorithms, the success rates showed satisfactory outcomes for both manipulation tasks as shown in Figure 3.3a and Figure 3.3b.

A notable drop in task performance was observed, particularly in the case of the BCQ algorithm for the pick-and-place of soda can. One possible explanation is that the BCQ algorithm extrapolates to unseen states to find a solution, making it challenging to accomplish complex manipulation tasks. Similar issue has been addressed for grid world problems as well [86]. From the results, we noticed that the IRIS algorithm requires more data as the task complexity increases to yield satisfactory results. One possible reason is that the IRIS algorithm, due to its limitation [86], demands every possible state of the object in the training dataset. Therefore, for complex manipulation tasks, more data is needed to cover as many states as possible in the case of IRIS algorithm.

Overall, according to the criterion of achieving the maximum success rate, the minimum number of demonstrations is 20 per group for the lift task, as shown in Figure 3.3a, while the minimum number is 30 per group for the relatively complex pick-and-place of soda can, as illustrated in Figure 3.3b. Both algorithms, HBC and IRIS, exhibited satisfactory performance. Given the application of the IRIS algorithm in handling diverse and suboptimal demonstrations [86], we selected the IRIS algorithm for our analyses.

3.5 Quality Features and Ranking of Demonstrations

In this section, we present the method followed for a series of experiments aiming at testing our hypotheses — high quality demonstrations will correspond to higher task performance (H1) and will attain an efficient task execution (H2). As a first step, we discuss here the ranking of demonstrations based on the quality features. For each

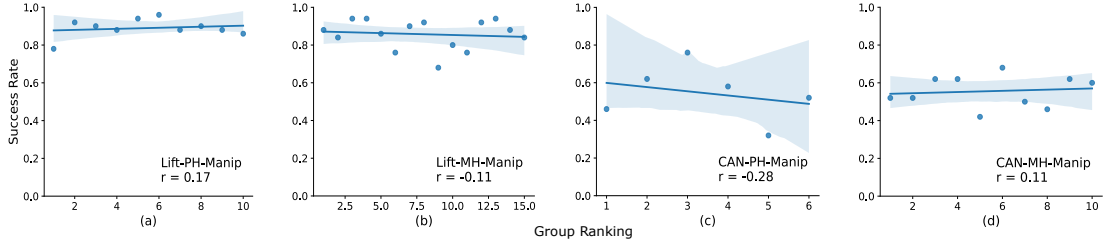


FIGURE 3.6: Correlation between input ranked groups based on the manipulability feature and the corresponding success rates

demonstration, considering both manipulation tasks provided by proficient and multi-human, we computed the metrics based on manipulability and jerk features using (3.2) and (3.4), respectively. Finally, we ranked all demonstrations from high to lower value, creating groups of 20 and 30 consecutively for both the lift and the pick-and-place of soda can tasks, respectively. The best group comprises top-ranked demonstrations while the worst group contains low-ranked demonstrations. To illustrate the quality diversity among the provided demonstrations, the mean and standard deviation based on the quality features are summarized in Table 3.1.

3.5.1 Manipulability

Considering the multi-human dataset, the manipulability curve $\vec{\omega}(D)$ of each demonstration for the lift task can be represented as shown in Figure 3.4. The best group, as illustrated in Figure 3.4a, comprises demonstrations with high manipulability measures, with average group α_G value 0.041. The term α_G represents the average of manipulability based metrics (α) of the corresponding group. As the value of α_G increases, the corresponding demonstrations in the group gradually fall to the lower manipulability measures as shown in Figure 3.4. The worst group consists of demonstrations with average group α_G value 0.066, including demonstration that contain zero values for some interval, representing a singular configuration, as shown in Figure 3.4d.

3.5.2 Joint-space Jerk

For the pick-and-place of soda can using proficient dataset, the jerky motion of multiple demonstrations with varying magnitude are depicted in Figure 3.5. The best demonstration showcases the lowest jerk magnitude for all joints of the robotic system, as illustrated in Figure 3.5a, with a β value 1179.89. Furthermore, the value of β increases with the jerk magnitude, where the worst demonstrations exhibiting high magnitude with β equals to 4208.18, as illustrated in Figure 3.5d.

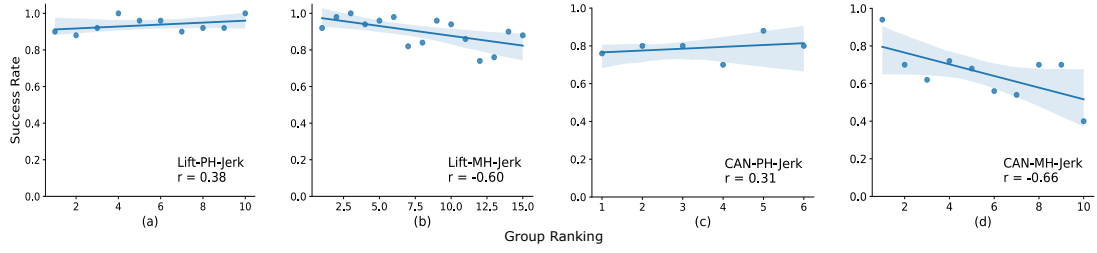


FIGURE 3.7: Correlation between input ranked groups based on the jerk feature and the corresponding success rates

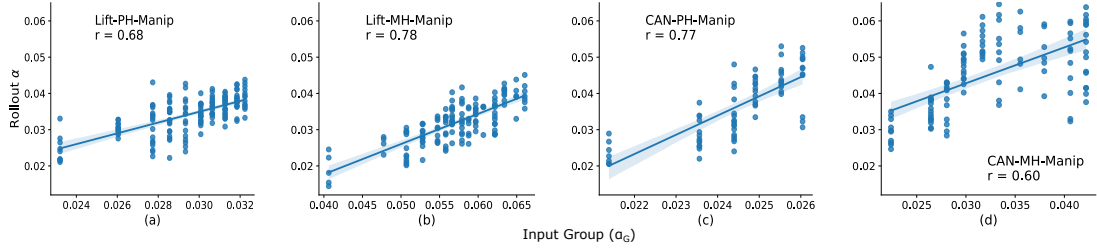


FIGURE 3.8: Correlation between input ranked groups based on the manipulability feature and the corresponding rollouts.

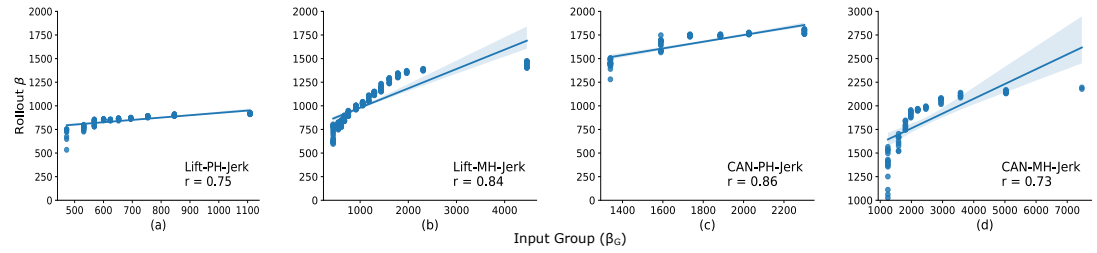


FIGURE 3.9: Correlation between input ranked groups based on the jerk feature and the corresponding rollouts.

3.6 Results

As per the results on the performance of various LfD algorithms on the lift and pick-and-place tasks (see Section 3.4), we choose to use IRIS in the following experiments. We trained a series of IRIS-LfD models for both manipulation tasks and analyzed the correlation between input ranked groups and corresponding task performance (H1), as well as the quality of task execution (H2). The findings of each study are presented in detail in the following subsections.

3.6.1 Study 01 (H1)

To evaluate the first hypothesis, we computed the correlation coefficients between ranked groups and the corresponding success rates (i.e. task performance). As depicted in Figure 3.6, there is no apparent impact of the manipulability feature on task performance in

both manipulation tasks as well as both datasets (P and MH). Moreover, the p-values exceed 0.05 in all cases. Regarding the jerk feature, the correlation coefficients also show no effect on task performance in the case of the proficient (P) dataset as shown in Figure 3.7a and Figure 3.7c. However, we found strong correlations for both manipulation tasks, lift ($r = -0.60$, $p = 0.019$) and pick-and-place of soda can ($r = -0.66$, $p = 0.036$), in the dataset provided by multi-human (MH), as illustrated in Figure 3.7b and Figure 3.7d. Overall, the impact of demonstration’s quality on task performance is significant when the demonstrations are of diverse-quality.

Findings — H1

The quality of demonstrations has a significant impact on robot task completion only in joint-space jerk under multi-human dataset.

3.6.2 Study 02 (H2)

Our second hypothesis aimed at assessing if quality features were implicitly learned by our LfD algorithm. To calculate the outcomes of task execution’s quality, we assessed each rollout using the same metrics as above, α and β . We then tried to see if there was any correlation between the quality ranking of each group of demonstrations used as input and the task execution quality produced after rollout. Furthermore, we used the actual values to provide a more realistic interpretation of the outcomes. The correlation analyses were performed for each feature across both types of datasets: proficient and multi-human.

Regarding the manipulability feature, there were strong correlations between input ranked groups and the corresponding rollouts as presented in Table 3.1. For the jerk feature, the coefficients also showed strong correlations between input-ranked groups and the corresponding rollouts as shown in Table 3.1. As compared to the manipulability feature, the correlation coefficients concerning the jerk feature are higher in each scenario. The overall correlation coefficients and the corresponding p-values are summarized in Table 3.1.

Findings — H2

For each quality feature, the coefficients show a strong correlation between the input-ranked groups and their corresponding rollouts.

3.7 Discussion

3.7.1 Role of Diversity (H1)

Using the two robotmimic v0.1 datasets (P and MH), we observed no distinct correlation between input ranked groups and the success rates; however, two cases showed strong correlation with respect to the jerk feature as shown in Figure 3.7b and Figure 3.7d.

Joint-Space Jerk: Regarding the above two cases, one of the possible reasons is that the multi-human dataset includes a diverse-quality of demonstrations. For instance, in the proficient-human dataset, the coefficient of variation (CV) for the lift and pick-and-place tasks is **30.94%** and **26.44%**, respectively. Conversely, in the multi-human dataset, the CV values for these manipulation tasks are substantially higher at **89.65%** and **61.62%**. Furthermore, the corresponding p-values are below 0.05, specifically **0.019** and **0.036** as shown in Table 3.1, respectively.

Manipulability: The CV values for the lift and pick-and-place tasks using the proficient-human dataset are **12.5%** and **25.71%**, while in the multi-human dataset, the corresponding CV values are **12.8%** and **26.19%**, respectively. Because the datasets were collected using a simulator, the data exhibits a skewed distribution, particularly concerning the manipulability feature.

Both diversity and high-quality demonstrations are important but can be in tension: increasing diversity may introduce lower-quality or inconsistent demonstrations, while focusing only on high-quality data may reduce coverage. This trade-off is also influenced by factors such as environmental stochasticity, variability in initial states, and task complexity, which affect how representative and consistent demonstrations need to be.

Overall, we hypothesize that H1 may hold if we have a diverse quality of demonstrations. This result somewhat aligns with [35], although their study considered a set of demonstrations rather than individual ones. For simple tasks, such as cube lifting, we expect minimal variations in manipulability on real robots, while noticeable joint-space jerk may occur due to the potential lack of smoothness from novice users. Due to dataset limitations, we plan to further investigate this hypothesis (H1) in our future work.

3.7.2 Learning Quality Features (H2)

Based on findings in study 02, the results support our second hypothesis (H2) as shown in Table 3.1. For each quality feature, the coefficients show strong correlations between input-ranked groups and the corresponding rollouts. All of the correlation coefficients

are higher than **0.60** in both manipulation tasks and types of datasets. While the p-value for each case is below 0.05 as shown in [Table 3.1](#).

In the multi-human dataset, as depicted in [Figure 3.9b](#) and [Figure 3.9d](#), outliers with input group (β_G) values around 4500 and 7500 slightly impact the results for both manipulation tasks. Additionally, the figures illustrate consistency among ranked groups, except for the last ones.

From the results, we posit that quality features can be transferred from given demonstrations to the corresponding LfD model. Alternatively, a set of demonstrations with high manipulability measure and low jerk values will yield an LfD model that executes tasks with nearly the same metrics as the given demonstrations.

3.8 Summary

In this work, we investigated the role of quality of individual demonstration in LfD. We quantified each demonstration’s quality based on the quality features and then evaluated the impact of demonstration’s quality on both task performance and task execution’s quality. This exploration involved the training of a set of LfD models using the robomimic v0.1 dataset. In the preliminary analysis, we selected the IRIS algorithm based on its performance, requiring 20 demonstrations for the lift task and 30 for the pick-and-place of soda can. In study 01, we computed the correlation coefficients between input-ranked groups and their corresponding success rates. The findings revealed a significant correlation in two instances, specifically when the demonstrations exhibited diverse qualities. Conversely, no evident correlations emerged in the cases where the demonstrations displayed less diverse quality. Regarding study 02, the results exhibited a strong correlation between the quality of input-ranked group of demonstrations and the corresponding rollouts for each quality feature.

This signify that not only the system can learn to perform the task (i.e. typically assessed through success rate) but also how to perform the task (through quality features). We hope that this work can lead to develop new ways to assess LfD models optimizing quality features rather than only looking at success rates. Furthermore, the proposed methodology for assessing the quality of demonstrations can be applied to a diverse range of tasks.

Although this study focused on two potential quality features using the IRIS algorithm, we aim to assess additional relevant features and algorithms in future research. We plan to develop a feedback system to make capable novice users to provide demonstrations of desired quality. This study enables to identify the cut-off regions beyond which the

quality of task execution becomes unsatisfactory as shown in [Figure 3.8](#) and [Figure 3.9](#). Eventually, a better execution function can be developed to guide novice users to provide effective demonstrations.

Next Study Motivation

The findings of this chapter demonstrate that the quality of demonstrations has a significant impact on both the learned robot policy and the quality of task execution. While high-quality demonstrations generally lead to better performance, expecting novice users to consistently provide such demonstrations is often unrealistic in real-world applications. As a result, robot learning methods must be able to cope with demonstrations that are incomplete, inconsistent, or unsuccessful, rather than assuming that all demonstrations are of high quality.

This observation motivates the next chapter, which shifts the focus from evaluating demonstration quality to improving the robustness of robot learning in the presence of imperfect demonstrations. Specifically, we propose a learning algorithm that exploits both failed and suboptimal demonstrations instead of discarding them. By learning not only from successful demonstrations but also from the information contained in imperfect examples, the proposed approach enables the robot to complete the task while producing trajectories that exhibit desirable motion quality characteristics.

Chapter 4

Quality-Aware Robot Learning from Failed and Suboptimal Demonstrations

Deploying robots in real-world human environments requires learning methods that remain effective even when teaching data are imperfect. Learning from Demonstration (LfD) is attractive because it enables users to teach robots through examples rather than explicit programming [11, 39]. Yet, in practice, non-expert users naturally provide a mixture of successful, failed, and suboptimal demonstrations (i.e., demonstrations that may achieve the task but with inefficient, unstable, or low-quality execution) due to variation in skill, task understanding, and teaching consistency [10, 37, 48, 49]. This mismatch between real human data and the common assumption of near-optimal demonstrations remains a central limitation in current LfD systems [34, 35, 174].

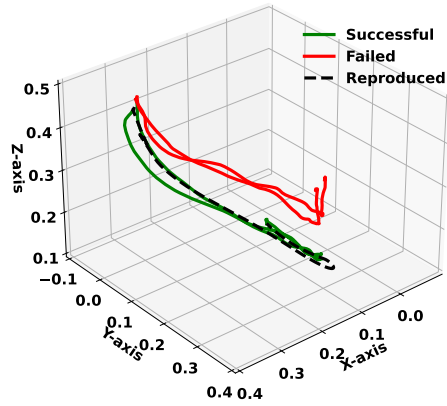
To address these limitations, various approaches have been proposed in literature [49, 99, 100]. For instance, tactile-feedback-based refinement typically starts with a policy learned from the initial demonstration set; when the robot reproduces the task and deviates toward an unsuccessful behavior, the human physically guides the robot back toward a feasible trajectory, and the corrected segments are then assigned higher importance during policy re-training [101]. In this way, failure-prone parts of the original demonstrations are not simply discarded; instead, they are iteratively corrected through interaction and converted into informative updates for the next policy. Other methods impose geometric or kinematic constraints to guide the learning process and reduce failure likelihood [102]. While these approaches improve the chance of task completion, they often pay limited attention to *how well* the task is executed, reducing performance assessment to a binary success-or-failure outcome rather than execution quality.

Many demonstration-driven imitation pipelines in LfD, including common behavior cloning [186] and probabilistic regression variants [187], are trained primarily on successful trajectories; these data indicate *what to imitate* but largely omit failed trajectories that reveal *what to avoid*. Failed demonstrations are often discarded [23, 188], despite their potential to reveal undesirable behaviors or implicit task constraints. Although some studies have attempted to incorporate failure information, they typically require explicitly labeled data or manually engineered features, which limits their scalability and generalization [49, 105, 189]. Moreover, certain approaches are designed to be robust to a small number of unlabeled suboptimal demonstrations, but still rely on a majority of expert examples to accurately identify anomalous behavior [190].

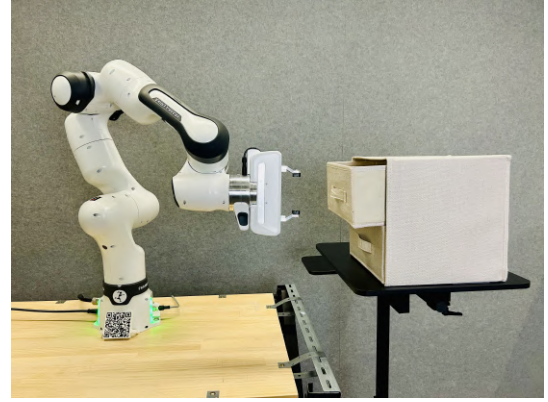
In this study, we propose a novel quality-aware LfD framework that leverages both failed and suboptimal demonstrations to learn an efficient LfD model. The proposed framework consists of two steps: policy learning and optimization step. The policy learning step learns a motion policy from the demonstration dataset, whereas the optimization step refines the generated trajectory to satisfy desired quality criteria or task constraints. Rather than discarding suboptimal data, we assign importance weights to each demonstration based on motion-related quality features [34], specifically, manipulability and joint-space jerk. Using these weighted demonstrations, our statistical learning algorithm steers the model toward high-quality demonstrations while actively repelling the influence of failed ones. Unlike conventional LfD approaches that focus solely on task completion, our framework explicitly optimizes task execution quality in an iterative manner by augmenting the training dataset with high quality trajectories until optimization convergence (i.e., no further improvements). We validated the effectiveness of our approach through extensive experiments in both 2D simulation with a 3R planar redundant robot and 3D real-world settings using a Franka 7-DoF panda robot.

4.1 Related Work

Several LfD methods have been developed to improve robot skill reproduction [49, 99–102]. For example, an iterative refinement approach was proposed, in which newer, more effective demonstrations are prioritized while older, less optimal ones are gradually down weighted [101]. A kinesthetic refinement method was introduced that allows users to physically correct demonstrations or reproductions, with the skill model subsequently updated using geometric constraints [102]. A probabilistic refinement framework was also presented, where demonstrations are weighted based on factors such as proximity to obstacles, thereby increasing the likelihood of successful task completion [103]. In contrast to these approaches, we proposed the weighting process based on quality of



(A) Demonstrations



(B) Drawer opening task

FIGURE 4.1: Learning from failed (red) and suboptimal (green) demonstrations enables the robot to successfully open the drawer by repulsing from failed demonstrations.

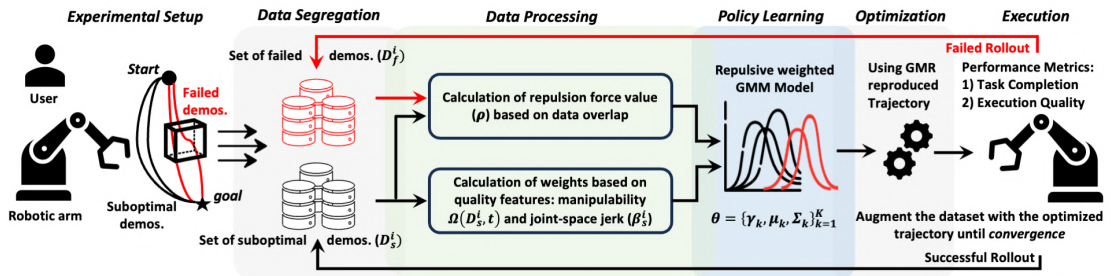


FIGURE 4.2: **Overview:** A set of demonstrations—including failed and suboptimal ones—are provided followed by computation of weights based on demonstration quality. The algorithm is designed to repel the GMM components away from failed demonstrations while attracting them toward high-quality, successful ones. The resulting GMR trajectory is then optimized using a non-linear cost function. Next, the optimized trajectory is augmented to the dataset iteratively until convergence is achieved.

demonstrations and then optimize the quality of task execution while ensuring robot task completion.

Another line of work combines LfD with reinforcement learning and failure-aware modeling [49, 104, 105, 190]. These methods typically initialize a policy from sub-optimal demonstrations and then refine it through either reward-driven interaction or explicit modeling of failed trajectories, such as GMM/GMR-based search away from failure regions [106]. While effective in specific settings, they often require either careful reward engineering, assumptions about the relationship between failed and successful demonstrations, or hand-crafted features and multiple labeled examples to generalize reliably. In addition, RL-based refinement can demand substantial environment interaction, limiting practicality in real-world deployment. In contrast, our framework directly integrates failed and suboptimal demonstrations through quality-aware weighting without relying on user-defined features or reward engineering.

Prior work has shown two major strategies for leveraging failed demonstrations: inverse-reward inference and direct statistical policy modeling. Inverse Reinforcement Learning from Failure (IRLF) extends maximum causal entropy IRL with a constrained convex formulation that incorporates both successful and failed demonstrations, improving learning when successful demonstrations are scarce [105, 191]. Complementary statistical approaches embed successful and failed trajectories into a shared skill representation and define objectives that simultaneously attract the policy toward successful behavior and repel it from failure regions [49, 192]. Collectively, these results highlight that failure data carry actionable structure rather than noise, reinforcing the value of explicitly modeling both successful and failed demonstrations during policy learning.

Research Question 02

How does an iterative refinement method improve robot task execution quality when training data includes suboptimal or failed demonstrations?

4.2 Features Formulation

This section presents the mathematical formulation of motion-related quality features, including manipulability and joint-space jerk. We also describe the ranking criteria used to evaluate the quality of the provided demonstrations.

4.2.1 Manipulability

Manipulability measure (Ω) quantifies the proximity of a robot's current configuration to a singularity, where motion in certain directions becomes limited [179]. We adopt this measure as it directly reflects the robot's dexterity, representing the quality of demonstration [10].

Consider a robot with n degrees of freedom, where the joint configuration is denoted by $\vec{q} = \langle q_1, q_2, \dots, q_n \rangle \in \mathbb{R}$. A demonstration $\mathcal{D}$ is defined as a joint-space trajectory given by $\mathcal{D} = \langle \vec{q}_0, \vec{q}_1, \dots, \vec{q}_T \rangle$ over a time interval $0 \leq t \leq T$, where $\vec{q}_t$ denotes the robot's joint configuration at time step t , and T is the total number of time steps in the demonstration. Let $\vec{\mathcal{D}} = \langle \vec{\mathcal{D}}_f, \vec{\mathcal{D}}_s \rangle$ denote the complete set of demonstrations, consisting of failed demonstrations $\vec{\mathcal{D}}_f$ and suboptimal demonstrations $\vec{\mathcal{D}}_s$, where $\vec{\mathcal{D}}_f = \langle \mathcal{D}_f^1, \mathcal{D}_f^2, \dots, \mathcal{D}_f^M \rangle$ and $\vec{\mathcal{D}}_s = \langle \mathcal{D}_s^1, \mathcal{D}_s^2, \dots, \mathcal{D}_s^N \rangle$. The superscripts M and N denote the number of failed and suboptimal demonstrations, respectively, i.e., $|\vec{\mathcal{D}}_f| = M$ and $|\vec{\mathcal{D}}_s| = N$.

For a task defined in an m -dimensional space, the Jacobian matrix $J(\vec{q}) : \mathbb{R}^n \rightarrow \mathbb{R}^{m \times n}$ is computed for a given joint configuration $\vec{q}$. The manipulability measure for demonstration $\mathcal{D}_s^i$ at time step t is then defined as:

$$\Omega(\mathcal{D}_s^i, t) = \sqrt{\det J(\vec{q}_t) J'(\vec{q}_t)} \quad (4.1)$$

$$\alpha_s^i = \frac{1}{|\mathcal{D}_s^i|} \sum_{t=0}^T (\Omega(\mathcal{D}_s^i, t)) \quad (4.2)$$

In the context of demonstration quality, higher α_s values indicate high-quality demonstrations, while lower values represents low-quality demonstrations, illustrating reduced flexibility and control.

4.2.2 Joint-space Jerk

For each robot joint $r \in \{1, 2, \dots, n\}$ in the demonstration $\mathcal{D}_s^i$, the jerk $\zeta(\mathcal{D}_s^i, r, t)$ is computed as the third derivative of the joint position over the time interval $0 \leq t \leq T$. The mean jerk value $\bar{\zeta}(\mathcal{D}_s^i, r) \in \mathbb{R}$ is then calculated for each joint, followed by the computation of an overall jerk-based metric $\beta_s^i \in \mathbb{R}$ for demonstration $\mathcal{D}_s^i$, which quantifies the extent of jerky motion at the joint-space level.

$$\bar{\zeta}(\mathcal{D}_s^i, r) = \frac{1}{T} \sum_{t=1}^T \zeta(\mathcal{D}_s^i, r, t) \quad (4.3)$$

$$\beta(\mathcal{D}_s^i) = \frac{1}{n} \sum_{r=1}^n \bar{\zeta}(\mathcal{D}_s^i, r) \quad (4.4)$$

The metric β_s^i is updated iteratively as each reproduced trajectory is added to the dataset, resulting in an augmented set denoted by $\mathcal{D}_s'$. Based on the computed β_s values, demonstrations are ranked such that higher values correspond to jerkier trajectories, while lower values indicate smoother joint-space motion, reflecting higher-quality demonstrations.

4.3 System Design

In this section, we present the technical details of the proposed framework shown in [Figure 4.2](#). We first collect demonstrations, including failed and suboptimal examples,

and partition them into separate datasets. The suboptimal set is then quantified using quality-related features, and a repulsion term is computed considering both types of demonstrations. Next, we train a repulsive weighted GMM and optimize the reproduced trajectory while preserving task completion. The optimized trajectory is evaluated on the robot and iteratively added to the training set until no further performance improvement is observed.

4.3.1 Collecting and Quantifying Demonstrations

Our approach begins with a set of demonstrations $\vec{\mathcal{D}} = \langle \vec{\mathcal{D}}_f, \vec{\mathcal{D}}_s \rangle$, comprising both failed demonstrations $\vec{\mathcal{D}}_f$ and suboptimal demonstrations $\vec{\mathcal{D}}_s$. For each suboptimal demonstration, an importance weight is computed based on the motion-related quality metrics defined in Equation 4.2 and Equation 4.4. Although demonstrations in our experiments are collected via kinesthetic teaching, the framework is compatible with other input modalities such as teleoperation. To ensure temporal consistency across demonstrations, we employ simple linear interpolation for alignment, although other methods such as Dynamic Time Warping (DTW) [193] could also be used. The weights for suboptimal (unlabeled) demonstrations are computed automatically based on the motion quality. In this way, all demonstrations are processed and weighted according to their motion quality, enabling the framework to leverage both types of data during learning.

4.3.2 Repulsive Weighted GMM for LfD

To enhance robustness in LfD, particularly when the dataset includes failed demonstrations, we propose a modified weighted-GMM that not only emphasizes high-quality demonstrations but also explicitly repels the influence of low-quality or failed ones.

Let $\mathcal{P} = \{z^j\}_{j=1}^P$ represent the full set of data points extracted from both suboptimal and failed demonstrations, where j represents the demonstration index in the respective failed and suboptimal dataset. The demonstrations are grouped into two categories: failed demonstrations, denoted by $\vec{\mathcal{D}}_f$, and suboptimal (but successful) demonstrations, denoted by $\vec{\mathcal{D}}_s$. We define $\mathcal{P}_f$ and $\mathcal{P}_s$ as the sets of indices corresponding to data points from $\vec{\mathcal{D}}_f$ and $\vec{\mathcal{D}}_s$, respectively. For each data point $z^j \in \mathcal{P}_s$, an importance weight $w_j \in \mathbb{R}^+$ is computed based on motion-quality metrics such as manipulability and joint-space jerk. In contrast, for data points $z^j \in \mathcal{P}_f$, a repulsion weight $\rho > 0$ is calculated to push the model away from regions associated with failed behaviors, while also accounting for the distribution of successful demonstrations.

This weighting scheme allows the GMM to be biased toward reproducing high-quality motions while discouraging imitation of failure-prone patterns, thereby improving both task performance and execution quality. We formulate the following penalized log-likelihood objective:

$$\mathcal{L}(\theta) = \sum_{j \in \mathcal{P}_s} w_j \log \left(\sum_{k=1}^K \gamma_k \mathcal{N}(z^j; \mu_k, \Sigma_k) \right) - \rho \sum_{j \in \mathcal{P}_f} \log \left(\sum_{k=1}^K \gamma_k \mathcal{N}(z^j; \mu_k, \Sigma_k) \right) \quad (4.5)$$

where $\theta = \{\gamma_k, \mu_k, \Sigma_k\}_{k=1}^K$ and ρ represent the GMM parameters and repulsion weight, respectively. When $\rho = 0$, the formulation reduces to the standard weighted-EM on successful demonstrations.

EM Algorithm with Repulsion

To robustly learn from both suboptimal and failed demonstrations, we derive an EM-style algorithm that mirrors standard GMM estimation but modifies the M-step updates to incorporate negative contributions from failure-prone data points.

E-Step: For each data point $z^j \in \mathcal{P}$, the soft assignment (responsibility) of component k is computed as:

$$p_{k,j} = \frac{\gamma_k \mathcal{N}(z^j; \mu_k, \Sigma_k)}{\sum_{\ell=1}^K \gamma_\ell \mathcal{N}(z^j; \mu_\ell, \Sigma_\ell)}. \quad (4.6)$$

M-Step: The dataset $\mathcal{P}$ is partitioned into two disjoint index sets, including $\mathcal{P}_s$ and $\mathcal{P}_f$. Each data point $z^j \in \mathcal{P}_s$ is assigned an importance weight $w_j \in \mathbb{R}^+$, while each data point $z^j \in \mathcal{P}_f$ is assigned a fixed repulsion weight $\rho > 0$. We define the following component-wise sufficient statistics: $E_k^+ = \sum_{j \in \mathcal{P}_s} w_j p_{k,j}$; $E_k^- = \sum_{j \in \mathcal{P}_f} \rho p_{k,j}$; finally, $\tilde{E}_k = E_k^+ - E_k^-$. The GMM parameters are then updated using the effective counts:

$$\gamma_k \leftarrow \frac{\tilde{E}_k}{\sum_{m=1}^K \tilde{E}_m} \quad (4.7)$$

$$\mu_k \leftarrow \frac{1}{\tilde{E}_k} \left(\sum_{j \in \mathcal{P}_s} w_j p_{k,j} z^j - \sum_{j \in \mathcal{P}_f} \rho p_{k,j} z^j \right) \quad (4.8)$$

$$\Sigma_k \leftarrow \frac{1}{\bar{E}_k} \left(\sum_{j \in \mathcal{P}_s} w_j p_{k,j} (z^j - \mu_k)(z^j - \mu_k)^\top - \sum_{j \in \mathcal{P}_f} \rho p_{k,j} (z^j - \mu_k)(z^j - \mu_k)^\top \right) \quad (4.9)$$

This formulation can be viewed as a generalization of the standard EM algorithm, where data points from suboptimal demonstrations exert a positive attractive influence on the GMM components, while failed data points exert a negative repulsive influence. As a result, the model is encouraged to prioritize high-quality behaviors while actively avoiding undesirable or failure-prone regions in the trajectory space.

Computation of Repulsion Weight

We define the repulsion weight ρ using the symmetric overlap $\mathcal{O}$ between the distributions of successful and failed demonstrations:

$$\mathcal{O} = \frac{1}{2} \left(\frac{1}{|\mathcal{P}_s|} \sum_{j \in \mathcal{P}_s} \mathcal{N}(z^j | \mu_f, \Sigma_f) + \frac{1}{|\mathcal{P}_f|} \sum_{j \in \mathcal{P}_f} \mathcal{N}(z^j | \mu_s, \Sigma_s) \right), \quad (4.10)$$

$$\rho = \bar{w}_s \exp(-\log(1 + \mathcal{O})), \quad (4.11)$$

where μ_s, Σ_s are the mean and covariance of the successful demonstrations, μ_f, Σ_f are those of the failed demonstrations, and $\bar{w}_s = \frac{1}{|\mathcal{P}_s|} \sum_{j \in \mathcal{P}_s} w_j$ is the mean importance weight of successful data points. Although alternative measures exist for computing the overlap ratio, such as the Kullback–Leibler divergence [194], this approach is known to be highly sensitive to the tails of the distributions [195]. In contrast, our overlap-based ρ focuses on the average PDF evaluated at actual data points, thereby reducing the influence of extreme tails.

4.3.3 Optimization

To refine the trajectory generated by GMR, we introduce a post-processing optimization step that minimizes a composite cost function subject to joint-space constraints, as defined in Equation 4.12. The joint-space trajectory is optimized to satisfy multiple objectives. The first term minimizes the Cartesian tracking error between the actual end-effector position, obtained via forward kinematics, and the resulting Cartesian trajectory using GMR, thereby ensuring task completion. The second term promotes dexterity by penalizing low manipulability, measured from the Jacobian at each configuration. The

third term enforces joint feasibility using a logarithmic barrier that penalizes deviations from the mid-range of the joint limits, thereby avoiding joint limit saturation (i.e., joint angles close to its limits) and preserving mobility. Finally, smoothness is encouraged by penalizing joint velocities, accelerations, and jerks, which are obtained from cubic spline interpolation of the trajectory. The resulting optimization problem is solved using the L-BFGS-B algorithm implemented via the *python-scipy* library. This procedure yields a smooth, feasible joint trajectory that satisfies physical constraints while closely following the GMR prediction and promoting high-quality execution. The overall cost function can be represented as:

$$\begin{aligned}
\mathcal{C}(\mathbf{q}_{1:T}) = & \sum_{t=1}^T \left[\lambda_{\text{track}} \left\| \mathbf{x}_t - \mathbf{x}_t^{\text{desired}} \right\|^2 + \lambda_{\text{manip}} \left(-\mu(\mathbf{J}_t) \right)^2 \right. \\
& \left. + \lambda_{\text{joint}} \sum_{i=1}^n \log \left(1 + \frac{|q_{t,i} - q_i^{\text{mid}}|}{q_i^{\text{range}}} \right) \right] \\
& + \lambda_{\text{vel}} \sum_{t=1}^{T-1} \|\dot{\mathbf{q}}_t\|^2 \Delta t + \lambda_{\text{acc}} \sum_{t=1}^{T-2} \|\ddot{\mathbf{q}}_t\|^2 \Delta t \\
& + \lambda_{\text{jerk}} \sum_{t=1}^{T-3} \|\dddot{\mathbf{q}}_t\|^2 \Delta t
\end{aligned} \tag{4.12}$$

Where λ 's represent the weighting coefficients that balance the relative importance of tracking accuracy, manipulability, joint limit regularization, and smoothness, respectively. Additionally, the terms $q_i^{\text{mid}} = (q_i^{\text{max}} + q_i^{\text{min}})/2$, and $q_i^{\text{range}} = (q_i^{\text{max}} - q_i^{\text{min}})/2$.

4.3.4 Performance Criteria

The primary performance criterion in our experiments is optimizing the quality of joint-level trajectory based on motion-related quality features considering the resulting Cartesian trajectory using GMR. Unlike conventional success/failure evaluation, our approach emphasizes not only task completion but also the quality of task execution. For instance, in the 2D navigation task, the robot needs to follow the resulting Cartesian trajectory with optimized joint trajectory with respect to the quality features. Similarly, in the drawer-opening task, the robot must successfully pull the drawer open, whereas in the insertion task, it must grasp the cuboid block and partially insert it into the designated slot with optimized joint-level trajectories. If the robot fails to complete the desired task successfully, the rollout or outcome adds into the failed dataset otherwise adds into the successful dataset as shown in [Figure 4.2](#).

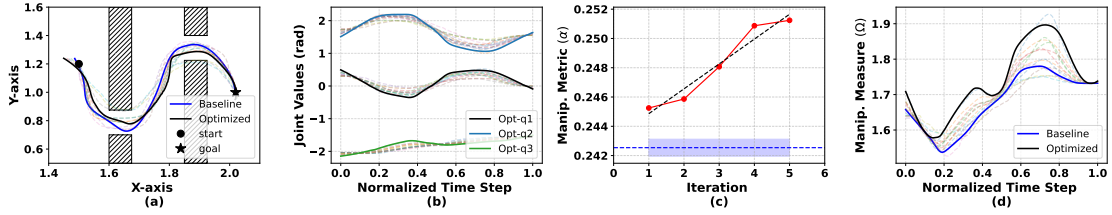


FIGURE 4.3: Subfigures (a) and (b) illustrate the Cartesian and joint-space trajectories, while (c) show the manipulability metric and its progress over successive iterations. The dotted horizontal line presents the mean of input demonstrations with variation showing by shaded region. In addition, the subfigure (d) shows the manipulability measures over iterations where the baseline corresponds to the initial GMR reproduction, while the optimized trajectory represents the final result after convergence.

4.4 Experiments

4.4.1 3R Planar Robot (2D Task)

We initially validated our approach in a controlled simulation environment using a redundant 3R planar robot, which features one additional degree of freedom relative to the 2D task space. For simplicity, each link was assigned a unit length. Demonstrations were collected using a cursor-based interface at a fixed sampling rate of 0.01 seconds, enabling a human operator to guide the end-effector along a desired path. To introduce clear distinctions between demonstration types, obstacles were placed within the workspace as shown in Figure 4.3a. Some demonstrations were deliberately guided through the obstacle and labeled as failed, while the remaining demonstrations avoided the obstacle and were treated as suboptimal but successful. This setup allowed the model to learn both *what-to-imitate* and *what-not-to-imitate*. Importance weights for suboptimal demonstrations were computed based on motion-related quality features while the repulsion weight is computed for the given failed and suboptimal demonstrations, encouraging the model to actively avoid reproducing undesirable behaviors. To determine the optimal number of mixture components in the GMM, we employed the Bayesian Information Criterion (BIC) [196], which balances model fit with complexity by penalizing overfitting.

Comparison with GMM/GMR-wEM

Figure 4.4 compares the proposed approach with the standard weighted GMM/GMR-wEM method [101] under different combinations of successful and failed demonstrations. As the number of successful demonstrations increases from one to three while keeping three failed demonstrations fixed, the proposed method consistently generates trajectories that avoid the obstacle and remain aligned with the successful demonstrations.

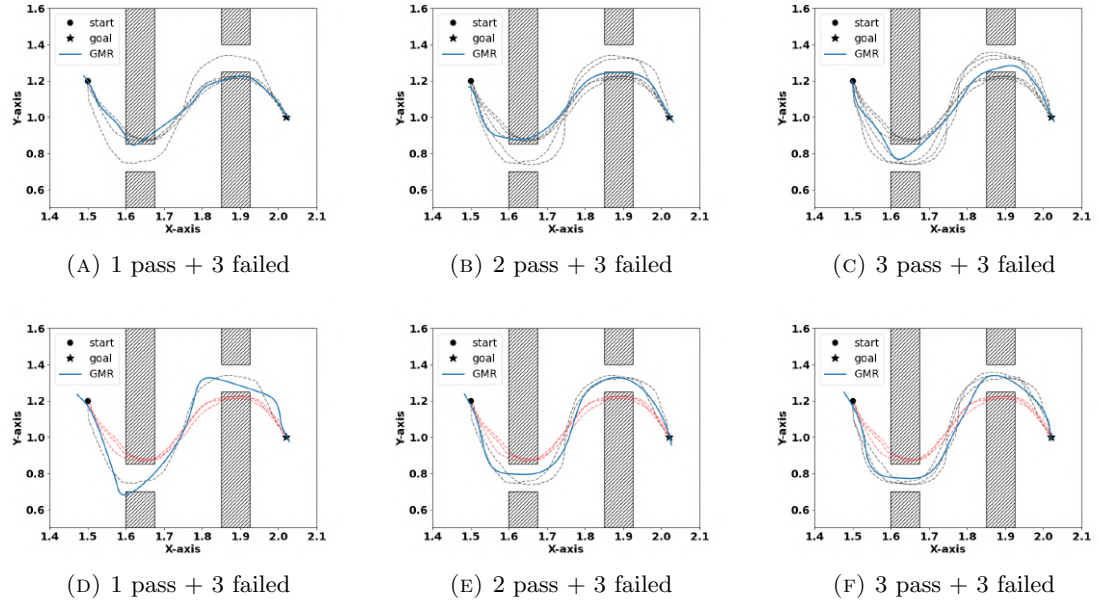
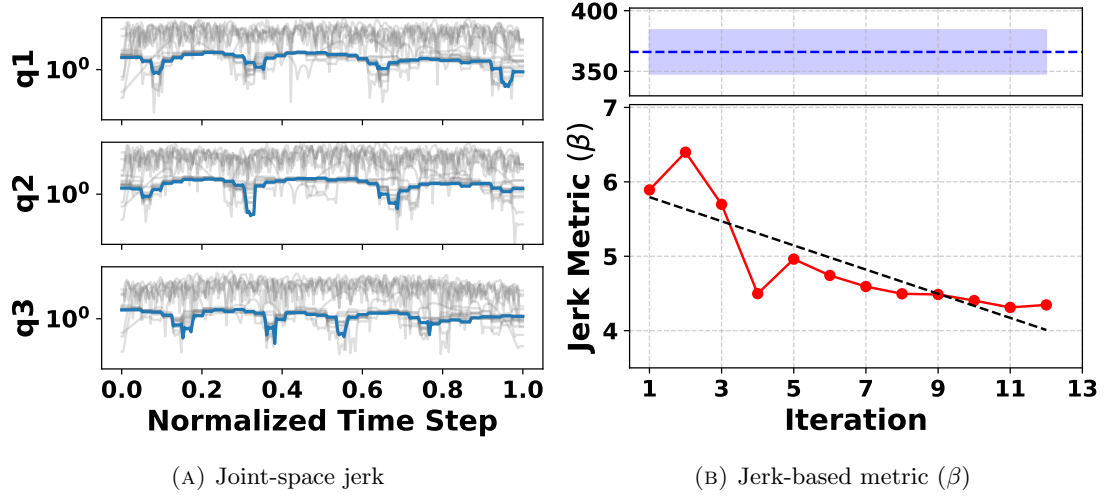
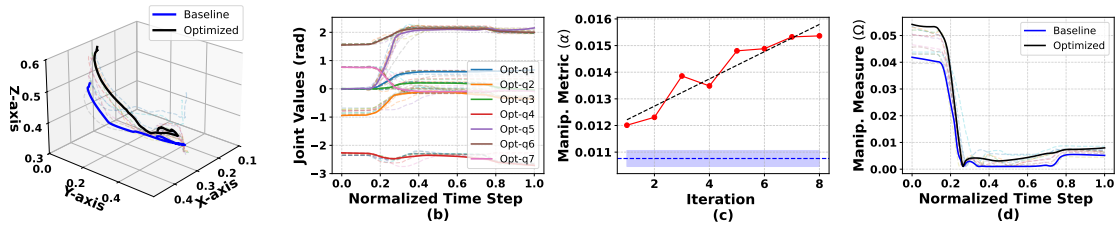


FIGURE 4.4: Comparison of the proposed approach with the standard weighted GM/GMR method [101] using different combinations of successful and failed demonstrations. Subfigures (A) and (D) compare the two methods using one successful and three failed demonstrations. Subfigures (B) and (E) present the results for two successful and three failed demonstrations, where the proposed approach successfully generates a collision-free trajectory, while the baseline method fails to avoid the obstacles. Subfigures (C) and (F) show the results for three successful and three failed demonstrations. Overall, the proposed approach consistently produces trajectories that avoid obstacles by effectively incorporating information from both successful and failed demonstrations.

In contrast, the standard weighted GMM/GMR-wEM approach is unable to effectively exploit information from failed demonstrations. Consequently, the generated trajectories either remain too close to the failed demonstrations or intersect the obstacle, particularly when only one or two successful demonstrations are available. These results demonstrate that explicitly incorporating failed demonstrations through the proposed repulsion mechanism enables the learned policy to better distinguish undesirable regions of the task space and generate safer trajectories, even when successful demonstrations are limited.

Manipulability

Figure 4.3 presents the evaluation of the learned trajectory starting from an initial configuration and reaching a predefined goal while navigating a constrained workspace with obstacles. The joint trajectories Figure 4.3b illustrate how the optimized trajectory adapts all three joint angles to meet task objectives, showing smooth transitions and

FIGURE 4.5: Joint-space jerk profiles and reproduced trajectory metric (β).FIGURE 4.6: Subfigures (a–b) show the Cartesian and joint-space trajectories, while (c–d) depict the manipulability metric (β) and its measures (Ω) over iterations.

distinct deviations from the baseline demonstrations (dashed lines). Notably, the optimized trajectory (solid curves) successfully tracks the desired Cartesian path [Figure 4.3a](#), avoiding collisions with the obstacle regions represented by hatched rectangles.

The sub [Figure 4.3c](#) subplot shows an overall increasing trend in the reproduced trajectory manipulability metric (α) across iterations, rising from approximately 0.242 to 0.250, suggesting progressive improvements in motion quality. The sub [Figure 4.3d](#) compares manipulability over time between the initial GMR-generated trajectory and the final optimized one. The optimized trajectory consistently exhibits higher manipulability values, indicating a more dexterous and stable configuration throughout execution.

Joint-space Jerk

[Figure 4.5a](#) presents a joint-wise comparison of the jerk profiles for the three revolute joints q_1 , q_2 , and q_3 . The plot contrasts the raw input demonstrations (shown in gray with reduced opacity) against the optimized trajectory (in blue). To quantitatively assess the reduction in joint-space jerk, the mean magnitude decreased substantially from

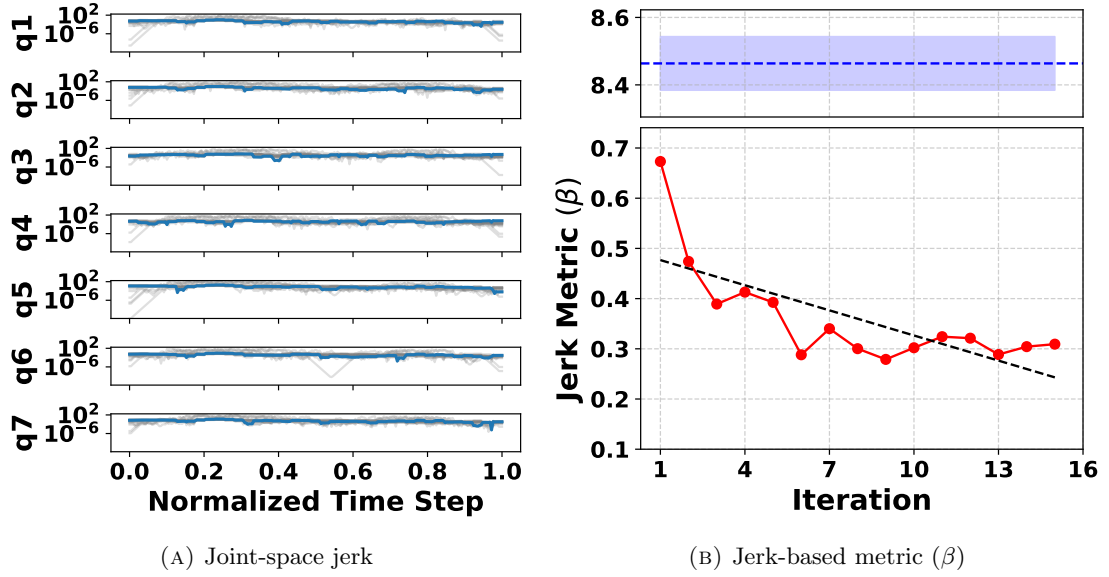
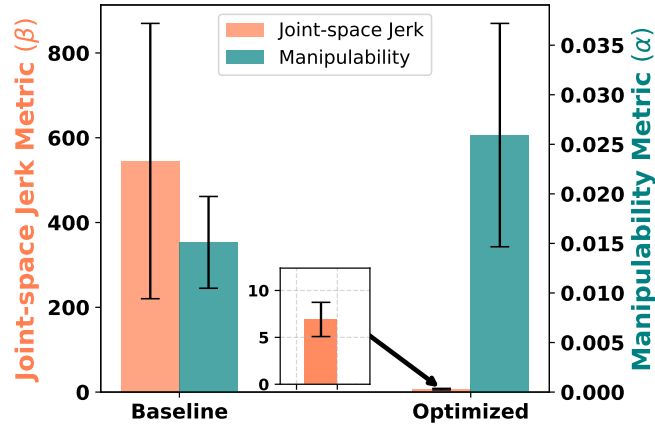
FIGURE 4.7: Joint-space jerk profiles and reproduced trajectory metric (β).

FIGURE 4.8: Baseline vs Optimized

355 ± 18 in the original demonstrations to 5 ± 0.67 in the optimized trajectory as shown in Figure 4.5b. Furthermore, the relative change per iteration was $-0.14(-2.86\%)$, while the correlation coefficient between iterations and jerk metric (β) was 0.70 with a p-value of 0.001.

It is evident that the optimized trajectory achieves a substantial reduction in jerk magnitude across all joints, particularly during motion segments where abrupt transitions were observed in the original demonstrations. This reduction in third-order derivatives reflects a smoother and dynamically consistent motion profile, achieved by explicitly penalizing jerk during trajectory optimization. We parameterize the joint-space trajectories using cubic spline control points instead of optimizing every time step directly. This reduces the dimensionality of the optimization problem and ensures smoothness

TABLE 4.1: Statistical summary of manipulability and joint-space jerk for all three tasks

	Manipulability			Joint-space Jerk		
	2D Task	Drawer Task	Insertion Task (N=18)	2D Task	Drawer Task	Insertion Task (N=18)
Relative Change per Iter.	0.001(0.60%)	0.001(3.42%)	$1.59 \pm 1.01(\%)$	$-0.14(-2.86\%)$	$-0.26(-7.22\%)$	$-12.57 \pm 4.32(\%)$
Regression Slope	0.002(0.68%)	0.001(3.65%)	$1.61 \pm 1.05(\%)$	$-0.16(-3.31\%)$	$-0.09(-4.64\%)$	$-9.88 \pm 3.21(\%)$
p -value	0.006	$\ll 0.0001$	—	0.001	0.002	—
R^2 (trend strength)	0.95	0.90	0.69 ± 0.34	0.70	0.53	0.61 ± 0.13
Baseline (mean $\pm$ std)	0.24 ± 0.001	0.01 ± 0.001	0.02 ± 0.01	366 ± 18	8.46 ± 0.08	545 ± 324
Optimized (mean $\pm$ std)	0.25 ± 0.002	0.02 ± 0.001	0.03 ± 0.01	5 ± 0.67	0.36 ± 0.09	6.9 ± 1.82

by construction, which significantly improves convergence and stability of the jerk minimization process.

4.4.2 Drawer Opening Task

Following the simulation study, we validated our approach on a real Franka 7-DoF Panda robot. Demonstrations were collected kinesthetically by physically guiding the robot through the desired motions. A total of four demonstrations were recorded: two successful demonstrations in which the robot successfully opened the drawer, and two failed demonstrations in which the drawer was not opened successfully. The experimental setup is shown in [Figure 4.1b](#).

Manipulability

To evaluate the effect of our optimization on kinematic quality, we compared manipulability values between the baseline and optimized trajectories on the 7-DoF Franka Panda robot. The optimized trajectory achieved a higher average manipulability (0.02 ± 0.001) compared to the baseline (0.01 ± 0.001). [Figure 4.6](#) presents the joint-space and Cartesian trajectories generated during the optimization process, along with the associated trajectory manipulability metric (β) and measure (Ω) values.

Joint-space Jerk

Similar to the simulation setup, we conducted a statistical comparison of joint-space jerk magnitudes to evaluate the improvement in motion smoothness achieved by our optimization. [Figure 4.7](#) illustrates the evolution of joint-space jerk across iterations, highlighting improvements in motion smoothness and the algorithm’s increasing emphasis on higher-quality reproductions. The mean of original demonstrations was 8.46 ± 0.08 , where it dropped to 0.36 ± 0.09 . The details are summarised in [Table 4.1](#).

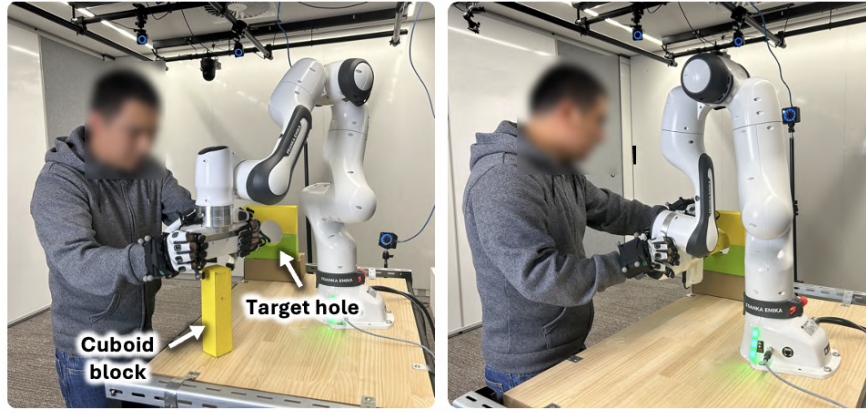


FIGURE 4.9: Cuboid block insertion task: (Left) user pick up the cuboid block; (Right) partially insert into the target hole.

4.4.3 Insertion Task (N=18)

We used an existing dataset collected from novice users ($N = 18$) [197]. The details of the dataset are provided in Section 5.4.2. The task involved teaching the robot to pick up a randomly positioned cuboid block (measuring $5 \times 5 \times 20$ cm) and partially insert it into a designated hole with a diameter of 10 cm, as shown in Figure 4.9. The dataset contains a total of 134 demonstrations from 18 users, including 120 successful and 14 failed attempts. In contrast to prior work [197], which considered only successful demonstrations, we applied our proposed optimization approach to integrate both successes and failures into the learning process.

Manipulability

As shown in Figure 4.8, manipulability improved over the course of iterations, increasing from 0.02 ± 0.001 to 0.03 ± 0.001 . Most participants showed statistically significant improvements ($p < 0.05$) in both metrics. For joint-space jerk, nearly all participants had p-values below 0.02, indicating consistently strong improvements. Manipulability also showed significant gains for many participants, though with greater variability, including a few cases with p-values above 0.05, suggesting non-significant changes in those instances. In addition, the relative change per iteration was $1.59 \pm 1.01\%$, with a linear regression slope of $1.61 \pm 1.05\%$, indicating a consistent upward trend in manipulability improvement.

Joint-space Jerk

As shown in Figure 4.8, the joint-space jerk measure decreased substantially across iterations, dropping from 545 ± 324 to 6.9 ± 1.82 . The statistical analysis further supports this improvement: p -values below 0.05 were observed for 17 out of 18 participants, confirming that the reduction is significant. The relative change per iteration was $-12.57 \pm 4.32\%$, and the linear regression slope was $-9.88 \pm 3.21\%$, highlighting a strong and consistent reduction in jerk over time.

4.5 Summary

In this study, we presented a quality-aware LfD framework based on a revised weighted GMM that incorporates both failed and suboptimal demonstrations. Unlike conventional approaches that focus solely on task success, our method explicitly optimizes motion quality through features such as manipulability and joint-space jerk. Results from simulation and real-world experiments on a Franka 7-DoF Panda robot demonstrate consistent improvements in motion quality, achieving smoother and more dexterous executions while maintaining task success.

However, the framework is inherently dependent on the choice of quality metrics. While the selected features proved effective, alternative or task-specific metrics may yield different or potentially improved outcomes. At the same time, incorporating additional features introduces trade-offs: higher-dimensional optimization can increase computational complexity, create conflicts between objectives, and reduce model interpretability. In some cases, redundant or poorly defined metrics may negatively impact performance.

Furthermore, although the approach is flexible and can be extended to other tasks and platforms, its scalability depends on how well the chosen metrics capture task-relevant behaviors. Future work should investigate automated methods for selecting and weighting quality features, as well as evaluate the framework across more diverse tasks and robotic systems. A quantitative comparison between incorporating all objectives as learning features versus post-learning optimization is beyond the scope of this thesis and represents an interesting direction for future work.

Overall, this work demonstrates that integrating quality-aware optimization into LfD enables more robust and refined robot behavior, highlighting the importance of moving beyond binary success criteria toward richer representations of task execution.

Next Study Motivation

This chapter improved the robustness of LfD by enabling robots to learn from failed

and suboptimal demonstrations. While this reduces the dependence on perfect demonstrations, the effectiveness of robot learning still depends on the quality of the demonstrations provided by the user. Understanding the factors that influence demonstration quality is therefore essential for designing LfD systems that are both effective and accessible to novice users.

This observation motivates the next chapter, which investigates how the robot’s degree of redundancy influences the demonstration process. Although kinematic redundancy provides greater flexibility by allowing multiple joint configurations to achieve the same task, it also increases the number of joints that users must manipulate during kinesthetic teaching. This introduces a potential trade-off between motion flexibility and the cognitive and physical demands placed on the demonstrator. To better understand this trade-off, we investigate how different degrees of redundancy affect human teaching performance, the quality of demonstrations, and the resulting robot learning performance.

Chapter 5

Investigating the Impact of Robot Degree of Redundancy on Learning from Demonstration¹

Learning from Demonstration (LfD) allows novice users to teach robots everyday tasks through demonstrations, eliminating the need of coding each specific task [11]. The efficacy of an LfD process can be enhanced by improving both its key elements: (1) the effectiveness of the robotic learning algorithms [28, 32, 33] and (2) the information content in the human-provided demonstrations [34, 35, 51]. In the literature, researchers have devoted considerable attention to the algorithmic approaches, with comparatively less emphasis on the human-centric aspect for enhancing LfD [32, 180, 198].

Among the available modalities for providing demonstrations [10], kinesthetic teaching offers an intuitive approach in which novice users physically guide the robot through the desired motion. This modality removes the need for additional sensing hardware, external input devices, or specialized interfaces, thereby lowering the barrier to entry for non-expert operators [10, 11]. In contrast, teleoperated teaching typically restricts user input to the end-effector motion, relying on inverse kinematics to compute the corresponding joint configurations [41, 199]. Kinesthetic teaching, however, directly captures user intent not only in Cartesian space (as in teleoperation) but also in the joint space, enabling the expression of preferences at a more granular level. As a result, it supports more comprehensive demonstrations of constrained tasks—particularly those

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involving joint limits, conceptual task constraints, or physical joint restrictions—because users can manipulate all robot joints to satisfy task requirements [45–47].

Kinematic redundancy—where a robot has more joint degrees of freedom than required for a given task—provides flexibility during kinesthetic demonstrations, since multiple joint configurations can achieve the same end-effector pose with different levels of efficiency and precision [200]. However, this flexibility also introduces complexity: novice users must navigate a larger set of feasible configurations, which can make it more difficult to select effective poses and provide successful demonstrations. At the same time, constraining redundancy (e.g., by limiting the number of joints to be manipulated) may reduce cognitive effort by narrowing the solution space, but may also reduce adaptability [201]. This trade-off highlights the importance of examining how redundancy shapes user strategies, demonstration quality, and downstream robot performance in LfD.

This study presents an exploratory investigation that has not been done before in the context of robot learning from demonstrations. Beyond a kinematic comparison of redundancy levels, it provides new insights into how users’ physical interaction patterns and demonstration quality influence the robot’s learning performance. To evaluate the impact of robot redundancy, we implemented a joint constraint, by locking one of the robot joints, to reduce redundancy by one joint degree. It is worth to highlight that this study only concerns the reduction of DoFs of the robotic mechanism (kinematic chain) and not the reduction of DoFs along “unwanted” task directions. Our experimental setup uses motion capture to record hand contacts with individual robot joints, enabling joint-level analysis that is, to our best knowledge, not available in current public datasets. The overall study pipeline is shown in Figure 5.1.

The key contributions of this study are as follows: (1) demonstrating tasks under constrained conditions imposes a higher mental workload on users; (2) task demonstration time increases as the DoR is reduced; (3) constrained condition leads to more failed attempts; (4) reduced redundancy results in increased physical interaction with the robot joints; and (5) demonstrations collected under reduced redundancy negatively affect subsequent robot performance. Furthermore, we collected a comprehensive dataset, *InteractLFD*, in this study for reproducibility and future research (available in the supplementary material).

Research Question 03

How does a robot’s degree of redundancy influence cognitive workload, demonstration time, number of failed attempts, and users’ hand placement and adjustment strategies during task demonstrations?

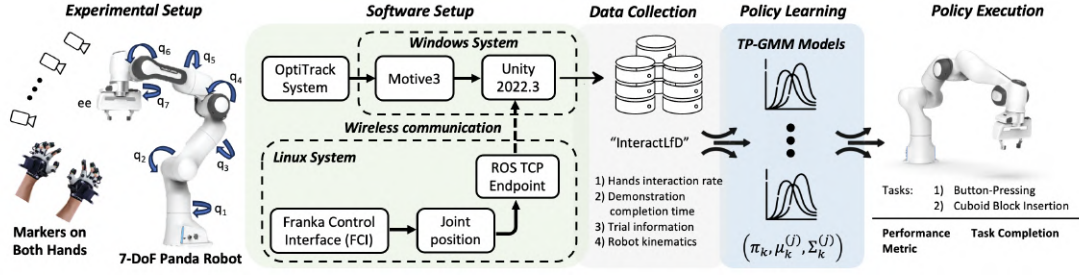


FIGURE 5.1: Study overview showing the robotic platform, motion-capture system, and experimental setup.

5.1 Background and Related Work

One of the key goals of LfD is to allow novice users to efficiently demonstrate desired tasks to the robotic systems [10]. Among different approaches of LfD [172], human-centric based approaches suppose that users are skilled and capable of providing optimal demonstrations [10]. However, this assumption overlooks the heterogeneous nature of human demonstrators, often resulting in suboptimal demonstrations [28, 48].

The quality of these demonstrations varies significantly among human demonstrators, influenced by their understanding of the task and their ability to deliver effective demonstrations [174]. In addition, the modalities of demonstration also play a crucial role [10, 43, 144], such as kinesthetic teaching shows promising results for constrained tasks [45, 202]. Eventually, LfD model learned using such demonstrations influences the robot performance [11, 34]. The quality of demonstration is shaped by several key factors as follows: first, the user’s mental workload must be considered, as higher mental workload can reduce their ability to focus, make precise adjustments, and ensure engagement during demonstrations [203]. Second, minimizing demonstration time is critical, as lengthy sessions can lead to user fatigue, reducing both the practicality and appeal of LfD for everyday tasks [43]. Third, previous research has shown that frequent failed attempts, demonstrations that do not accomplish the task, can undermine user confidence, further affecting demonstration quality [204, 205]. Finally, users physical interaction with the robotic system needs to be investigated, focusing on their dynamic interactions with the robot joints during demonstrations [44].

Learning from demonstrations under constraints presents key challenges that reflect real-world scenarios. For instance, robots with high DoF often exploit redundancy to perform given tasks efficiently, extending its operation within physical, environmental, or task-specific limitations [119, 120]. To address these challenges, the concept of conceptual constraints was introduced to complement low-level data with high-level abstract information, facilitating successful task execution [46]. Subsequently, Mueller et al. [47] automated the generation of conceptual constraints and incorporated them into

human-aware motion planning algorithms. One such approach is to use a combination of positive and negative demonstrations to learn the constraints of a task [118]. By combining these two types of demonstrations, the algorithm can learn the constraints of the task more effectively. Another approach is to use a two-step learning framework to infer continuous constraint functions from demonstrations [121]. While it is important to develop algorithmic methods to incorporate the environmental or task constraints, it remains crucial to also investigate how robot-side constraints (i.e., different degrees of redundancy) influence human actions and, in turn, robot learning outcomes.

During demonstrations, Fischer et al. [176] identified three key challenges faced by novice users: singular robot configurations, self-collisions, and excessive force applied to the end-effector. Additionally, Hirschmanner et al. [164] examined various user factors, including trust and workload, in the LfD process. Recently, Aliasghari et al. [44] observed that successful human teachers tend to use both hands and frequently adjust their hand placement on the robot’s arm during demonstrations, suggesting that dynamic physical interaction is crucial for effective demonstrations. Our work extends these efforts by examining the impact of robot-side constraints—specifically, the robot’s DoR—on human–robot interaction, an aspect that remains underexplored. We investigate how a robot’s DoR affects cognitive workload, demonstration time, the number of failed attempts, and users’ hand placement and adjustment strategies during task demonstrations. In addition, we examine how the robot’s DoR influences robot task performance. Based on these considerations, we formulate the following hypotheses:

Hypotheses

- H1 Robot DoR affects the time required to provide a demonstration.
- H2 Robot DoR affects the number of failed attempts.
- H3 Robot DoR affects user interaction with the robot joints.

5.2 Selection of Tasks

We considered two manipulation tasks, button-pressing and cuboid block insertion, to collect demonstrations and evaluate the given hypotheses. These tasks have been used in the literature [35, 144] for their generic nature, ensuring they did not require domain-specific expertise.

Button Task: As illustrated in Figure 5.2(a) and Figure 5.2(b), two buttons, B1 and B2, were installed at the centers of adjacent sides of a cardboard box, which was positioned such that the robot’s end-effector could reach both buttons. Participants

were required to teach kinesthetically the robot to press B1 and subsequently B2 using its end-effector tip. This task required at least five joint DoF to sequentially press both buttons. The task difficulty were defined by two factors: (1) avoiding collisions with the cardboard box as well as self-collisions of the robot, and (2) ensuring the robot’s end-effector remained perpendicular to the plane of each button during pressing.

Insertion Task: In this task, participants were required to teach the robot to pick up a randomly positioned cuboid block (measuring $5 \times 5 \times 20$ cm) and partially insert it into a designated hole with a diameter of 10 cm, as shown in Figure 5.2(c) and Figure 5.2(d). Due to the block orientation, vertically standing, completing this task required at least six joint DoF. The difficulty of this task were defined by two aspects: (1) avoiding collisions with the hole’s edges during insertion and self-collisions of the robot, and (2) the requirement to lift the cuboid block vertically, ensuring the z -axis of the robot’s end-effector remained aligned with the block’s z -axis (upright position).

5.3 Policy Learning

For the learning algorithm, we used the classic task-parametrized Gaussian Mixture Models (TP-GMM) to train LfD models from the data collected [206]. Due to its ability to generalize well from a limited number of demonstrations, TP-GMMs are widely used in LfD research [32, 207], where task modeling is performed across multiple coordinate frames. The robot task learning begins after the user provided a set of M demonstrations, where each demonstration m comprises T_m state data points. Each state n is represented as $\xi = (t_n, x_n^T, \omega_n^T)$, where t_n , x_n^T , and ω_n^T represent the time, end-effector position, and orientation in quaternion form, respectively.

In a D -dimensional space, each frame j is defined by an orientation matrix $A \in \mathbb{R}^{D \times D}$ and an origin vector $b \in \mathbb{R}^D$, relative to the global frame. A K -component GMM is fitted to the data in each local frame, with parameters $(\pi_k, \mu_k^{(j)}, \Sigma_k^{(j)})$ representing the prior probabilities, means, and covariance matrices for each component k , respectively. These parameters are estimated using the Expectation-Maximization (EM) algorithm. For trajectory generation, the local models are linearly transformed into the global frame and combined using a product of Gaussian mixtures. Gaussian Mixture Regression is then applied at each time step to retrieve the corresponding trajectory point. For a detailed explanation, refer to Calinon’s work [206].

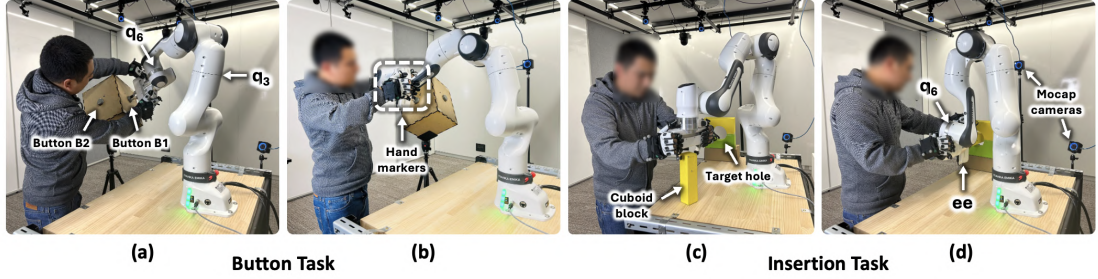


FIGURE 5.2: Subfigures (a–b) show a participant kinesthetically demonstrating the button task. Subfigures (c–d) depict the insertion task, with (c) showing block pickup and (d) partial insertion.

5.4 Experimental Design

5.4.1 Experimental Setup

Robotic System: In this study, we used the 7-DoF Panda robot, a collaborative robotic manipulator, where the robot joints are represented as $q_1, q_2, \dots, q_7$ as shown in Figure 5.1. This robot arm is equipped with torque sensors in all 7 joints, making it safe and suitable for human-robot interaction applications. We used the default gravity compensation controller to counteract the robot’s inherent weight, allowing participants to kinesthetically manipulate the robot’s joints without additional force/torque.

Motion Capture System: We employed the OptiTrack motion capture (MoCap) system to track physical interactions of participants in real time. This MoCap system has sub-millimeter accuracy. It allowed us to track the position and orientation of the participants’ hands and infer which of the robot’s joints were manipulated during the demonstrations. The setup comprised nine Prime-13W cameras: five mounted on a ceiling-attached rig and four positioned on tripods around the experimental area. These cameras were strategically spaced and focused to cover the workspace between the participants and the robot. To ensure accurate tracking, we used five sets of markers arranged in unique geometries to monitor the robot’s origin (1x), participants’ hands (2x), the box with buttons (1x), and the insertion bar (1x), as depicted in Figure 5.2. The combination of trackers and the camera setup was intended to reduce tracking errors. Although minor data losses could occur, we estimate that they would be minimal and did not influence the study’s outcomes. To further limit tracking noise, reflective objects were removed from the experimental area.

Joint Constraint: Each task was performed under two conditions: unconstrained (U) and constrained (C). In the unconstrained condition, all of the robot’s joints were free to be manipulated without any mobility restrictions. In contrast, under the constrained condition, the 6th joint was locked at a default value of 1.571 radians. The goal of locking

a joint was to reduce redundancy while keeping the feasibility of 3D motions—hence, reducing the number of available solutions for the given tasks.

To determine which joint to constrain, we first conducted a pilot study ($N = 6$) and complemented it with kinematic analysis. The pilot results showed that the proximal joints, particularly the shoulder joints (1–3), play a major role in defining the robot’s reachable workspace. Constraining any of these joints would drastically reduce reachability and render tasks such as pick-and-place infeasible. For this reason, we focused on the distal joints (5th, 6th, and 7th) as more suitable candidates for reducing redundancy while preserving task feasibility. In typical 6- or 7-DoF manipulators, the wrist joints form a spherical configuration where the last three axes intersect to realize roll–pitch–yaw orientations. A rotation of approximately 90° around the roll joint can interchange the functional roles of pitch and yaw, enabling similar orientations to be achieved with two joints instead of three. Consequently, constraining the mid-wrist joint can reduce redundancy without compromising workspace coverage. Finally, our pilot study also indicated that the 5th and 7th wrist joints should remain unconstrained during the block-insertion task to facilitate grasping blocks with random orientations. Based on the insights from pilot study and kinematic analysis, we selected the 6th joint to be locked during the constrained condition.

5.4.2 Participants Recruitment

We conducted a prior power ($f = 0.3$, $power = 0.85$, $groups = 2$, $measures = 3$) analysis with G* Power to calculate the sample size [208], and then recruited a total of 24 participants (6F/18M, age = 26.75 ± 4.22) through advertisements on social media and notice boards across our university campus. Along with demographics questions, we asked two additional questions: *Have you had experience with research/industrial robots before?* and *Do you feel uncomfortable or scared when you are around robots?*. The first question served to filter participants based on prior experience with robots while the second one assessed participants’ level of comfort in the presence of a robot. A positive answer to either question would make the person ineligible to participate. The University’s IRB approved the research ethics application (ID 2024-24636-51920-3).

5.4.3 Experiment Protocol

Upon the participant’s arrival, the experimenter explained the study’s purpose, introduced the MoCap system and robotic manipulator, and described the tasks. Each participant signed a consent form before starting the experiment. Markers were attached to the back of participants’ hands to track their kinesthetic interaction with the robot’s

joints. With the gravity compensation controller activated, participants kinesthetically interacted with the robot for few minutes to familiarize themselves with its joints and explore its workspace. The experimenter instructed participants not to move around the experimental setup or manipulate the robot's joints using only their fingertips. Instead, they were advised to interact using their hands, with palms in contact with the robot, to ensure accurate tracking by the OptiTrack system. The participants were instructed to avoid transient or unintentional contacts to ensure that detected manipulations reflected deliberate interactions.

Participants were required to teach two tasks sequentially, button-pressing task followed by insertion task, under both conditions (*constrained* and *unconstrained*). Based on the order of conditions, participants were divided into two groups. We used a counterbalanced design where participants were exposed to both conditions in different sequences. The first group, *constrained* to *unconstrained* (*C2U*), comprised 12 participants who began demonstrating tasks with the constrained condition, followed by demonstrating tasks with unconstrained condition. The second group, *unconstrained* to *constrained* (*U2C*), also included 12 participants but completed the tasks in the reverse constraint order, starting with the unconstrained condition and transitioning to the constrained condition. This allowed us to control for order effects. Observation from pilot study data revealed only a negligible effect of task type and its interaction with the constraint mode; therefore, we kept a consistent task order (button task followed by insertion task) across both groups (*C2U* and *U2C*). This consistency ensured comparability between conditions, although it did not allow us to explore potential learning or interaction effects.

Participants were instructed to provide five successful demonstrations across both tasks and conditions, resulting in four sessions, one for each combination of task and condition. Following each session, the participant completed a questionnaire to reflect their observations on the task and constraint condition. The experiment took on average one and half hour per participant.

5.4.4 Performance Metrics

We assessed the human performance using four key measures: (1) mental workload, measured using the NASA-TLX questionnaire; (2) demonstration time, recorded as the total duration of each demonstration; (3) number of failed attempts; and (4) joints manipulation frequency distribution, quantifying the user's physical interaction with the robot. This last measure is calculated by dividing the number of hand touches on the

TABLE 5.1: Statistical results summary for demonstration time (H1) with standardized coefficients. The terms *CC* and *U2C* represent Constrained-Condition and Group Unconstrained-to-Constrained.

Button Task				
		Estimate [95% CI]		Standardized Coeff.
Demo. Time (sec)	<i>CC</i>	26.34 ($p < 0.001$)	[16.94, 35.73]	0.78 [0.50, 1.06]
	<i>U2C</i>	2.85 ($p = 0.73$)	[−13.27, 18.99]	0.08 [−0.40, 0.57]
	<i>CC:U2C</i>	1.46 ($p = 0.83$)	[−11.74, 14.71]	0.04 [−0.35, 0.44]
Insertion Task				
		Estimate [95% CI]		Standardized Coeff.
Demo. Time (sec)	<i>CC</i>	33.22 ($p < 0.001$)	[25.45, 40.98]	1.17 [0.89, 1.44]
	<i>U2C</i>	−0.24 ($p = 0.98$)	[−11.51, 11.01]	−0.0085 [−0.41, 0.39]
	<i>CC:U2C</i>	−3.71 ($p = 0.51$)	[−14.69, 7.24]	−0.13 [−0.52, 0.26]

robot’s joints by the demonstration duration, indicating how frequently a user engaged with each specific joint during the demonstration.

In addition to the human performance, we evaluated the robot performance based on task completion, defined as the robot’s success in carrying out the task using the learned model. We used each participant data under each *tasks* $\times$ *conditions*, and then tested on real robot once per model. For the button-pressing task, each button is modeled as a sphere with a 3 cm radius based on the button center. A trial is considered successful if the robot’s end-effector tip reaches the target position (i.e., within the goal sphere) without self-collision or collision with the box. For the insertion task, the success is defined as the robot accurately picking up a cuboid block and performing a partial insertion without colliding with itself or the experimental setup.

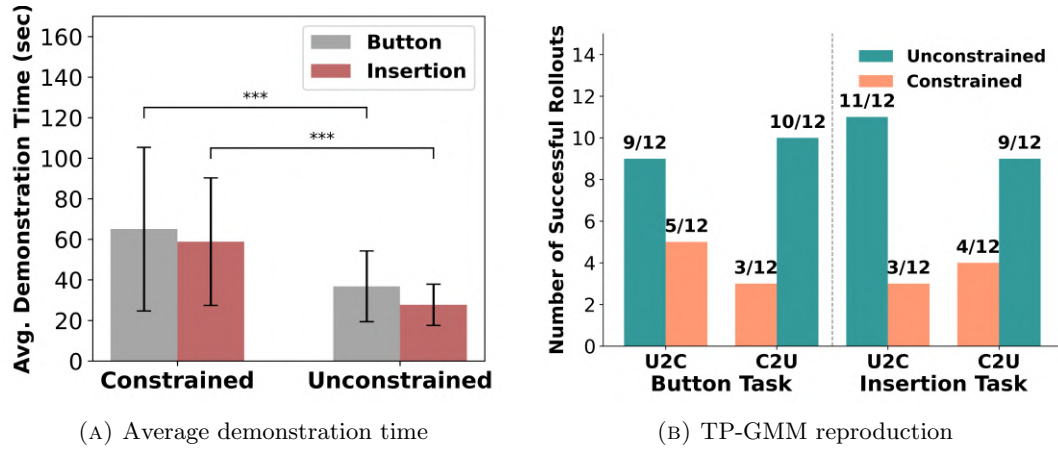


FIGURE 5.3: Performance metrics of demonstration time and reproduction results using learned model.

5.4.5 Data Collection

We utilized OptiTrack’s Motive 3 software on a Windows machine to track participants’ hand positions and orientations in real time. We employed Unity 2022.3 to retrieve this tracking data via the OptiTrack-Unity plugin and compute interactions with the robot’s joints. This was achieved by accessing the robot’s joint positions via a ROS TCP Endpoint, enabling seamless real-time data transfer between Unity on Windows and the robot controller on a Linux system.

Motion Capturing System: To track the origin position and orientation of the 7-DoF Panda robot, we placed a marker set 17 cm behind the robot’s base center at a height of 1 cm (see Figure 5.2). This setup allowed precise tracking of hand positions and orientations relative to the robot’s origin. Using Motive 3 software, we transmitted this data to Unity at an average speed of 400 KB/s, ensuring an average frame rate above 30 fps.

Unity Integration: A digital model of the 7-DoF Panda robot was incorporated into Unity to visualize hands-robot interactions. Unity natively computed both hands positions and orientations with respect to the robot’s origin, defined in a left-handed coordinate system. Hands positions were recorded in x, y , and z coordinates, while orientations were captured as quaternions (x, y, z, w) . In order to determine which robot joint was touched at a given time, we used Unity’s collision detection feature. The hands tracking in the MoCap system was calibrated at the start of each session by instructing participants to place their hands at specific positions on the table where the robot was located. Additionally, the positions and orientations of task-related components, such as the two buttons on the box and the insertion bar, were recorded using additional markers. Data collection of the joint manipulation occurred at a rate of 30 fps to ensure smooth and detailed tracking.

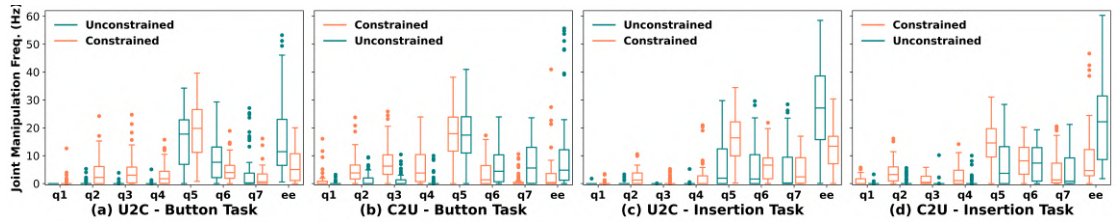


FIGURE 5.4: Subfigures (a–b) show joint manipulation frequency for both $U2C$ and $C2U$ groups during the button-pressing task. Subfigures (c–d) present the corresponding results for the insertion task.

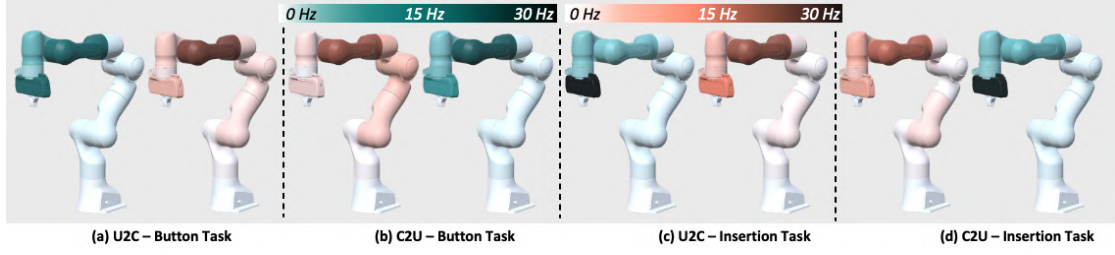


FIGURE 5.5: Average joint manipulation frequency; teal and coral indicate unconstrained and constrained condition, respectively.

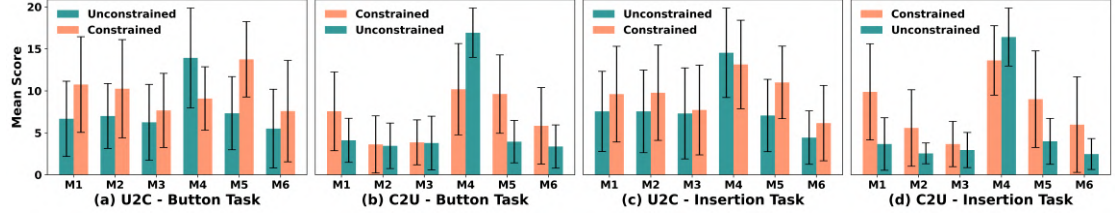


FIGURE 5.6: NASA-TLX scores across six dimensions: Mental (M1), Physical (M2), Temporal (M3), Performance (M4), Effort (M5), and Frustration (M6). Subfigures: (a–b) Button task; (c–d) Insertion task for U2C and C2U groups.

5.5 Results

This section presents results for both tasks, showing how joint constraint affect user demonstration’s ability and robot performance. We used Cumulative Link Mixed Models (CLMM) and Linear Mixed Models (LMM) to analyze the impact on demonstration time, failed attempts, and physical interaction with robot joints. The CLMM approach was used for ordinal response variables, such as the number of attempts, while the LMM approach was used for continuous response variables, such as the time to complete the demonstrations. These statistical approaches also allowed us to assess the interaction effects among joint constraint and groups, providing insights into potential order or learning effects.

5.5.1 Demonstration Time (H1)

Button Task: Participants in the *U2C* and *C2U* groups provided a total of 125 and 122 successful demonstrations, respectively. Using LMM with the formula: $Time \sim Constraint * Group + (1 - Participant)$, we analyzed 247 observations from 24 participants. The results showed that the constrained condition significantly increased the demonstration time by an estimated 26.34 seconds ($p < 0.001$; see [Table 5.1](#)).

Insertion Task: *U2C* and *C2U* participants completed 121 and 120 demonstrations, respectively, yielding 241 observations. The constrained condition led to a significant increase of 33.22 seconds in demonstration time ($p < 0.001$).

TABLE 5.2: Statistical results summary (Estimate [95% CI]) for failed attempts (H2).

	Constrained-Condition	Group U2C	Constrained : Group U2C
Button Task	2.49 ($p < 0.001$) [1.81, 3.17]	0.92 ($p = 0.013$) [0.20, 1.64]	-1.07 ($p = 0.018$) [-1.96, -0.18]
Insertion Task	0.40 ($p = 0.21$) [-0.22, 1.03]	0.56 ($p = 0.11$) [-0.12, 1.24]	-0.72 ($p = 0.14$) [-1.61, 0.17]

Figure 5.3(a) shows the average demonstration times across both tasks and conditions, with significantly longer demonstrations observed in the constrained condition. Since demonstrations were temporally aligned using DTW, we anticipate longer input durations resulted in proportionally longer robot completion times.

5.5.2 Number of Failed Attempts (H2)

Button Task: In the *U2C* group, participants had 18 (21.68%) failed attempts under the unconstrained condition and 50 (45.47%) under constrained. In contrast, the *C2U* group had 48 (44.41%) failures under constrained and only 02 (3.12%) under unconstrained. CLMM results showed a significant increase of 2.49 (counts) failed attempts under constrained condition ($p < 0.001$). Additionally, the *U2C* group had 0.92 more failures under unconstrained condition than the *C2U* group ($p = 0.013$), as shown in Table 5.2. The *U2C* group’s failure rate under *constrained condition* was significantly higher, with the group effect negatively moderated (see Table 5.2), indicating more failures compared to *C2U* in the same condition.

Insertion Task: In the *U2C* group, participants had 15 (19.73%) failures under unconstrained and 12 (16.67%) under constrained. The *C2U* group showed 15 (20%) failures under constrained and 06 (9.09%) under unconstrained. CLMM results showed no significant difference in failure rates between task conditions ($p = 0.21$). The order of joint constraint (*C2U* vs. *U2C*) had no significant impact under constrained, indicating no interaction effect.

5.5.3 User Interaction with Robot Joints (H3)

Button Task: Joint manipulation frequency across both groups is shown in Figure 5.4(a) and Figure 5.4(b), with joint q_5 having the highest and joint q_1 the lowest interaction rates. As detailed in Table 5.3, under constrained condition, participants interacted more with joints $q_1 - q_4$, particularly q_3 ($est. = 6.1$, $p < 0.001$), while interactions with q_6 , q_7 , and the end-effector decreased, notably q_7 ($est. = -7.6$, $p < 0.001$). Order of constrain condition significantly affected joint interaction. In the *U2C* group, participants interacted less with q_3 ($est. = -1.9$, $p = 0.03$) and q_4 ($est. = -2.4$, $p = 0.003$) than in the *C2U* group under constrained condition. Interaction patterns, particularly

TABLE 5.3: Statistical results summary for user interaction (joint manipulation frequency) with the robot’s joints (H3). q_n and ee represent n^{th} robot’s joint (7-DoF) and end-effector, respectively.

Button Task									
Constrained-Condition			Group U2C				Constrained : Group U2C		
	Estimate [95% CI]	Std. Coeff.	Estimate [95% CI]	Std. Coeff.			Estimate [95% CI]	Std. Coeff.	
q_1	1.0 ($p < .001$) [0.48, 1.53]	0.6 [0.3, 0.9]	-0.1 ($p = 0.8$) [-0.9, 0.7]	-0.06 [-0.5, 0.4]			-0.6 ($p = 0.1$) [-1.3, 0.14]	-0.35 [-0.8, 0.08]	
q_2	3.8 ($p < .001$) [2.61, 4.9]	0.9 [0.6, 1.2]	-1.1 ($p = 0.3$) [-2.9, 0.8]	-0.24 [-0.7, 0.2]			-0.2 ($p = 0.8$) [-1.9, 1.5]	-0.05 [-0.4, 0.4]	
q_3	6.1 ($p < .001$) [4.8, 7.4]	1.2 [0.9, 1.5]	-1.1 ($p = 0.3$) [-3.3, 1.1]	-0.2 [-0.7, 0.2]			-1.9 ($p = 0.03$) [-3.8, -0.2]	-0.4 [-0.8, -0.04]	
q_4	5.3 ($p < .001$) [4.2, 6.4]	1.2 [0.9, 1.5]	-0.6 ($p = 0.5$) [-2.6, 1.3]	-0.15 [-0.6, 0.3]			-2.4 ($p = 0.003$) [-3.9, -0.8]	-0.55 [-0.9, -0.2]	
q_5	1.4 ($p = 0.3$) [-1.6, 3.9]	0.15 [-0.13, 0.4]	-0.7 ($p = 0.8$) [-6.1, 4.8]	-0.07 [-0.7, 0.5]			2.3 ($p = 0.2$) [-1.3, 5.8]	0.24 [-0.14, 0.6]	
q_6	-2.1 ($p = 0.02$) [-4, -4]	-0.4 [-0.6, -0.1]	2.2 ($p = 0.2$) [-0.9, 5.2]	0.4 [-0.15, 0.9]			-1.5 ($p = 0.2$) [-3.9, 0.9]	-0.25 [-0.7, 0.2]	
q_7	-6.2 ($p < .001$) [-8, -5]	-1.0 [-1.3, -0.8]	-3.4 ($p = 0.07$) [-6.9, 0.15]	-0.55 [-1.1, 0.03]			4.5 ($p < 0.001$) [2.4, 6.6]	0.74 [0.4, 1.1]	
ee	-7.6 ($p < .001$) [-10, -5]	-0.6 [-0.8, -0.4]	4.6 ($p = 0.3$) [-3.3, 12.5]	0.4 [-0.27, 1.0]			-2.1 ($p = 0.3$) [-5.7, 1.6]	-0.17 [-0.5, 0.1]	
Insertion Task									
q_1	0.8 ($p < .001$) [0.7, 1.1]	0.9 [0.6, 1.2]	-0.04 ($p = 0.8$) [-0.5, 0.4]	-0.05 [-0.5, 0.4]			-0.6 ($p = 0.001$) [-1.0, -0.3]	-0.7 [-1.1, -0.3]	
q_2	3.8 ($p < .001$) [2.9, 4.6]	1.2 [0.9, 1.5]	-0.4 ($p = 0.5$) [-1.7, 0.9]	-0.13 [-0.6, 0.3]			-1.6 ($p = .007$) [-2.7, -0.4]	-0.5 [-0.9, -0.1]	
q_3	1.2 ($p < .001$) [0.8, 1.7]	0.9 [0.6, 1.2]	-0.24 ($p = 0.4$) [-0.8, 0.3]	-0.2 [-0.6, 0.2]			-0.8 ($p = 0.01$) [-1.4, -0.1]	-0.5 [-0.9, -0.1]	
q_4	1.7 ($p < .001$) [0.9, 2.9]	0.6 [0.3, 0.9]	-0.5 ($p = 0.5$) [-2.3, 1.2]	-0.2 [-0.7, 0.3]			0.6 ($p = 0.4$) [-0.8, 2.0]	0.2 [-0.2, 0.6]	
q_5	7.4 ($p < .001$) [5.3, 9.5]	0.8 [0.5, 1.0]	-0.8 ($p = 0.9$) [-6.3, 4.7]	-0.08 [-0.8, 0.5]			2.2 ($p = 0.2$) [-0.7, 5.2]	0.2 [-0.08, 0.5]	
q_6	0.8 ($p = 0.4$) [-1.2, 2.8]	0.13 [-0.2, 0.45]	-1.2 ($p = 0.5$) [-4.3, 1.9]	-0.2 [-0.7, 0.3]			-0.4 ($p = 0.7$) [-3.3, 2.4]	-0.07 [-0.5, 0.4]	
q_7	-0.2 ($p = 0.8$) [-2.3, 1.9]	-0.03 [-0.4, 0.3]	1.2 ($p = 0.5$) [-1.9, 4.2]	0.2 [-0.3, 0.7]			-1.1 ($p = 0.4$) [-4.1, 1.8]	-0.2 [-0.6, 0.3]	
ee	-13.7 ($p < .001$) [-17, -10]	-0.9 [-1.1, -0.7]	5.1 ($p = 0.2$) [-2.8, 13.1]	0.3 [-0.2, 0.9]			-1.4 ($p = 0.6$) [-6.3, 3.5]	-0.09 [-0.4, 0.2]	

for q_7 , also reversed depending on order of condition. Full data is summarized in [Table 5.3](#).

Insertion Task: Under unconstrained condition, the end-effector had the highest interaction frequency across both groups, while joint q_5 saw the most interaction under constrained condition. Under constrained, participants engaged more with joints $q_1 - q_5$ and significantly less with the end-effector (see [Table 5.3](#)). In the $U2C$ group, interaction with joints $q_1 - q_3$ was reduced under constrained condition. Without considering order, we observed a general increase in interaction with these joints.

[Figure 5.5](#) shows the color-scale representation of joint manipulation frequency across tasks and conditions.

5.5.4 Mental Workload

To evaluate the perceived difficulty level of each task with both conditions, participants responded to two items using a 7-point Likert scale. The first item was: “Rate how easy you think this task was”. After completing the same task under the alternate constraint condition (i.e., following $U2C$ or $C2U$), the second item was: “Rate the difficulty level of this task compared to the previous one”. Additionally, we employed the NASA-TLX (Task Load Index) [203] to assess various aspects of participants’ experiences. The findings, as shown in [Figure 5.6](#), consistently indicate that participants perceived the tasks as more difficult under the constrained condition, highlighting the impact of joint constraint on perceived difficulty and workload.

5.5.5 Policy Learning

Successful demonstrations were used to train the TP-GMM models. As a first step, we analyzed each set of demonstrations (i.e., each participant demonstrations data under each $task \times condition$) by evaluating lengths of each demonstration, completion time, and ensuring task completion. We removed those demonstrations that failed (self-collision, joint limits, excessive force), incomplete, or unusually long (2x avg. duration). Next, we applied a moving average filter to smooth the trajectories. After pre-processing, we employed the dynamic time warping technique to align the trajectories temporally [193]. TP-GMM models were then trained on these filtered set of demonstrations for the given task parameters (i.e., matrix A and vector b) and tested on the real robot to evaluate robot performance. The number of Gaussian components (K) was selected using Bayesian Information Criteria, with diagonal covariance regularization ($1e-5$) and EM parameters (min 5, max 100 steps, tolerance $1e-5$). The Franka ROS2 default joint trajectory controller was used.

Figure 5.3(b) illustrates that the presence of a joint constraint has a notable impact on the robot’s performance. Each bar in the plot represents the number of successful models or rollouts out of a total of 12, corresponding to demonstrations provided by 12 individual users in each group. The results indicate a notable difference between the two experimental conditions. For both manipulation tasks, the number of successful attempts was consistently higher when the model was trained on demonstrations collected under the unconstrained condition, whereas the success rate decreased when demonstrations were collected under the constrained condition.

5.6 Discussion

This work examined how robot redundancy shapes novice users’ experiences during kinesthetic teaching. Rather than treating redundancy as a purely kinematic property, our findings highlight its role as an interaction design variable that directly influences how users explore, understand, and demonstrate robot motion. Across tasks, higher degrees of redundancy supported more efficient and less demanding demonstrations, while also reshaping users’ physical engagement with the robot and the consistency of their demonstrations.

Taken together, the results suggest that providing users with greater kinematic freedom can lower barriers to effective demonstration by reducing physical and cognitive effort. However, this same flexibility introduces new challenges related to coordination, collision avoidance, and execution fidelity. In the following section, we discuss how redundancy

influenced user physical interaction, demonstration diversity, and failure patterns, before reflecting on the broader implications for LfD.

User Physical Interaction: User physical interaction provided insight into how redundancy affected participants’ reasoning about the demonstration through their hands. Under the constrained condition, participants primarily manipulated the lower joints (q_2 – q_5), with q_5 receiving the most interaction. Participants who started with the constrained condition initially expected the unconstrained condition to require similar full-joint manipulation; however, they learned that focusing on joints closer to the end-effector enabled more efficient task completion, leading to a gradual shift in hand contact points toward these joints (see Figure 5.4). The change in user interaction patterns over time has also been reported by Aliasghari et al. [44]. We also observed significant differences in hand contact sequences between failed attempts and successful demonstrations.

Demonstrations Diversity: In our previous work [34], we observed that demonstration diversity influenced the robot task performance. Under the constrained condition, greater variability in user demonstrations made learning more challenging for the robot, requiring additional demonstrations to achieve satisfactory performance. In contrast, demonstrations collected under the unconstrained condition were more consistent, resulting in improved robot task performance. Although it may appear counterintuitive that reduced redundancy leads to greater demonstration diversity, limiting redundancy increases task difficulty for novice users by reducing valid solutions and obscuring the desired configuration. These results are echoing the importance of previous LfD research [111]. In addition, our work suggests a possible design pathway for diverse demonstrations [161].

Learning Through Practice: Learning effects significantly influenced the number of failed attempts in the button-pressing task: participants who began with the easier, unconstrained condition demonstrated improved performance when transitioning to the more challenging constrained condition. Most failures stemmed from joint-limit violations, excessive force application, and self-collisions. In contrast, no notable differences were observed in the insertion task, likely because participants became progressively familiar with the robot’s joint configuration and task requirements over the course of the study, consistent with prior findings [43, 44].

Challenges of Higher Robot Redundancy: Our user study revealed two key challenges associated with higher robot redundancy during kinesthetic teaching. First, *higher redundancy increased the likelihood of violating inherent robot constraints*—such as self-collision—because additional links introduced more opportunities for *interference during demonstration* (e.g., we observed frequent collisions between links 5 and 2 during the

insertion task). Second, novice users found it more difficult to manage singular configurations due to the larger configuration space. These observations highlight the need for assistive feedback systems to support novice users during kinesthetic demonstrations of redundant robots. We also observed implications for execution, as reproducing demonstrations in redundant systems required closer alignment in both task and joint spaces. These findings suggest that, although redundancy offers flexibility, it also introduces coordination and modeling challenges that affect both teaching and execution in LfD. These results are supported from the literature [42, 209].

Limitations

All participants were university students, with a predominantly male sample, reflecting our focus on recruiting LfD novices rather than achieving gender balance. Future work will include industry professionals, such as caregivers and healthcare workers, to broaden the applicability of our findings. In addition, while order effects were observed in the button task but not the insertion task (H2), studies with larger participant samples are needed to better assess learning effects.

We evaluated two manipulation tasks—button pressing and cuboid block insertion—chosen for their comparability with prior work [35, 144]. Future work will explore additional constraint types [47, 210], such as maintaining an upright cup during transport, to evaluate and compare their effects against the current results.

Finally, our study was conducted using a commercially available 7-DoF Franka Emika Panda robot, whose shoulder–elbow–wrist morphology is representative of many collaborative manipulators. However, our findings are based on a single redundancy level and may not fully generalize to robots with fewer or substantially more degrees of freedom. While similar effects may arise in lower-DoF systems for sufficiently constrained tasks, or be amplified in highly redundant robots, future work is needed to systematically examine how redundancy-related effects scale across different robot morphologies.

5.7 Summary

This work examined how robot kinematic redundancy shapes both novice users’ kinesthetic teaching strategies and the resulting robot learning performance. Across tasks, constraining redundancy increased users’ mental workload, prolonged demonstration time, and led to more frequent failures. In contrast, providing higher redundancy encouraged users to interact more locally around the end-effector, resulting in more efficient

demonstrations and consistently stronger robot performance. Together, these findings demonstrate that kinematic redundancy is not merely a mechanical property, but a key interaction factor that influences how users explore the robot's configuration space and how effectively robots learn from human demonstrations.

Key Takeaways:

- Providing novice users with higher robot redundancy (more DoF than strictly required) helps them identify suitable joint configurations within a larger feasible solution space.
- The sequence of hand contact points contains valuable information that can be used to anticipate and prevent potential failures.

Next Study Motivation

This chapter demonstrated that the robot's degree of redundancy significantly influences the demonstration process, affecting users' mental workload, physical interaction, and ultimately the quality of demonstrations and robot learning performance. While these findings provide valuable insights into how robot embodiment shapes human teaching, novice users still face challenges in understanding and adapting to the robot's kinematic capabilities during kinesthetic demonstrations.

This observation motivates the next chapter, which focuses on supporting users during the teaching process through real-time feedback. Rather than expecting users to infer the robot's internal constraints, we investigate how key robot information can be communicated intuitively using augmented reality. Specifically, we first identify effective visualization strategies for robot constraints through a focus group study and then develop and evaluate an AR-based feedback system that provides real-time guidance during demonstrations. By improving users' awareness of robot constraints, the proposed system aims to reduce cognitive workload, improve demonstration quality, and ultimately enable more effective robot learning.

Chapter 6

Design and Evaluation of AR-Based Real-Time Feedback System for Kinesthetic Robot Teaching ¹

Despite advances in robotics, their adoption in workplaces such as SMEs remains limited due to concerns about usability, accessibility, and the impact of automation on workers [3–6]. A worker-centered perspective argues that robots should empower workers by adapting to their practices rather than requiring specialized technical expertise [211]. Learning from Demonstration, particularly kinesthetic teaching, addresses this challenge by enabling users to teach robots through physical guidance instead of programming, making robot adaptation more accessible to non-experts [10, 39, 212].

However, effective robot teaching remains challenging. Novice users often provide demonstrations that are incomplete, inconsistent, or poorly suited to the robot’s learning process [48, 52]. Existing approaches typically filter out poor demonstrations rather than addressing why they occur [49, 213]. A key challenge is that users have limited insight into the robot’s internal state and learning progress, making it difficult to judge whether the robot has understood their demonstrations [55]. This mismatch often leads to ineffective teaching and frustration.

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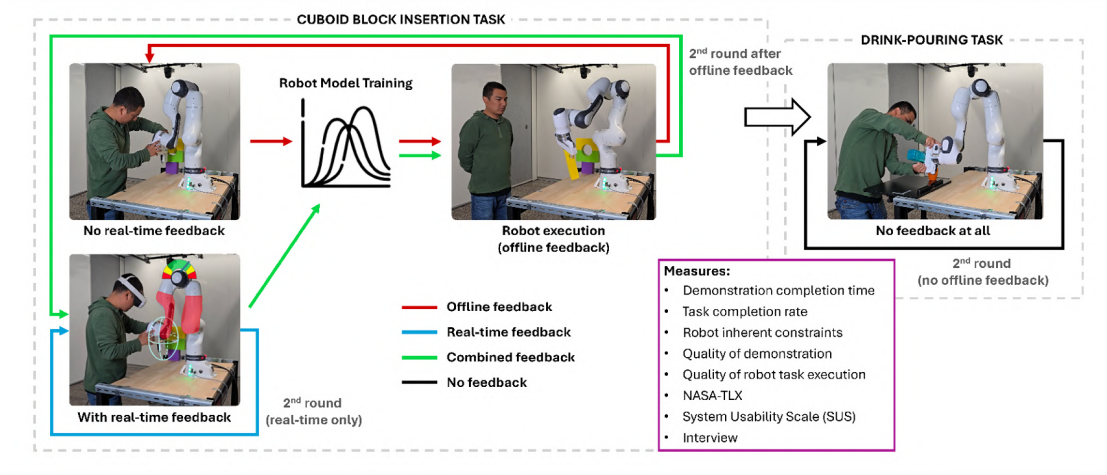


FIGURE 6.1: **Overview:** The study compares three feedback conditions during kinesthetic robot teaching: No real-time feedback (red), Real-time feedback (blue), and Combined feedback (green). Participants first performed a cuboid block insertion task by providing three kinesthetic demonstrations to a 7-DoF robot, followed by a second demonstration round depending on the feedback condition. A second drink pouring task was then performed with no feedback for any group, enabling assessment of participant’s learning across tasks.

This challenge is rooted in the opaque nature of the robot’s learning process. When teaching other humans, people adjust their instruction based on visible cues such as questions, hesitation, or mistakes. When teaching robots, these cues are largely absent, turning teaching into a one-sided activity rather than a mutual exchange. To address this gap, robot learning must be treated as a dialogue in which the robot actively communicates its current state to the teacher through clear and accessible feedback—such as visual, haptic, or auditory signals [214–216]. Prior work suggests that when learning and communication are tightly connected in this way, teaching becomes more effective, trust is strengthened, and both humans and robots can better adapt to one another [79, 217–219]. From a design perspective, this highlights the need for new communication interfaces that help robot “students” express what they understand, what they do not, and when they need help, making teaching a shared and responsive process rather than a guessing game [56, 57, 220, 221].

Augmented Reality (AR) technology has emerged as a significant advancement in human–robot interaction (HRI), as it enables the integration of three-dimensional virtual graphics directly into the user’s real-world field of view [169, 222, 223]. By visually enriching the physical environment, AR not only improves the user’s perceptual awareness of robotic systems but also provides intuitive spatial feedback during interaction. Building on these capabilities, AR-based tools have been successfully deployed in industrial environments, where they help reduce cognitive load, facilitate robot training, and support calibration and inspection processes for complex assembly tasks [224–226].

In this work, we address the overarching question: *How can robots communicate internal constraints to users in human–robot Learning from Demonstration interactions?* To explore this question, we conducted a focus group ($N = 9$) with AR/XR developers and roboticist as design stakeholders to generate interaction concepts for communicating a robot’s internal constraints (e.g. joint limits, potential self-collisions, and singular configurations). We derive design findings that surface opportunities and tensions around safety, alerting mechanisms, cognitive load, and opportunities for customization.

Research Question 04

How can robots communicate internal constraints to users in human–robot Learning from Demonstration interactions?

Building on these insights, we developed an AR-based feedback system to make robot constraints perceptible in real time. To evaluate its effectiveness, we conducted a two-task between-subjects user study ($N = 36$). Our findings show that novice users who received real-time AR feedback not only performed better on the initial task but also retained these benefits when the feedback was removed in the second task, demonstrating measurable learning. Moreover, the feedback supported users in developing a better understanding of the robot’s motion constraints, leading to improved robot task performance even without visual assistance. Figure 6.1 illustrates the overview of experimental study.

6.1 Background and Related Work

One of the key goals of LfD is to enable novice users to efficiently and effectively demonstrate tasks to robotic systems [10, 11]. A growing body of research has focused on understanding and supporting the human teaching process, including ways to measure, evaluate, and improve user demonstrations [34, 50, 161, 227]. However, most existing approaches rely on *offline* feedback, where users observe the robot’s reproduced motion after completing demonstrations to refine or correct their guidance [217, 218, 228]. While such feedback can improve robot learning outcomes, it does not provide guidance during the demonstration itself, highlighting an opportunity to explore *real-time feedback mechanisms*, such as those leveraging AR technology, to support users during task demonstration.

6.1.1 Teaching Robot Interfaces

Kinesthetic teaching allows users to guide a robot through physical interaction, offering an intuitive starting point for demonstration-based learning [10, 202]. However, this interaction becomes challenging when robots have many joints or operate under physical constraints [42]. During teaching, users must implicitly account for issues such as limited joint ranges, complex postures, or impending collisions, even though these factors are often invisible or difficult to perceive through touch alone. As a result, demonstrators may unintentionally guide the robot into problematic configurations, producing demonstrations that are hard for the robot to interpret or reproduce [42, 175].

Augmented reality offers an opportunity to address these challenges by making aspects of the robot's internal state visible during teaching [144]. When AR cues are anchored to the robot, they can communicate information such as approaching limits, constrained motion, or regions of reduced flexibility directly within the user's field of view [222]. This added transparency can help users better understand how the robot experiences the task, enabling more informed and effective demonstrations. At the same time, careful design is required to avoid overwhelming the demonstrator, as visualizing too many signals at once may increase cognitive load, distracting users from the physical task and reducing teaching quality [144, 229]. From a human-centered perspective, it is therefore important not only to support successful task demonstrations, but also to consider user-side measures such as perceived mental workload and the ease of providing demonstrations without violating the robot's physical constraints.

6.1.2 AR for Teaching Robots

Prior work has explored AR systems in human-robot interaction, with most applications focusing on robot teleoperation or programming interfaces rather than supporting users during kinesthetic demonstrations [230, 231]. For instance, AR has been used to specify goal poses, preview robot motion, and define task trajectories through virtual overlays or haptic input devices [124, 232]. Other studies have investigated tablet- or headset-based AR interfaces for commanding or programming robots, reporting effects on user workload and task efficiency depending on the task and platform [230, 233, 234]. By providing spatially aligned visual feedback within the user's field of view, AR allows users to perceive robot motion, limitations, and task-relevant information, without interrupting the flow of the demonstrations [169, 222, 235]. Recently, researchers have investigated how additional sensory cues—including haptic feedback and AR visualizations—can improve user awareness, guidance, and performance during human-robot interaction [79, 236]. However, these AR-based systems have focused on remote interaction with robots,

with comparatively less attention on examining how these design principles translate to hands-on kinesthetic teaching with physical robots [223, 231], enabling the robot to learn more effectively and execute tasks more successfully [34, 35].

Despite the promise of augmented reality for making robot status visible during interaction, there is still limited understanding of how AR-based feedback can be integrated into kinesthetic teaching in a way that meaningfully supports users without overloading them [144, 230, 231]. In particular, little is known about how such feedback can help users navigate constrained teaching situations—such as avoiding collisions, respecting motion limits, or steering clear of problematic postures—while maintaining a smooth and efficient teaching experience. This study builds on prior work by designing and evaluating an AR-based feedback system that communicates robot status in real time during kinesthetic demonstrations. Rather than treating AR merely as a programming aid, we examine how in-situ visual feedback can support users during teaching, how it shapes demonstration quality and efficiency, and how it compares to more conventional forms of offline or post-hoc feedback. To this end, we pose the following research questions:

RQ1a: What design considerations and trade-offs emerge when envisioning real-time communication of robot’s internal states during LfD?

RQ2a: How does the presence of AR-based constraint feedback influence how users demonstrate tasks to a robot, including their effort, strategies, and accommodation of robot constraints?

RQ3a: How does AR-based constraint feedback influence robots’ ability to learn from demonstrations and successfully execute the demonstrated tasks?

6.2 Study 1: Focus Group

We conducted a focus group study to explore how users interpret and prefer to receive real-time robot state information during kinesthetic teaching. The focus groups were designed as an exploratory, ideation-driven method to complement the literature findings by eliciting user perspectives in a concrete LfD context. Rather than replacing prior knowledge, this step allowed us to ground established HCI and AR design principles in the specific constraints of physical human–robot interaction and kinesthetic teaching scenarios.

To explore the design space for communication during a demonstration task, we held an in-person focus group with roboticists and AR/VR application developers. This

TABLE 6.1: Participant Demographic Information

ID	Gender	Age	Experience Type		Yrs. of Exp	Expertise Description
P1G1	Male	29	AR/VR Robotics	and	8	Developed AR applications; completed master's and ongoing PhD in XR/AR; participated in robotics projects since undergraduate years.
P2G1	Female	28	AR/VR Robotics	and	7	PhD in Mechanical Engineering and HRI; VR data processing, robot building, probabilistic robotics, and HCI experience.
P3G1	Male	26	Robotics		3	Experience with bipedal robot, lower-limb exoskeleton, and robotic manipulators for HRI.
P4G2	Male	33	AR/VR Robotics	and	2	Conducted research on robot teleoperation using AR glasses.
P5G2	Male	30	AR/VR		2	Working on AR/VR-based architectural heritage reconstructions.
P6G2	Male	28	AR/VR		1	Visualized physics simulation data in VR.
P7G3	Male	23	AR/VR		1	Conducted XR-related research.
P8G3	Male	29	AR/VR		3	Background in kinematic motion synthesis and soft-rigid body simulation.
P9G3	Male	26	AR/VR Robotics	and	6	VR scene design experience; 4 years experience working with NAO and TIAGo robots.

mix was intentional: roboticists provided insight into robot's capabilities, while AR/VR developers contributed expertise on AR/VR interaction possibilities. Participants were paired such that, collectively, each group encompassed expertise in both AR/VR and robotics, enabling cross-domain discussions [237]. Through the focus group, we examine **RQ1a**.

6.2.1 Participants Recruitment

A total of 9 participants ($F = 1$, $M = 8$; $M_{age} = 28$ years, $SD_{age} = 2.82$) were recruited for the study (see Table 6.1). Their backgrounds included AR/VR application development, robot teleoperation, robotic manipulation, exoskeleton control, XR research, physics-based simulation, and probabilistic robotics. Participants' experience levels ranged from 1 to 8 years across AR/VR, robotics, or both domains.

6.2.2 Procedure

The study was a structured, multi-stage session conducted in our School User Experience Lab. Upon arrival, each participant was provided with a written consent form and a Plain Language Statement outlining the study's purpose and procedures. Participants then completed a demographic questionnaire, followed by a brief icebreaker activity.

Participants were assigned to small groups of three and introduced to the concept of LfD. The facilitator provided an overview of LfD, the typical modalities for demonstrations, and the challenges commonly faced by novice users. This ensured a shared baseline understanding before participants began generating design concepts.

Next, participants engaged in a hands-on interaction session with a Franka Emika 7-DoF robot². They were encouraged to explore predefined interaction features on the robot to build familiarity with its capabilities and limitations.

Following the exploration phase, participants engaged in a structured ideation activity inspired by the rapid brainstorming (crazy 4's method) [238]. The activity focused on three fundamental robot constraints: joint limits, self-collision, and manipulability. In each ideation round, participants individually generated four ideas for AR visualizations and their associated interaction features, each centered on the specific robot constraint (e.g., joint limits). After completing each round, participants reconvened within their groups to present their ideas, engage in discussion, and collaboratively select two ideas that best reflected their collective perspectives. During this selection process, participants were permitted to cherry-pick and combine features from multiple ideas, merging them where appropriate. This iterative process of divergence and convergence was designed to promote both a broad exploration of ideas and their subsequent refinement.

After completing all three rounds, groups prepared and delivered short pitches of their top ideas for each feature. Each group presented its final proposed concepts to all participants, enabling cross-group comparison and discussion. Participants were then allowed to make final refinements or clarifications before voting. In this final stage, all participants voted for the strongest idea for each feature, based on clarity, usefulness, and potential to support AR-mediated robot learning.

²<https://franka.de/franka-research-3-arm>

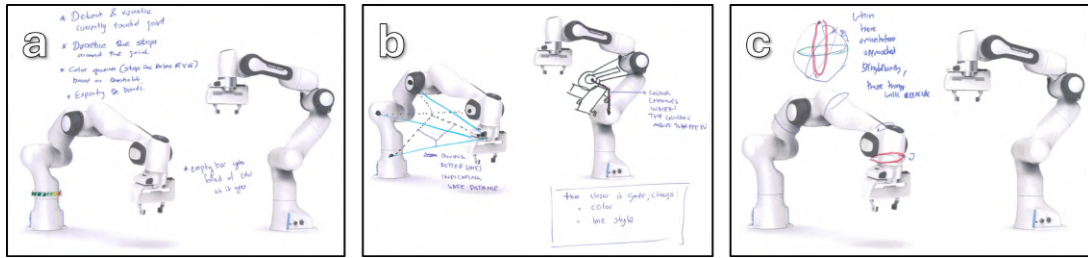


FIGURE 6.2: Final visualization designs of each feature from the focus group study: (a) Joint Limits, (b) Self-Collision, and (c) Manipulability.

6.2.3 Analysis

First the audio recording were transcribed using Otter.ai³, and then reviewed and corrected by the first author through close listening to the recordings. Personally identifiable information, was removed and replaced with pseudonyms following a (participant ID–group ID) convention (e.g., P1G1). The anonymized transcripts and participant drawings were then imported into a shared Miro⁴ board to support collaborative analysis. Two coders independently familiarized themselves with the dataset through repeated engagement with the transcripts and sketches, generating preliminary inductive codes [239]. During the first meeting, they conducted a line-by-line review of the Group 1 transcripts, compared their initial codes, and collaboratively developed a shared code set. In the second meeting, the coders jointly reviewed and coded the Group 2 transcripts and sketches, refining the code set and consolidating preliminary themes. Throughout this process, all iterations of the code set were systematically documented as tables in a Miro board to ensure transparent version control. After coding was completed, the coders reconvened to generate candidate themes, refine them individually, and finalize them through discussion. As a final validation step, the second author verified all selected reporting quotes against the original audio recordings prior to deleting the raw audio data.

6.2.4 Findings

The findings from the focus group were then interpreted in relation to existing literature in HCI and AR/XR. Overall, the results were largely consistent with established principles, particularly regarding the importance of reducing cognitive load, avoiding visual clutter, and providing intuitive and minimally intrusive feedback. Participants' emphasis on clear, immediate, and spatially aligned visual cues aligns with prior work on effective visual feedback design in AR systems. At the same time, some insights

³<https://otter.ai>

⁴<https://miro.com>

highlighted domain-specific considerations unique to kinesthetic teaching, such as the need to balance continuous feedback with physical task execution demands. This comparison between user findings and prior literature helped refine the design guidelines and ensured that the proposed AR feedback system is both theoretically grounded and empirically validated within the context of LfD.

In this section, we present the findings from the focus group, organized into three holistic categories based on joint limits, self-collision and manipulability.

Joint Limits

Our group findings show that novices consistently struggle with joint limits as these constraints are rarely made explicit during LfD. Joint limits are defined as angle values for each joint, yet users are typically expected to know them in advance or infer them through repeated failures, such as when the robot abruptly stops or becomes stuck. Participants often focused on the task goal rather than individual joint angles, making it difficult to understand which joint caused the failure and why. This gap between the user’s mental model and the robot’s hidden angular constraints highlights the need for clearer communication of joint-specific limits. In following section we discuss the findings from focus group suggested to elevate these issues in a bottom-up hierarchical order (see [Figure A1](#) for figures).

Color-Based Awareness of Joint Limits: Participants repeatedly emphasized the value of immediate, visually intuitive indicators for joint-limits awareness, with color emerging as the preferred modality for quickly communicating joint-safety states. Many explicitly gravitated toward a stoplight-inspired system, where green denotes safe movement and red signals unsafe or constrained states, because of its familiarity and low cognitive demand. As one participant summarized, “If it is okay they’re green if they are not okay they become red” (P9G3).

Beyond discrete states, participants also discussed the value of progressive visual feedback as a joint approaches its limit. Several described *color transitions* or *expansion of the visual cue* as an effective way to convey urgency without abrupt interruption. For example, one participant envisioned a warning that escalates visually as the limit is neared: “when I turn further, that red color strip will broaden up further and then like it will start blinking [showing] you have reached the limit” (P1G1). Similarly, smoother color gradients were seen as beneficial for expert users who may seek more fine-grained feedback, as illustrated by another participant’s description of a continuous transition: “green means that it’s good and then it might change to like a yellow and then an orange and then a red and then it’s like a smoother transition” (P2G1).

Quantifying Proximity to Joint Limits: Beyond simple colored awareness, participants expressed a strong need for explicit quantification mechanisms that communicate how close a joint is to its mechanical limit reflecting both proximity and directionality of the limit. This need was particularly salient for tasks requiring precise control or slow, incremental adjustments during demonstration. Discretized visual representations were repeatedly favored because they provide an immediate sense of the remaining range without requiring mental calculation. As one participant noted, “Discretized version would be better because it would give a sense of ... how much you’ve got to go” (P3G1). Visual elements aligned with the axis of movement, such as arrows that progressively change color, were seen as particularly intuitive. One participant explained how such cues could dynamically encode constraint information: “once you get closer to a particular limit in one direction you can have the parts of the arrows ... more red ... and if it’s far from the joint limits ... more green” (P6G2). These representations were perceived as mapping the pathway from the current posture to the boundary condition in a visually legible manner.

Participants also discussed the role of numerical information in conveying joint-limit proximity, acknowledging that numerical labels could be valuable for expert users seeking high precision, while cautioning that such information might overwhelm novice users if presented by default. As one participant suggested, numerical values are most useful when they can be consulted deliberately: “present the actual information to the user so like the numerical rotation degrees ... they can actually see if it’s close to it or not” (P6G2).

Integrated and Dynamic Feedback: Participants emphasized that joint-limit information must remain spatially grounded within the XR scene in order to minimize cognitive load and preserve attentional focus during interaction. Rather than relying on external or detached interfaces, they consistently favored visual overlays that are co-located with the robot itself—such as rings, bars, or localized annotations placed directly on or adjacent to the relevant joint. These embedded representations were perceived as more intuitive and less disruptive to the demonstration workflow. One participant, for example, described a design in which constraint feedback is tightly coupled to the joint geometry: “*Since the joints are circular, I was thinking of putting rings, and whenever they get close to the limit, they become different color and the bars move*” (P4G2).

To further reduce visual clutter, particularly in robots with many degrees of freedom, participants advocated for context-sensitive presentation of joint-limit feedback. Several proposed that the system should automatically identify the currently manipulated or active joint and display constraint information only for that joint. As one participant succinctly suggested, “How about like AR automatically detects the joint that we are

rotating” (P1G1). This approach was seen as a way to hide unnecessary complexity while still providing critical feedback at the moment it is needed. As another participant explained, selective disclosure helps simplify the experience for users who do not wish to engage with full system detail: “So the people can see this ... this is if you want to hide all of the complexities ... from the users” (P6G2).

Finally, while localized feedback was preferred during active manipulation, participants acknowledged the value of optional global summaries for users who desire a broader overview. Participants suggested these could take the form of side panels or compact dashboards that aggregate joint-limit information without competing for attention in the main scene. As one participant noted, such summaries can exist alongside in-situ cues without needing to be visually dominant: “We can actually have a list of ... ideas next to it in summary ... even if we don’t draw them exactly how we want” (P2G1).

Escalating Alerts Near Critical Limits: Because users must react quickly when approaching or exceeding joint limits, participants emphasized the need for feedback escalation that is both clear and non-intrusive. Early warnings were generally expected to remain subtle, relying on gradual visual changes, such as bands becoming more prominent or colors shifting, to signal increasing risk without disrupting the task flow. As one participant described, escalation should begin with visual emphasis rather than abrupt alerts: “as you reach like a problem state, the band becomes more prominent ... you are in the red zone” (P2G1). This approach allows users to adjust their motion proactively while maintaining focus on the manipulation.

More disruptive cues were viewed as appropriate only at critical thresholds, such as when a hard limit is imminent or has already been reached. Blinking or rapid animation, in particular, was discussed as a high-salience signal that should be used sparingly to avoid distraction or visual fatigue. One participant noted that blinking is effective precisely because it demands attention, but only when something has gone wrong: “I do like blinking in the sense that ... it causes more awareness ... probably something wrong” (P3G1). Another participant similarly associated blinking with limit violations rather than early warnings: “when I reach the limit it starts blinking rapidly as well” (P1G1).

While our workshop focused on visual cues (i.e., AR), some participants suggested optional auditory cues as a complementary channel, particularly in visually dense or high-attention tasks. These audio signals were framed as redundant warnings rather than primary feedback and were expected to remain configurable to suit different users and environments. As one participant explained, audio could reinforce color-based feedback near critical states: “if it is going to the end it goes to red if it is okay it’s green ... can also make like an audio that it is going to be limited” (P9G3).

Customization and User-Dependent Interactions: Recognizing the diversity of user backgrounds, expertise levels, and task objectives, participants emphasized the importance of customizable joint-limit visualizations that can adapt to individual needs. Simpler, high-level cues were generally seen as more appropriate for novice users, while expert users were thought to benefit from more granular or on-demand information. Several participants described interaction techniques that allow users to explicitly request additional detail only when needed, thereby avoiding unnecessary clutter. One participant, for instance, proposed a direct interaction that reveals minimal, joint-specific feedback: “click on a joint and a stoplight for that joint specifically appears ... very binary” (P2G1). Another echoed this preference for conditional visibility, noting that automatically triggered but context-aware displays would be acceptable as long as they remain unobtrusive: “If we could make it automatic ... pops up ... first of all it’s not annoying” (P3G1).

Participants further emphasized that advanced information should remain hidden by default and be surfaced only through deliberate user action, such as tapping or pinching a joint. This approach was viewed as particularly valuable for maintaining focus during complex demonstrations. As one participant explained, selective disclosure empowers users to control both content and complexity: “... tap/pinch joint ... info is not there unless you request it ... pick three and almost like-the info is not there unless you request it” (P1G1).

Self-collision

Participants reported that the self-collision was particularly challenging for novices because it operates outside the user’s primary focus on the end-effector. As self-collision constraints are not directly communicated and are only exposed through execution failure, users struggled to diagnose the cause of errors. Our findings suggest that novices lack awareness of the robot’s full-body geometry, add we report following finding to elevate these issues (see [Figure A2](#) for figures)

Collision Proximity Awareness: Participants emphasized that the foundation of effective self-collision support in XR is the ability to immediately perceive how close different robot components are to making contact. This baseline awareness was described as essential for both novice and expert users, particularly because self-collisions can emerge gradually and unintentionally during complex demonstrations. Participants favored perceptual cues that directly visualize proximity between robot links. One participant illustrated this idea through an expanding geometric marker that encodes both distance and urgency: “...it starts with a small cross, and the cross starts growing bigger as you

move towards it ... green, and then you change the color to orange, and then to red ..." (P1G1).

Color-coded escalation emerged as a consistent strategy for communicating collision risk, with transitions from green (safe) through intermediate warning states to red when contact is imminent or has occurred. Participants emphasized that these cues should be visually grounded on the robot itself and applied only to the components at risk, rather than globally. As one participant explained: *"if close to self-collision, the two limbs that are anticipated to collide will both become highlighted, and maybe with exclamation points or something as a warning"* (P2G1).

Beyond simple highlighting, participants proposed additional visual forms—such as arrows or translucent meshes—that dynamically appear as distance decreases, reinforcing spatial understanding of the impending collision. For example, one participant described a directional cue: *"... when two links are getting closer to each other, an arrow pops up ... yellow when still in transition ... red when actually collide"* (P3G1).

Directional Guidance for Recovery After Collision Avoidance: Once users are made aware that a collision risk exists, participants emphasized that feedback must move beyond awareness and provide actionable guidance for correcting the demonstration. Directional feedback cues were therefore framed as a critical complement to proximity and urgency indicators. As one participant explained, effective support should intervene at the moment of risk by suggesting an alternative path: *"... at the point that I am touching, giving me the cues ... you can move to the other side [cue]"* (P1G1).

Several participants further suggested that motion or animation could strengthen the interpretability of directional cues, making corrective actions more intuitive. Animated arrows that "push" or flow in a particular direction were viewed as especially effective in communicating intent. As one participant noted, *"... the arrows animated to move ... showing like, go that way, go that way ... makes it clearer"* (P2G1). Others emphasized that this directional guidance should be tightly coupled with existing color-coded urgency cues, allowing users to simultaneously assess risk severity and corrective options. As one participant summarized, directional indicators should *"... point the user to which way you can correct this collision ... shows how to resolve the collision"* (P9G3).

3D Volume and Mesh Representations: Beyond planar or symbolic cues, participants proposed volumetric feedback that visually represents the spatial boundaries within which collisions may occur in three-dimensional space that explicitly occupy volume—such as mesh shells, translucent envelopes, or spherical casings around joints or links. One participant described this idea as moving beyond flat color cues toward an embodied spatial representation: *"... instead of just using color, actually having almost a casing"*

compress ... kind of like those things with the spikes ... you almost can see an extended version ...” (P1G1).

Participants further suggested that these volumetric elements should behave dynamically, deforming or materializing as robot components approach collision. Progressive appearance or compression of meshes was viewed as an intuitive way to convey increasing risk without abrupt interruption. As one participant explained: “... *mesh that shows around there ... if it’s coming closer, the mesh starts to appear*” (P5G2). Another similarly proposed joint-level volumetric indicators that activate only when relevant: “... *basically have meshes for each joint ... whenever they get closer, they start to appear*” (P4G2).

Distinguishing Normal and Danger States: Participants stressed the importance of clearly distinguishing between normal, low-risk proximity and genuinely dangerous configurations in order to minimize cognitive overload. They emphasized that when the robot is operating safely, visual feedback should remain largely passive, consisting of subtle or neutral elements that do not compete for attention. As one participant explained, under normal conditions the system should remain visually quiet, only becoming noticeable as risk begins to emerge: “... *normal situation ... nothing appears ... as you move, you’re starting to see something happening ...*” (P3G1).

More intrusive feedback—such as strong color shifts, directional arrows, or explicit alerting elements—was viewed as appropriate only after a predefined danger threshold has been crossed. One participant described this hierarchy explicitly, noting that certain visual elements are intentionally non-triggering, while others are reserved for intervention: “... *the dots are not meant to be a trigger ... arrows are meant to be like, hey, you need ... color scale giving inkling you might start having a problem ... here’s the thing that’s red*” (P2G1).

Manipulability

Our findings also reveal that manipulability is one of the least intuitive constraints for novice users. Participants were often unaware that certain demonstrated configurations limited the robot’s ability to move freely, even though the pose appeared correct. Since manipulability depends on the combined configuration of all joint angles and not on a single visible constraint, users had difficulty understanding why the robot could not smoothly reproduce a demonstrated motion. To communicate these limitation properly following suggestions were made during the focus group (see [Figure A3](#) for figures)

Global vs. Local Visualization of Singularity: Participants described manipulability as a fundamentally multi-scale problem that requires both global and local awareness in XR.

They emphasized that users must be able to distinguish whether a risk arises from an individual joint becoming constrained or from the overall robot configuration approaching a singularity. Singularities, in particular, were framed as whole-body phenomena rather than localized issues, motivating the need for global feedback mechanisms. As one participant articulated, “... *singularity is less a local like joint thing, but more like a global whole robot thing. So, like a cloud around the robot or something that turns the color of the robot to red, yellow ...*” (P3G1). To communicate these global manipulability states, participants proposed visualizations such as clouds, envelopes, or workspace bubbles that surround the entire robot and dynamically reflect whether the system remains within a safe operational region.

At the same time, participants cautioned that global indicators should not obscure the underlying source of risk. When manipulability degradation originates from specific joints, localized highlighting was preferred so users can take targeted corrective action. Participants further suggested that manipulability degradation could be communicated by highlighting the robot components most implicated in the instability, rather than applying uniform feedback across the entire system. Localized color highlighting on specific joints or limbs was seen as helping users understand where corrective action might be needed. One participant explained that proximity to singularities could be conveyed through graded color changes on the affected elements: “...*those actual limbs or joints... get color highlighted... if you’re getting close to a singularity, red or yellow*” (P2G1). Such cues were seen as particularly useful during learning-from-demonstration, where users may be exploring the boundaries of feasible motion.

Manipulability Awareness Alerts: Building on the need for global and local manipulability awareness, participants emphasized the importance of making approaching manipulability limits perceptible through intuitive and immediately interpretable alerts. Color coding was the most frequently proposed mechanism, with transitions toward yellow or red used to signal degrading stability or an increased risk of entering a singular configuration. These color changes were viewed as a direct and easily recognizable indication that certain motions were becoming unsafe or unavailable. As one participant described, once a limit is reached, visual feedback should clearly communicate loss of controllability: “... *joint gets technically disabled ... red color strip is shown, like you cannot move it further*” (P1G1).

Beyond color alone, participants discussed augmenting visual alerts with other perceptual cues to reinforce diminishing manipulability without overwhelming the user. Subtle light-based effects and optional auditory signals were proposed as complementary modalities that could increase salience in visually complex scenes. One participant likened this multi-modal feedback to familiar automotive systems: “... *lights can get to the red ...*

also can make a sound ... slow sound, then faster ... like a car when you want to park it” (P9G3).

Directional Guidance and Preventive Feedback: While awareness-oriented feedback helps users recognize that a manipulability issue exists, participants emphasized that such information alone is insufficient for effective intervention. Knowing that the robot is approaching a singularity does not inherently communicate how to avoid it. As a result, participants strongly advocated for XR systems that provide directional or preventive guidance to actively steer users away from unstable configurations. One participant explicitly highlighted the need for prescriptive feedback near singularities: “... *if you were close to a singularity ... suggesting what to do to avoid it ... arrows could be a cool idea*” (P2G1).

Directional guidance was frequently framed in terms of arrows or animated cues that indicate safer directions of movement in real time. These cues were expected to adapt dynamically as the robot configuration evolves, effectively translating abstract manipulability constraints into actionable motion advice. As one participant noted, effective systems should not only inform but actively intervene: “... *this tells you what to do to avoid [singularity] ... or prevents you from letting it happen*” (P6G2).

In addition, some participants proposed anticipatory or predictive feedback based on simulation, allowing the system to detect and communicate impending singularities before they occur. Such approaches were seen as enabling more robust avoidance strategies by suggesting alternative joint usage or motion paths. One participant described this predictive logic explicitly: “... *calculate ... simulation ... check the singularity ... if singularity occurs, notify alert ... use other joints and avoid until these joints avoid the singularity*” (P8G3).

6.2.5 Discussion

Based on our findings, we derive the following implications for the design of XR interfaces supporting robot learning from demonstration under kinematic constraints.

Implication 1: *Risk and Constraints Should Be Color-Coded* Joint limits, self-collision risk, and manipulability should be communicated through color coding. Stoplight-style mappings (green–yellow–red) should be used to indicate urgency with minimal cognitive effort. Continuous color gradients should convey proximity to constraints and support anticipation of risk.

Implication 2: *Awareness Cues Should Be Layered with Actionable Guidance* Constraint feedback should be layered. Color changes and visual highlights should signal

the presence of risk. When thresholds are exceeded, directional and animated cues should provide explicit guidance for corrective action, such as indicating safe movement directions.

Implication 3: *Constraint Feedback Should Escalate to Avoid Visual Overload* Constraint feedback should remain visually minimal during safe operation and escalate only when constraints become relevant or dangerous. Feedback should be spatially anchored to affected robot components, limited to active joints, and clearly distinguish normal from danger states to reduce clutter in XR environments.

Implication 4: *Feedback Granularity Should Adapt to User Expertise* The level of detail in constraint feedback should adapt to user expertise and task demands. Novices should be supported with simple, binary indicators, while experts should have access to continuous gradients and numerical values. Detailed information should be revealed on demand to maintain simplicity during routine interaction.

6.3 Study 2: Lab Study Evaluation

Based on the design implications obtained in the focus group, we implemented a set of visualization to assess their effect human-robot interaction during two LfD tasks, addressing research questions **RQ2a** and **RQ3a**. For that we conducted a lab study with novice demonstrators. The study was approved by our university ethics committee (2025-24636-71568-5).

6.3.1 Robot Key Features Visualization

We developed an XR application to visualize the three key components for LfD with robotic manipulators: (1) joints limits, (2) self-collision, and (3) manipulability. We used the latest Unity engine (Unity 6) and the Meta Quest 3 headset. We selected this headset for its passthrough capabilities and broad platform support, which help lower barriers to adoption and improve replicability.

The visualization design followed the implications identified in the focus group study. First, we implemented **recognizable color-coded cues**, using green–yellow–red for joint limits and self-collision, and blue–purple–red for manipulability. Second, we provided **actionable guidance**, particularly for joint limits, to help users respond appropriately. Third, to **reduce information overload** [240], visualizations are not continuously displayed; instead, each component appears only when its magnitude approaches the respective limit, with color transparency (α) gradually increasing as the limit is

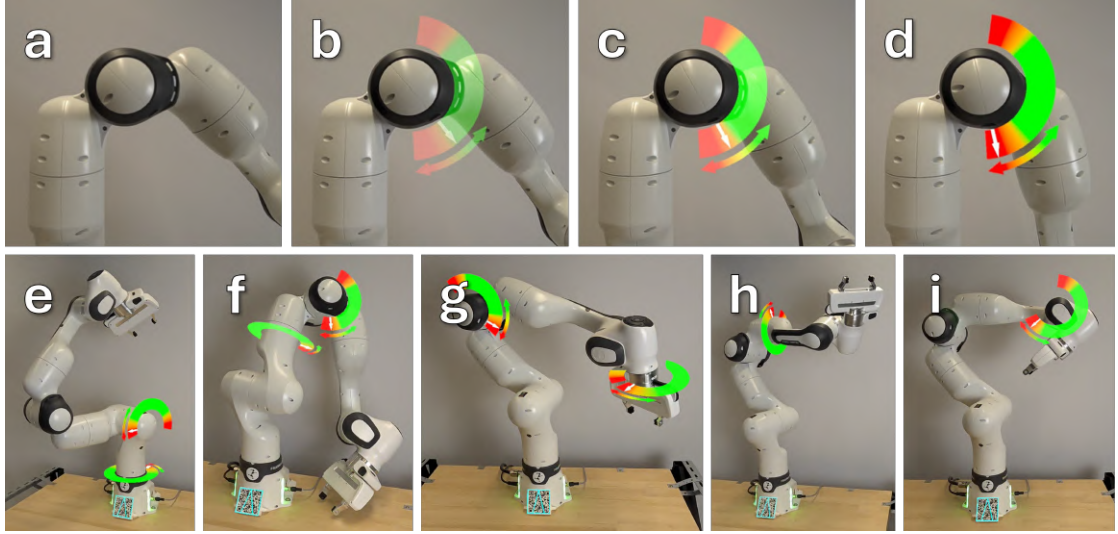


FIGURE 6.3: First row shows joint limit visualizations on joint 4 based on the current absolute angular distance from the joint limit (δ): a) no visualization, $\delta > 30^\circ$, color $\alpha = 0$, b) $20 < \delta < 30^\circ$, $0 < \alpha < 0.5$, c) $10 < \delta < 20^\circ$, $0.5 < \alpha < 1$, d) $\delta < 10^\circ$, $\alpha = 1$. Second row shows the joint limit visualizations on the 7 joints (J1 to J7): e) J1 and J2, f) J3 and J4, g) J4 and J7, h) J5, and i) J6.

approached. Finally, in line with the implication that **feedback granularity should adapt to user expertise**, we designed the visualizations for simplicity, providing only the basic indicators suitable for novice users. Detailed or numerical feedback was not implemented, as our lab study focused exclusively on novice participants. Further details are provided below for each visualization component.

Joints Limits

We chose an arc overlay design that displays the actual angular range of motion for each joint, as shown in Figure 6.3. These arcs have a fixed thickness of 5 cm and use three colors to indicate proximity to the joint limits at either extreme. Angles within 10 degrees of a limit are shown in red, angles between 10 and 20 degrees are shown in orange, and angles beyond 20 degrees are shown in green, with smooth gradients between colors. Each arc included a pointer, similar to clock hands, to indicate the current joint angle. In addition, we employ dynamic transparency: the arc is completely invisible when the joint is within a safe range (angular distance from the limit above 30 degrees) and becomes progressively more opaque as the joint approaches either limit (Figure 6.3-a to d).

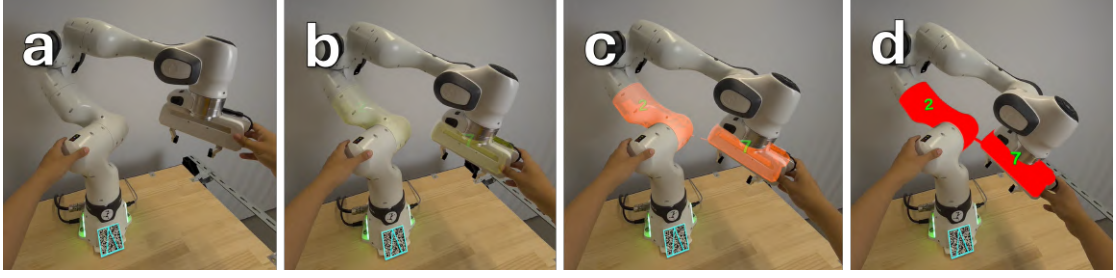


FIGURE 6.4: Self collision visualizations between links 2 and 7 based on the surface-to-surface distance between them: a) very safe distance above 15 cm (no visualization), b) safe distance below 15 cm with links highlighted in green with high transparency, c) warning distance with links in orange, and d) danger distance with links in solid red.

Self-Collision

For self-collision visualization, we compute in real time the distances between all possible pairs of robot links and identify the pair with the smallest distance. Only this closest link pair is visualized at any given time. The two links are highlighted and connected by a dashed line to indicate their proximity. To facilitate quick identification, we display a label for each highlighted component (a number for links and RF/LF for the right and left fingers of the end effector, respectively). Similar to the joints limits visualization, we employ dynamic transparency: no visualization is shown when the links are at a safe distance from each other (greater than 20 cm). As the distance decreases, the visualization progressively becomes more opaque, transitions from green to red, and the connecting line changes from dashed to solid, as shown in Figure 6.4.

Manipulability

We visualize robot manipulability using a manipulability ellipsoid, which represents the set of achievable end-effector velocities for unit-norm joint velocities at a given configuration [179]. The ellipsoid's principal axes correspond to the singular vectors of the Jacobian, with axis lengths proportional to the associated singular values [241]; larger volumes indicate higher manipulability, while smaller, flattened ellipsoids indicate reduced manipulability.

The ellipsoid is rendered at the end effector to provide immediate feedback during execution. Its color transitions from blue to red as manipulability decreases, with transparency varying between 0 and 0.5 (Figure 6.5); ellipsoids exceeding a volume threshold of 0.2 cm^3 are hidden to reduce clutter. Figure 6.6 shows all visualization components simultaneously, including joint-limit and self-collision indicators, along with a QR-based anchor at the robot base to ensure accurate AR alignment.

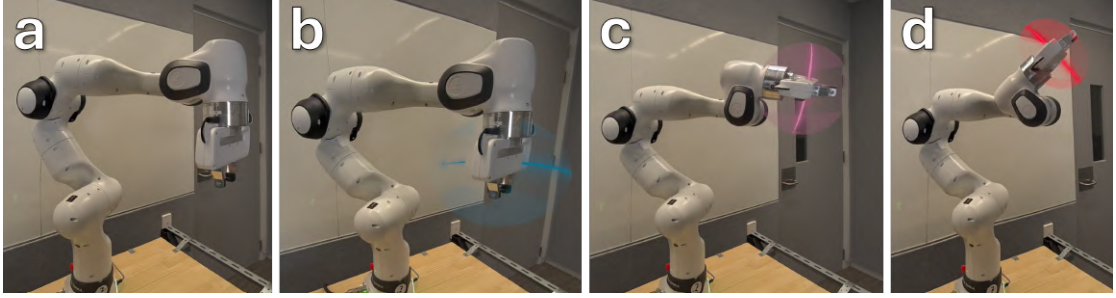


FIGURE 6.5: Manipulability ellipsoid anchored to the robot’s end effector: (a) high manipulability, represented by a large ellipsoid (not visualized), (b) medium manipulability with reduced volume shown in blue, (c) low manipulability with further volume reduction indicated in purple (blue-red transition), and (d) very low manipulability represented by a small ellipsoid shown in red.

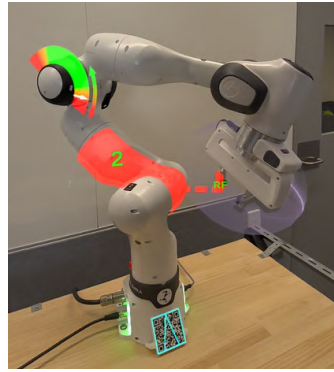


FIGURE 6.6: All three visualization features present simultaneously on the robot: joint limit on J4, self collision between link 2 and right finger (RF), and manipulability spheroid on the end effector. We used a QR-code-based spatial anchor placed in link 0.

6.3.2 Experiment Setup

The study took place in a lab environment equipped with the Frank Emika 7-DoF Panda robot, a collaborative robotic manipulator (see [Figure 6.8](#)). This robot arm is equipped with torque sensors in all 7 joints, making it safe and suitable for human-robot interaction applications. We used the default gravity compensation controller to counteract the robot’s inherent weight, allowing participants to kinesthetically manipulate the robot’s joints without additional force.

For the XR rendering, we employed the Meta Quest 3 as the head-mounted display (HMD) to visualize key robot features, using its passthrough capabilities.

6.3.3 Experiment Conditions

This study investigated whether augmenting demonstrations with AR-based feedback improves user performance compared to traditional offline feedback, whether real-time

AR feedback alone provides measurable benefit, and whether combining real-time and offline feedback further enhances interaction. We designed three conditions:

Offline Feedback — Participants demonstrated tasks without real-time guidance. After the first three demonstrations, they observed the robot’s reproduced execution and used this offline feedback to adjust subsequent demonstrations.

Real-Time Feedback — Participants received continuous AR feedback during demonstrations, showing key task-relevant states to guide movements in real time. No execution playback was provided between rounds.

Combined Feedback — Participants received both real-time AR cues and offline execution playback, enabling immediate guidance during demonstrations and reflection on the robot’s reproduced motion afterward.

6.3.4 Study Goals

Study 2 builds on the design insights identified in Study 1. Specifically, the study investigates how real-time, AR-based constraint feedback derived from focus group findings on awareness, escalation, and actionable guidance, influences key aspects of interaction during LfD. We examine how different feedback modalities shape (1) the quality and efficiency of human demonstrations, (2) how users accommodate inherent robot constraints such as joint limits, self-collisions, and reduced manipulability, and (3) the resulting quality and reliability of robot task execution. In addition, we assess how these feedback designs affect users’ subjective experience, including perceived workload and system usability.

By comparing offline, real-time, and combined feedback conditions across tasks of varying complexity, the study aims to understand not only whether real-time feedback is beneficial, but how and under what interactional circumstances it supports learning, confidence, and effective teaching. This exploratory framing allows us to surface patterns and design implications that inform the role of feedback timing and modality as interaction design variables in embodied robot teaching.

6.3.5 Experimental Tasks

We evaluated two manipulation tasks—cuboid-block insertion and pouring—commonly used in prior work [144, 242] for their generic nature, requiring no domain-specific expertise.

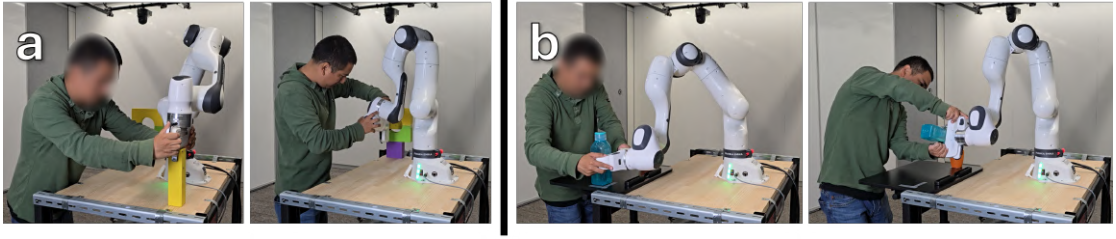


FIGURE 6.7: Lab study evaluation tasks: a) Cuboid block insertion and b) water pouring.

Insertion Task

Participants taught the robot to pick up a cuboid block ($5 \times 5 \times 20$ cm) and partially insert it into a 10 cm-diameter hole as shown in Figure 6.7(a). The task required avoiding collisions with the hole edges and the robot itself, while aligning the block perpendicularly to the hole's surface.

Pouring Task

Participants taught the robot to grasp a bottle and move it toward a cup to simulate pouring, as seen in Figure 6.7(b). Success required careful joint coordination to maintain bottle orientation, avoid collisions, and ensure a natural pouring trajectory. Real liquid was not used for safety, given the novice participants and sensitive equipment.

The study used a between-subjects design in which participants were assigned to one feedback condition. Each participant provided multiple demonstrations over time, allowing us to account for changes across demonstrations while controlling for individual differences between participants.

6.3.6 Participants Recruitment

We conducted an a prior power analysis using G*Power [208] to determine the required sample size, after which we recruited 36 participants (21M/15F; age = 26.34 ± 4.02) through social media advertisements and campus notice boards. To maintain consistency across experimental conditions, participants were evenly distributed such that each group comprised 7 males and 5 females (i.e., 12 participants per group). In addition to demographic questions, we used the questionnaire from [243] to ensure that all participants were novice users. Following guidance from our University ethics committee and to avoid any risk for participants, we also included two screening questions: *Do you experience motion sickness when using Augmented Reality (AR) devices?* and *Do you*

feel uncomfortable or scared when you are around robots? The first question served to filter participants based on prior experience with AR headset while second assessed participants' level of comfort in the present of robot. A positive answer to either question would make the person ineligible to participate.

6.3.7 Experiment Procedure

Participants provided informed consent, completed a demographic questionnaire, and were introduced to the robot, including safe kinesthetic guidance in compliant mode. AR headset features were demonstrated for real-time and combined groups. Participants then completed a brief familiarization session.

Tasks were performed sequentially, starting with the easier insertion task followed by pouring, to allow participants to first become familiar with the robot before attempting the more challenging task. Participants were randomly assigned to one of three feedback conditions (offline, real-time, or combined; 12 per group) and instructed to provide six successful demonstrations per task. After the insertion task, SUS and NASA-TLX questionnaires were completed, with NASA-TLX repeated after the pouring task. A brief debriefing interview concluded the one-hour procedure.

For offline participants, feedback consisted of robot execution playback between demonstration rounds. Real-time participants received AR visual cues during demonstrations but no playback. Combined participants received both AR cues and playback. During the pouring task, no feedback was provided for any group, enabling assessment of transfer learning and participants' ability to generalize learned constraints independently.

Objective Measures

To quantify how each experimental condition affects both human performance and the resulting robot performance, we define a set of objective metrics based on the robot's kinematic state. Specifically, consider a robot with n degrees of freedom, whose joint variables are denoted by $\mathbf{q}(t) = [q_1(t), \dots, q_n(t)]$, where $q_i \in \mathbb{R}$. A demonstration D is defined as a discrete joint-space trajectory

$$D = \langle \mathbf{q}(t_0), \mathbf{q}(t_1), \dots, \mathbf{q}(t_T) \rangle,$$

where $\mathcal{T}_m = \{t_0, \dots, t_T\}$ denotes the execution interval of m^{th} demonstration, and $|\mathcal{T}_m|$ is the total number of recorded time steps. Let $\mathcal{D} = \langle D_1, D_2, \dots, D_M \rangle$ depicts the set of all given demonstrations, where the total number of demonstrations is represented as

$\mathcal{M} = |\mathcal{D}|$. All metrics below are computed only over successful demonstrations $\mathcal{D}_s \subset \mathcal{D}$.

Task completion rate — defined as the proportion of demonstrations that successfully achieved the task objective, computed as the ratio of successful demonstrations $|\mathcal{D}_s|$ to the total number of demonstrations $\mathcal{M}$.

Active Demonstration Time — To quantify the effective duration of human demonstrations, we computed the *active demonstration time* by excluding periods in which the robot remained idle. Joint velocities were obtained by finite differencing the recorded joint positions at 30 Hz. At each timestep, the robot was classified as idle if the maximum absolute joint velocity across all joints was below a motion threshold (0.017 rad/s), reflecting encoder noise bounds and the minimum intentional guiding force under impedance control. Idle segments shorter than 0.2 s were discarded to avoid spurious detections. The total idle duration was subtracted from the total demonstration duration to yield the active demonstration time.

Distance to Joint Limits — To assess how closely the robot operated to its mechanical limits, we computed a normalized distance-to-limit measure. For each joint i with limits $q_{i,\min}$ and $q_{i,\max}$, the normalized clearance at time t is

$$d_i(t) = \frac{\min(q_i(t) - q_{i,\min}, q_{i,\max} - q_i(t))}{q_{i,\max} - q_{i,\min}}. \quad (6.1)$$

We then average across all n joints to obtain a demonstration-level profile

$$d_{\text{joint}}(t) = \frac{1}{n} \sum_{i=1}^n d_i(t), \quad (6.2)$$

and compute the mean clearance over the full trajectory:

$$d_{\text{joint}}^{(m)} = \frac{1}{|\mathcal{T}_m|} \sum_{t \in \mathcal{T}_m} d_{\text{joint}}(t), \quad (6.3)$$

where higher values of $d_{\text{joint}}^{(m)}$ indicate that the robot operated farther from its joint limits.

Distance to Self-Collision — We quantified the safety margin of robot motion with respect to internal collisions using the MoveIt 2 collision environment. For each configuration $\mathbf{q}(t) \in D^m$, the robot state was inserted into the planning scene and signed distances were computed between all non-ignored link pairs. For links i and j , let $d_{ij}(t)$ denote their signed Euclidean separation, where negative values indicate penetrating

(self-colliding) configurations and positive values indicate collision-free separation. The instantaneous minimum separation is

$$d_{\min}(t) = \min_{(i,j)} d_{ij}(t). \quad (6.4)$$

A conservative safety margin for demonstration m is obtained as the minimum distance over its execution interval:

$$d_{\text{self}}^{(m)} = \min_{t \in \mathcal{T}_m} d_{\min}(t), \quad (6.5)$$

where larger values of $d_{\text{self}}^{(m)}$ correspond to safer kinematic configurations during the demonstration.

Quality of Demonstration (Manipulability) — To evaluate the manipulability of demonstrated configurations, we used the measure [179], which reflects proximity to kinematic singularities and has been linked to demonstration quality in learning-from-demonstration settings [10]. For a x -dimensional task with Jacobian $J(\mathbf{q}(t)) \in \mathbb{R}^{x \times n}$, the manipulability at time t for demonstration m is

$$\Omega(D^m, t) = \sqrt{\det(J(\mathbf{q}(t))J^\top(\mathbf{q}(t)))}. \quad (6.6)$$

The demonstration-level manipulability score is obtained by averaging over the trajectory:

$$\alpha^{(m)} = \frac{1}{|\mathcal{T}_m|} \sum_{t \in \mathcal{T}_m} \Omega(D^m, t). \quad (6.7)$$

Higher $\alpha^{(m)}$ values indicate more dexterous and higher-quality demonstrations, whereas lower values reflect reduced motion flexibility.

6.3.8 Policy Learning

For the learning algorithm, we used a standard Gaussian Mixture Model (GMM) to train LfD models from the collected data [174, 206]. This approach is widely used in LfD research due to their ability to capture the underlying distribution of demonstrated trajectories from a limited number of demonstrations with minimal learning time [32, 50].

The robot task learning begins after the user provides a set of $\mathcal{M}$ demonstrations, where each demonstration m consists of T_m state data points. Each state s is represented as $\xi_s = (t_s, x_s^T, \omega_s^T)$, where t_s , x_s , and ω_s correspond to the time, end-effector position, and orientation in quaternion form, respectively.

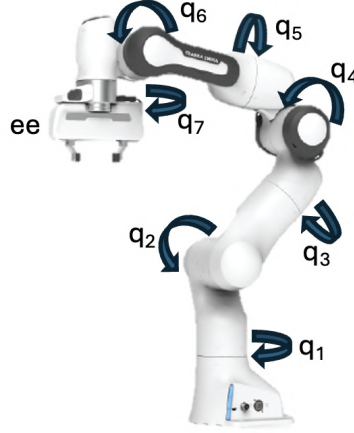


FIGURE 6.8: Franka Emika Panda Robot

TABLE 6.2: NASA-TLX Questionnaire Results (21-point Scale) — $M \pm SD$ with 95% Confidence Intervals

		Offline Feedback	Real-Time Feedback	Hybrid Feedback
Insertion	Mental Demand	6.92 ± 3.63 [4.61, 9.22]	6.17 ± 4.76 [3.14, 9.19]	6.67 ± 1.67 [5.61, 7.73]
	Physical Demand	8.50 ± 3.80 [6.08, 10.92]	8.00 ± 4.51 [5.13, 10.87]	6.67 ± 1.44 [5.75, 7.58]
	Temporal Demand	6.75 ± 3.41 [4.58, 8.92]	5.83 ± 3.35 [3.70, 7.96]	6.67 ± 2.06 [5.36, 7.98]
	Performance	12.33 ± 5.53 [8.82, 15.85]	16.00 ± 2.83 [14.20, 17.80]	16.75 ± 2.86 [14.93, 18.57]
	Effort	8.83 ± 3.13 [6.85, 10.82]	6.67 ± 3.52 [4.43, 8.91]	6.50 ± 1.93 [5.27, 7.73]
	Frustration	6.75 ± 4.09 [4.15, 9.35]	4.42 ± 3.63 [2.11, 6.72]	5.50 ± 2.11 [4.16, 6.84]
Pouring	Mental Demand	12.25 ± 4.59 [9.33, 15.17]	6.08 ± 5.02 [2.90, 9.27]	6.42 ± 4.06 [3.84, 8.99]
	Physical Demand	11.00 ± 4.37 [8.22, 13.78]	5.92 ± 5.20 [2.62, 9.22]	7.92 ± 3.82 [5.49, 10.35]
	Temporal Demand	7.75 ± 4.05 [5.18, 10.32]	4.83 ± 3.35 [2.70, 6.96]	6.67 ± 2.93 [4.80, 8.53]
	Performance	8.75 ± 5.15 [5.48, 12.02]	15.50 ± 3.85 [13.05, 17.95]	15.33 ± 3.03 [13.41, 17.26]
	Effort	14.00 ± 4.75 [10.98, 17.02]	6.17 ± 4.95 [3.02, 9.31]	8.83 ± 3.86 [6.38, 11.28]
	Frustration	11.67 ± 4.54 [8.78, 14.55]	4.00 ± 4.20 [1.33, 6.67]	4.75 ± 4.43 [1.93, 7.57]

After temporally aligning the demonstrations, a K-component GMM is fitted to all demonstration data, with parameters (π_k, μ_k, Σ_k) representing the prior probabilities, means, and covariance matrices for each component k . The number of components K is selected using the Bayesian Information Criterion (BIC) to balance model complexity and data fit. For trajectory generation, Gaussian Mixture Regression (GMR) is applied to predict the expected end-effector position and orientation at each time step. This approach allows the robot to reproduce smooth and generalized trajectories from the given demonstrations [206].

System Usability Scale (SUS) — To evaluate system usability in each condition, we used the SUS [244]. The SUS consists of ten items rated on a 5-point Likert scale and produces a single overall usability score. Its strength lies in being a widely adopted, bench-marked instrument across diverse interface types and applications, enabling us to interpret the usability of our system relative to established norms beyond AR or robotics [245].

Task Load — We employed the NASA-TLX [203] to measure participants’ subjective workload across feedback conditions. The NASA-TLX consists of six sub-scales each rated on a 21-point scale.

In addition to these quantitative metrics, we asked participants about their experience at the end of the experiment in a semi-structured interviews:

- Q1) What changes did you make after observing the robot’s execution?
- Q2) What adjustments did you make after failed demonstrations (if any)?
- Q3) Did you understand the key qualities (or what features are you paying attention to) that make a demonstration effective?
- Q4) What would you improve if you were to repeat the task?
- Q5) How did you perceive the impact of your demonstrations on the robot’s performance?
- Q6) How did the feedback system help you in performing the second task effectively?

6.3.9 Statistical Analysis

We used linear mixed-effects models to analyze the effects of feedback condition and demonstration number on outcome measures:

$$y \sim \text{condition} + \text{demo_index} + \text{condition} : \text{demo_index} + (\text{demo_index} | \text{participant}),$$

where y is the dependent variable, *condition* the experimental condition, and *demo_index* the demonstration number. Random slopes for demonstration number accounted for individual performance trajectories. Post-hoc comparisons used estimated marginal means with Tukey adjustments.

For single-observation measures per participant (e.g., SUS, NASA-TLX, task execution), Welch’s ANOVA tested group differences, with Games-Howell post-hoc contrasts adjusting for unequal variances and sample sizes.

6.3.10 Qualitative Analysis

Audio recordings were transcribed using Otter.ai and reviewed by the first author, with personally identifiable information replaced by pseudonyms (e.g., P_29, offline). Two coders independently familiarized themselves with 25% of the transcripts, generating preliminary inductive codes [239]. Codes and relevant quotes were imported into a shared Miro board, and the coders collaboratively developed a unified code set. Remaining transcripts were jointly coded, refining the codes and consolidating themes. All

iterations were documented for transparent version control. Candidate themes were then generated, refined, and finalized through discussion, with the second author verifying reporting quotes against the original audio before raw data deletion.

6.4 Results

6.4.1 Task Completion Rate

Descriptive statistics indicated clear differences in completion rates across conditions and tasks. For the insertion task, completion rates increased from offline ($M = 0.78, SD = 0.14$) to real-time ($M = 0.86, SD = 0.13$) and combined ($M = 0.94, SD = 0.09$). A similar pattern was observed for the pouring task, where the offline condition showed substantially lower performance ($M = 0.49, SD = 0.10$) compared to real-time ($M = 0.83, SD = 0.14$) and combined ($M = 0.82, SD = 0.12$), which were comparable. Welch's ANOVA confirmed a significant effect of group for insertion ($F(2, 20.90) = 5.72, p = 0.010$), with Games–Howell post-hoc tests showing that combined significantly outperformed offline ($p = 0.010$), while other comparisons were not significant. For pouring, a strong group effect was observed ($F(2, 21.47) = 33.53, p < 0.001$), with both combined and real-time significantly outperforming Offline (both $p < 0.001$), and no significant difference between combined and real-time ($p = 0.986$).

6.4.2 Active Demonstration Time

To investigate how the feedback conditions influenced demonstration time, we first analyzed all six successful demonstrations for each task across all conditions. We then conducted separate analyses on the initial and final three demonstrations to examine changes over time. All results reported here focus on active demonstration time (i.e. the periods during which participants were actively guiding the robot). Figure 6.9 shows the active demonstration time means for both tasks.

Insertion Task — A mixed-effects analysis of active demonstration time revealed a significant main effect of demonstration number ($\chi^2(1) = 6.39, p = 0.01$), indicating changes as participants progressed through repetitions. No significant effects of condition ($\chi^2(2) = 0.56, p = 0.75$) or condition-by-demonstration interaction ($\chi^2(2) = 0.37, p = 0.83$) were observed. Estimated marginal means showed comparable active times across conditions (offline: $M = 16.7$, real-time: $M = 17.5$, combined: $M = 15.8$; all $p > 0.73$). Analyses of the initial three demonstrations showed a significant effect of demonstration number ($\chi^2(1) = 11.1, p < .001$), while no significant effects were found

TABLE 6.3: **Between Conditions** analysis results of the linear mixed-effects (LME) model, estimated marginal means (EMMs), and Tukey-adjusted pairwise contrasts for **demonstration time (seconds)**.

		Insertion Task			Pouring Task
		Initial (three demos)	Last (three demos)	All (six demos)	All (six demos)
LME	Condition	1.067 ($p = 0.59$)	3.193 ($p = 0.21$)	0.561 ($p = 0.75$)	12.43 ($p = 0.002$)**
	Demonstration	11.1 ($p < .001$)***	0.639 ($p = 0.42$)	6.39 ($p = 0.01$)*	19.02 ($p < .001$)***
Condition : Demonstration		1.072 ($p = 0.58$)	3.822 ($p = 0.14$)	0.372 ($p = 0.83$)	0.157 ($p = 0.93$)
EMMs	Offline Feedback	17.7 [13.7, 21.6]	13.4 [10.2, 16.7]	16.7 [13.5, 19.9]	20.2 [17.8, 22.6]
	Real-time Feedback	18.9 [15.0, 22.8]	15.6 [12.4, 18.7]	17.5 [14.2, 20.7]	11.8 [9.51, 14.1]
	Hybrid Feedback	17.0 [13.1, 20.9]	14.5 [11.3, 17.7]	15.8 [12.6, 19.0]	11.5 [9.21, 13.8]
Contrasts	Offline — Real-time	-1.24 ($p = 0.89$)	-2.14 ($p = 0.60$)	-0.74 ($p = 0.94$)	8.35 ($p < .001$)***
	Offline — Hybrid	0.656 ($p = 0.97$)	-1.03 ($p = 0.88$)	0.923 ($p = 0.91$)	8.66 ($p < .001$)***
	Real-time — Hybrid	1.889 ($p = 0.77$)	1.11 ($p = 0.87$)	1.664 ($p = 0.74$)	0.309 ($p = 0.98$)

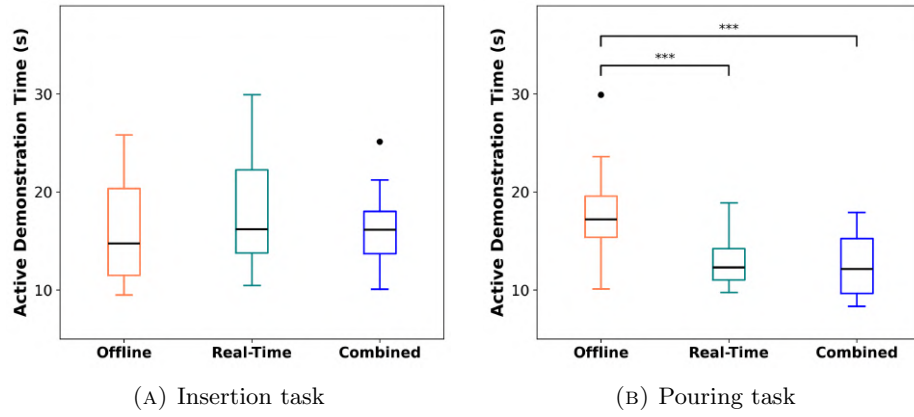


FIGURE 6.9: Active demonstration time (sec) for both tasks (a) insertion task, and (b) pouring task.

in the final three demonstrations (all $p > 0.15$), suggesting early learning effects that stabilized over time.

Pouring Task — For the pouring task, significant main effects of condition ($\chi^2(2) = 12.43$, $p = 0.002$) and demonstration number ($\chi^2(1) = 19.02$, $p < .001$) were observed, with no significant interaction ($\chi^2(2) = 0.15$, $p = .92$). Estimated marginal means indicated longer active times in the offline condition ($M = 20.2$) compared to both real-time ($M = 11.8$) and combined ($M = 11.5$) conditions. Tukey-adjusted comparisons confirmed significant differences between offline and both feedback conditions (both $p < .001$), with no difference between real-time and combined ($p = .97$). These results indicate robust condition-dependent differences alongside a general repetition-related effect. The results are summarized in Table 6.3.

TABLE 6.4: **Between Conditions** analysis results of the linear mixed-effects (LME) model, estimated marginal means (EMMs), and Tukey-adjusted pairwise contrasts for **distance to joint-limits**. Values are normalized to the corresponding joint ranges, where smaller values indicate proximity to the joint limits (and larger values indicate being farther away).

		Insertion Task			Pouring Task
		Initial (three demos)	Last (three demos)	All (six demos)	All (six demos)
<i>LME</i>	<i>Condition</i>	9.17 ($p = 0.01$)*	0.396 ($p = 0.82$)	8.15 ($p = 0.01$)*	1.17 ($p = 0.55$)
	<i>Demonstration</i>	1.230 ($p = 0.26$)	0.392 ($p = 0.53$)	3.88 ($p = 0.04$)*	2.25 ($p = 0.13$)
	<i>Condition : Demonstration</i>	0.468 ($p = 0.79$)	1.473 ($p = 0.47$)	1.67 ($p = 0.43$)	1.21 ($p = 0.54$)
<i>EMMs</i>	<i>Offline Feedback</i>	0.330 [0.322, 0.339]	0.337 [0.330, 0.347]	0.331 [0.323, 0.340]	0.345 [0.338, 0.353]
	<i>Real-time Feedback</i>	0.340 [0.331, 0.349]	0.342 [0.334, 0.350]	0.341 [0.333, 0.350]	0.340 [0.333, 0.347]
	<i>Hybrid Feedback</i>	0.349 [0.341, 0.359]	0.351 [0.342, 0.358]	0.349 [0.341, 0.358]	0.334 [0.326, 0.341]
<i>Contrasts</i>	<i>Offline — Real-time</i>	−0.009 ($p = 0.29$)	−0.004 ($p = 0.78$)	−0.009 ($p = 0.24$)	0.005 ($p = 0.57$)
	<i>Offline — Hybrid</i>	−0.019 ($p = 0.01$)*	−0.012 ($p = 0.12$)	−0.018 ($p = 0.01$)*	0.011 ($p = 0.08$)
	<i>Real-time — Hybrid</i>	−0.009 ($p = 0.26$)	−0.007 ($p = 0.37$)	−0.008 ($p = 0.35$)	0.006 ($p = 0.42$)

6.4.3 Distance to Joint-Limits

Insertion Task — The analysis revealed significant main effects of condition ($\chi^2(2) = 8.15$, $p = .01$) and demonstration number ($\chi^2(1) = 3.88$, $p = .04$), with no significant condition-by-demonstration interaction ($\chi^2(2) = 1.67$, $p = .43$). Estimated marginal means showed smaller joint-limit clearance in the offline condition ($M = 0.331$) compared to the real-time ($M = 0.341$) and combined ($M = 0.349$) conditions, with a significant difference between offline and combined ($p = .01$). Figure 6.10a shows the normalized distance to joint limits for each condition. Analysis of the initial three demonstrations showed a significant effect of condition ($\chi^2(2) = 9.17$, $p = .01$), whereas no significant effects were observed in the final three demonstrations (all $p > 0.39$), indicating that early group differences in joint-limit clearance diminished over repetitions. The results are summarized in Table 6.4.

Pouring Task — For the pouring task, no significant main effects of condition ($\chi^2(2) = 1.17$, $p = .55$) or demonstration number ($\chi^2(1) = 2.25$, $p = .13$) were observed, and the interaction was also non-significant ($\chi^2(2) = 1.21$, $p = .54$). Estimated marginal means indicated comparable joint-limit distances across offline ($M = 0.345$), real-time ($M = 0.340$), and combined ($M = 0.334$) conditions, with no significant pairwise differences. These results indicate consistent joint-limit clearance across conditions throughout the pouring task. The results are summarized in Table 6.4, and Figure 6.10b presents the normalized distance to joint limits for each condition.

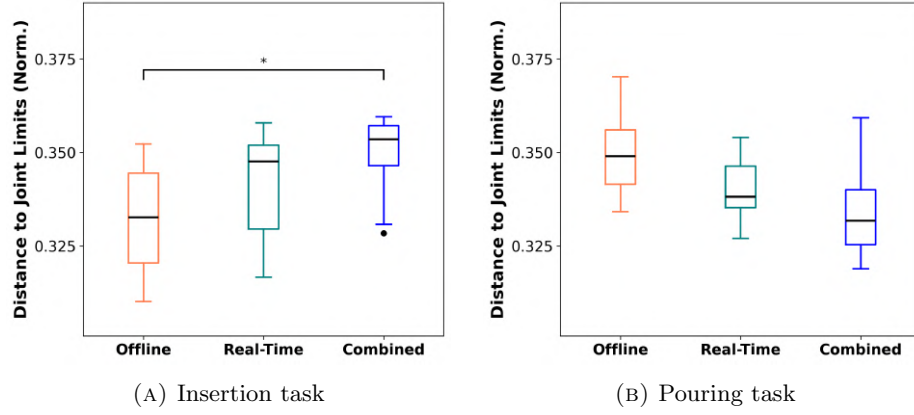


FIGURE 6.10: Distance to joint-limits for both tasks (a) insertion task, and (b) pouring task.

6.4.4 Distance to Self-Collision

Insertion task — No significant main effects of condition ($\chi^2(2) = 2.25$, $p = .32$) or demonstration number ($\chi^2(1) = 0.77$, $p = .37$) were observed, and the condition-by-demonstration interaction was not significant ($\chi^2(2) = 1.46$, $p = .48$). Estimated marginal means indicated comparable self-collision distances across conditions, with a small but significant difference between offline and real-time conditions ($p = .04$). Analyses of the initial three demonstrations revealed no significant effects (all $p > 0.35$), whereas for the final three demonstrations the effect of demonstration number approached significance ($\chi^2(1) = 3.81$, $p = .05$), and a significant difference emerged between offline and real-time conditions ($p = .01$). These results suggest largely stable self-collision margins with a modest late divergence between conditions. Figure 6.11a shows the distance to self-collision for each condition.

Pouring task — For the pouring task, a significant main effect of condition was observed ($\chi^2(2) = 7.95$, $p = .019$), while the effects of demonstration number ($\chi^2(1) = 1.27$, $p = .25$) and the interaction term ($\chi^2(2) = 1.62$, $p = .44$) were not significant. Estimated marginal means showed larger self-collision clearance in the offline condition ($M = 0.124$) compared to the real-time condition ($M = 0.107$), with a significant pairwise difference ($p = .005$); no other contrasts were significant. These findings indicate condition-dependent differences in self-collision safety margins during the pouring task, with offline demonstrations maintaining the greatest clearance. The results are summarized in Table 6.5.

TABLE 6.5: **Between Conditions** analysis results of the linear mixed-effects (LME) model, estimated marginal means (EMMs), and Tukey-adjusted pairwise contrasts for **distance to self-collision (m)**.

		Insertion Task			Pouring Task
		Initial (three demos)	Last (three demos)	All (six demos)	All (six demos)
<i>LME</i>	<i>Condition</i>	2.087 ($p = 0.35$)	4.145 ($p = 0.13$)	2.25 ($p = 0.32$)	7.95 ($p = 0.01$)*
	<i>Demonstration</i>	0.044 ($p = 0.83$)	3.814 ($p = 0.05$)*	0.77 ($p = 0.37$)	1.27 ($p = 0.25$)
	<i>Condition : Demonstration</i>	0.730 ($p = 0.69$)	0.740 ($p = 0.70$)	1.46 ($p = 0.48$)	1.62 ($p = 0.44$)
<i>EMMs</i>	<i>Offline Feedback</i>	0.084 [0.074, 0.095]	0.075 [0.063, 0.087]	0.082 [0.071, 0.092]	0.124 [0.116, 0.131]
	<i>Real-time Feedback</i>	0.099 [0.089, 0.109]	0.101 [0.089, 0.112]	0.099 [0.089, 0.110]	0.107 [0.099, 0.114]
	<i>Hybrid Feedback</i>	0.092 [0.082, 0.102]	0.093 [0.081, 0.106]	0.092 [0.082, 0.103]	0.114 [0.107, 0.121]
<i>Contrasts</i>	<i>Offline — Real-time</i>	−0.014 ($p = 0.11$)	−0.026 ($p = 0.01$)*	−0.018 ($p = 0.04$)*	0.017 ($p = 0.005$)**
	<i>Offline — Hybrid</i>	−0.008 ($p = 0.53$)	−0.018 ($p = 0.09$)	−0.011 ($p = 0.32$)	0.009 ($p = 0.16$)
	<i>Real-time — Hybrid</i>	0.007 ($p = 0.61$)	0.007 ($p = 0.64$)	0.008 ($p = 0.57$)	−0.007 ($p = 0.28$)

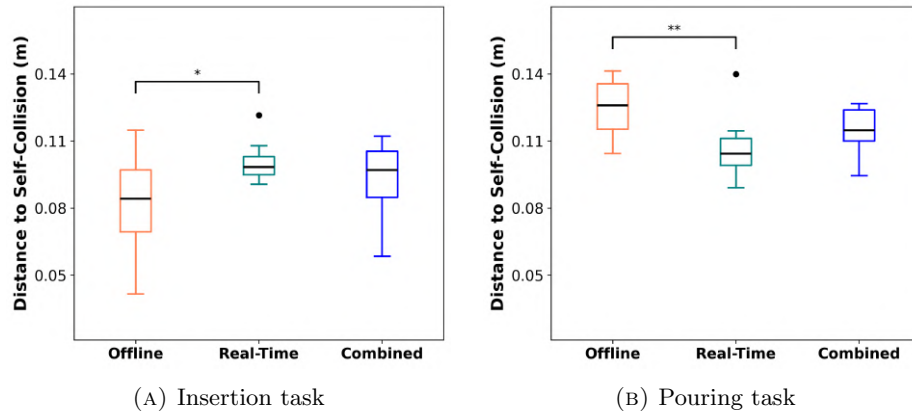


FIGURE 6.11: Distance to self-collision for both tasks (a) insertion task, and (b) pouring task.

6.4.5 Quality of Demonstration

To assess the quality of the demonstrated configurations, we analyzed manipulability as an indicator of how dexterous and well-conditioned the robot’s posture was during demonstrations. Figure 6.12a shows the manipulability measure of each condition and task.

Insertion Task — No significant main effects of condition ($\chi^2(2) = 1.75$, $p = .42$) or demonstration number ($\chi^2(1) = 0.47$, $p = .49$) were observed, and the condition-by-demonstration interaction was not significant ($\chi^2(2) = 0.84$, $p = .66$). Estimated marginal means were similar across offline ($M = 0.0497$), real-time ($M = 0.0512$), and combined ($M = 0.0493$) conditions, with no significant pairwise differences (all $p > 0.57$). Separate analyses of the initial and final three demonstrations likewise revealed no significant effects or pairwise differences (all $p > 0.11$), indicating stable manipulability across conditions and repetitions.

Pouring Task — For the pouring task, the mixed-effects model revealed no significant main effects of condition ($\chi^2(2) = 0.99$, $p = .61$) or demonstration number ($\chi^2(1) = 1.74$,

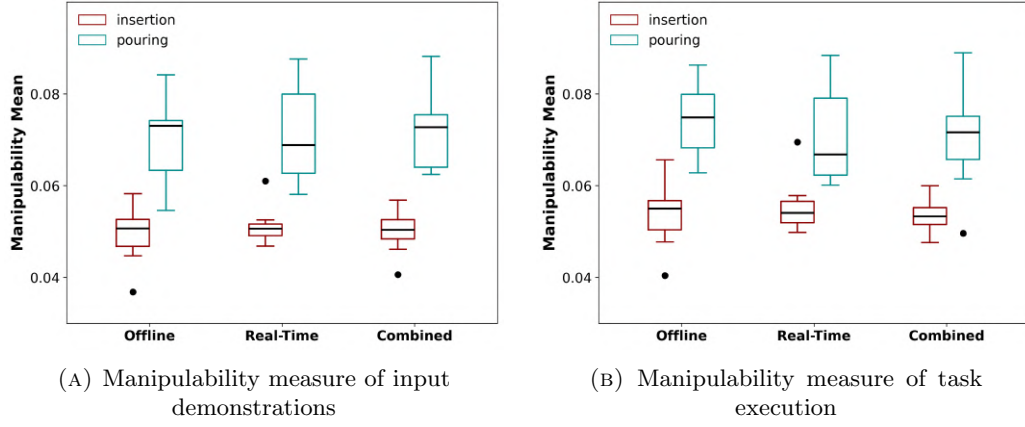


FIGURE 6.12: Manipulability measure of (a) input demonstrations, and (b) robot task execution.

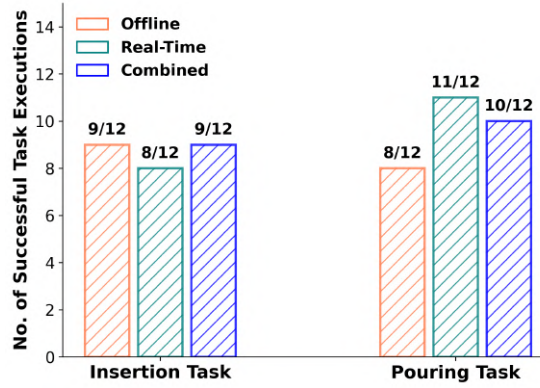


FIGURE 6.13: Robot task executions

$p = .19$), and no significant interaction ($\chi^2(2) = 1.61$, $p = .44$). Estimated marginal means were comparable across offline ($M = 0.0684$), real-time ($M = 0.0705$), and combined ($M = 0.0725$) conditions, with no significant pairwise differences. These results indicate consistent robot manipulability across conditions throughout the pouring task.

6.4.6 Robot Performance

Task performance differed across conditions. For the insertion task, success rates were similar across offline (9/12), real-time (8/12), and combined (9/12) conditions. In contrast, the pouring task showed clearer differences, with higher success under real-time (11/12) and combined (10/12) conditions compared to offline (8/12). Successful autonomous task executions are shown in [Figure 6.13](#).

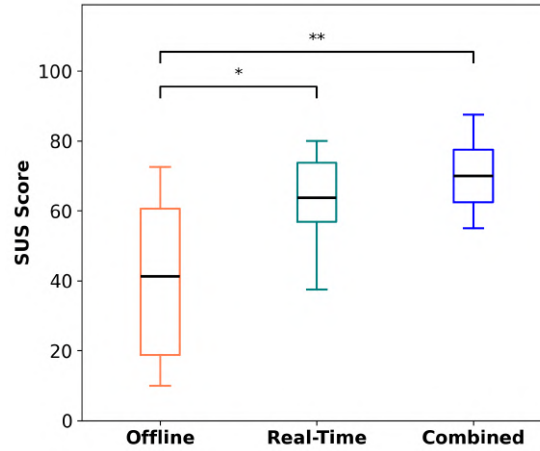


FIGURE 6.14: System Usability Score (SUS)

6.4.7 Quality of Task Execution

Figure 6.12b shows the summary of manipulability measure across conditions and tasks.

Insertion Task — A Welch ANOVA found no significant effect of condition on task execution quality ($F(2, 20.47) = 0.332, p = 0.72$), with no significant pairwise differences between any conditions (Games–Howell, all $p \geq 0.69$).

Pouring Task — Likewise, no significant effect of condition was observed ($F(2, 21.62) = 0.510, p = 0.51$), and post-hoc comparisons revealed no significant differences between conditions (all $p \geq 0.57$).

6.4.8 System Usability

A Welch’s ANOVA revealed a significant effect of condition on system usability scores, $F(2, 20.35) = 8.71, p < 0.001$. offline condition reported the lowest usability scores ($M = 40.62, SD = 22.67, 95\% CI = [26.22, 55.03]$), followed by real-time condition ($M = 63.33, SD = 14.03, 95\% CI = [54.42, 72.25]$) and combined condition ($M = 70.83, SD = 10.19, 95\% CI = [64.36, 77.31]$). Games–Howell post-hoc tests showed that offline condition scored significantly lower than real-time condition ($p = 0.02$) and combined condition ($p = 0.002$). The difference between real-time and combined conditions was not significant ($p = 0.31$). Figure 6.14 shows the summary of SUS scores across conditions.

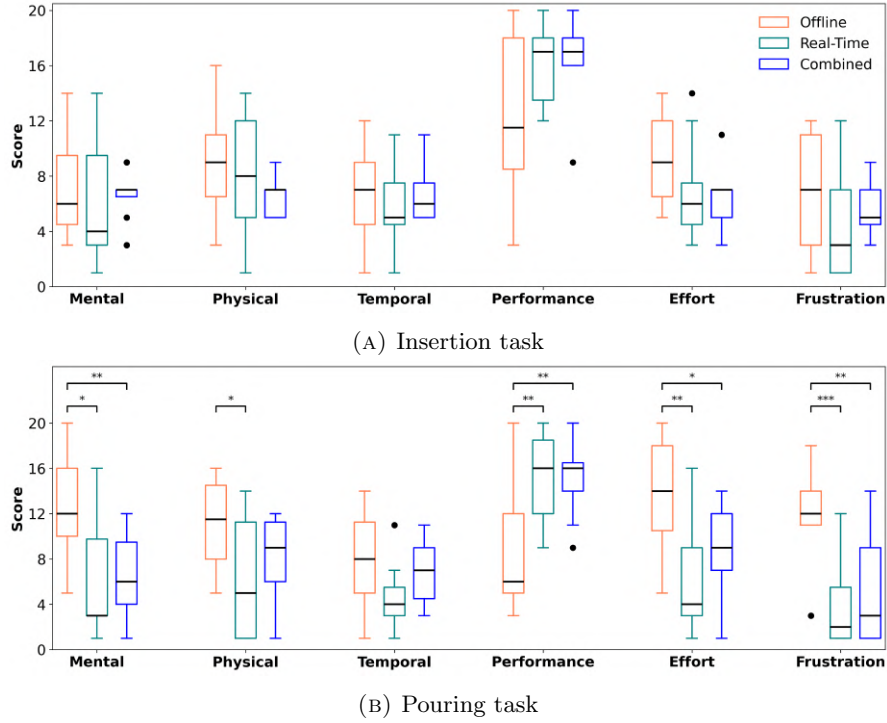


FIGURE 6.15: Participants' responses to the NASA-TLX questionnaire (21-point scale).

6.4.9 NASA-TLX

Welch's ANOVAs were conducted to examine differences across the three feedback conditions for all NASA-TLX sub-scales in the *insertion* and *pouring* tasks. For the insertion task, no significant condition effects were observed on any TLX dimension (all $p > 0.05$). In contrast, for the pouring task, significant condition effects were found for Mental Demand ($F(2, 21.83) = 6.69, p = 0.005$), Physical Demand ($F(2, 21.68) = 3.48, p = 0.049$), Performance ($F(2, 21.11) = 7.98, p = 0.003$), Effort ($F(2, 21.71) = 8.03, p = 0.002$), and Frustration ($F(2, 21.98) = 10.50, p < 0.001$), while Temporal Demand was not significant ($p = 0.17$).

Post-hoc Games-Howell tests for the pouring task showed that the offline condition reported significantly higher workload than both real-time and combined conditions across Mental Demand, Performance, Effort, and Frustration (all $p \leq 0.021$), and higher Physical Demand than the real-time condition ($p = 0.043$). No significant post-hoc differences were found for the insertion task. Overall, these results indicate pronounced workload differences during the pouring task, with the offline condition consistently associated with higher perceived workload, whereas no meaningful between-condition differences emerged for the insertion task (see Table 6.2).

6.4.10 Qualitative Results

Adjustments After Failed Demonstrations

Proactive Avoidance of Joint Limits: Participants across all conditions reported modifying their demonstrations to avoid exceeding joint limits, though the source and immediacy of this awareness differed by feedback modality. In the offline feedback condition, participants described learning joint constraints retrospectively through failure and playback, leading to cautious adjustments such as moving more carefully or choosing alternative directions. For example, one participant noted they “*tried to move the arm carefully not to go beyond its joint’s limit... because that was the problem*” (P_1, offline), while another stated they “*learned about the joint limits... and tried to make moves where I have more space to move*” (P_5, offline). In contrast, participants in the Real-time Feedback condition explicitly relied on real-time AR cues to guide their adjustments. One participant described “*following the limit given by every joint’s information... basically I just follow anything that AR information gives to me*” (P_17, real-time). Others reported actively selecting safer approaches to avoid failures, stating they would “*find an approach where the robot does not reach its limitations*” (P_14, real-time). Participants with *combined feedback* articulated a more integrated understanding of joint behavior, describing intentional adaptation of inputs based on learned constraints: “*I learned about each joint’s movement... and then I adjust my actions for the robot to move as efficient and effective as possible*” (P_27, combined).

Re-sequencing and Decomposition of Movements Following failed demonstrations, many participants altered the order in which they moved joints, though the strategy used varied by condition. In the offline feedback condition, participants reported broad changes in sequencing, such as “*completely changing the starting position... and making it go in other direction*” (P_6, offline) or “*adjust[ing] the link joints before adjusting the gripper*” (P_4, offline). Participants in the Real-time Feedback condition described more structured sequencing strategies informed by the AR visualization. One participant explained they preferred to “*rotate the bigger... lower joint first and then go up... the bottom one is more certain*” (P_16, real-time), while another noted adapting the sequence to avoid known limits: “*there is a limit for upward movement, so I have to move it downward and twist it*” (P_18, real-time). In the combined condition, participants reported decomposing actions at an even finer granularity, such as “*change the sequence and control the joint... only changed one joint at one time*” (P_29, combined).

Conservative and Simplified Motion Strategies After failing to record a successful demonstration, participants frequently adopted more conservative control strategies, including

slowing down and simplifying movements. Participants in the offline condition emphasized general caution, stating they would be “*more cautious of the limits... and keep the movements fluid*” (P_3, offline) or “*try to make single directions compared to diagonal ones*” (P_8, offline). Participants in the real-time feedback condition linked conservative behavior to specific failure events detected through AR. One participant recalled a failure due to “*a collision of the block with the table*” and another due to “*rushing too hard to get the block*” (P_15, real-time), prompting more controlled subsequent attempts. In the combined condition, slowing down was sometimes associated with increased cognitive demand, as one participant noted they “*tried to do it slowly*” because the feedback “*was kind of making me a little bit miserable*” (P_35, combined).

Understanding Key Qualities of an Effective Demonstration

Across all conditions, participants showed understanding of the robot’s physical constraints, particularly joint limits and ranges of motion, as the most fundamental quality of an effective demonstration. In the offline condition, this understanding emerged primarily through reflection on failures and robot rollouts. Participants emphasized avoiding problematic configurations, noting that “*the joints were the main problem... I had to move it in such a way that it did not go beyond its limit*” (P_1, offline) and recognizing that “*there is some angle that it cannot be worked in that way*” (P_7, offline).

Participants in the both real-time Feedback conditions described AR as making these constraints immediately visible and actionable. One participant explained that seeing “*the movement limit of every joint... helps me a lot to determine the extent... I should move every joint*” (P_17, Real-time Feedback).

Learning and Internalization of Robot Capabilities Participants in conditions with real-time AR feedback more explicitly described learning about the robot’s capabilities rather than simply correcting errors. They reflected on gaining familiarity with the joints’ behavior over time, noting that after initial failures they “*already know the limitation of the robot itself and know how to move the robot more efficient[ly]*” (P_18, real-time). Participants also described effective demonstrations as those that supported learning through direct, embodied interaction with the robot, rather than abstract or theoretical instruction. One participant emphasized that understanding emerged through experience: “*getting a hands-on... experiencing it myself... if I was told... theoretically, I don’t think I would understand it*” (P_14, real-time). Others noted that real-time feedback supported early familiarization and confidence building, with one participant reporting greater confidence during tasks performed with AR compared to those without it.

Perceived Impact of Demonstrations on Robot Performance

Demonstration Quality Directly Shapes Robot Effectiveness and Efficiency: Participants commonly perceived a strong link between the quality of their demonstrations and the robot's resulting performance, particularly in terms of efficiency, smoothness, and task success. Participants across conditions emphasized that demonstrations did not merely teach the robot what to do, but how well to do it. One participant reflected that effective teaching involved ensuring the robot learned *"the most efficient movement as possible... not only just to solve the challenge, but... in the most perfect way"* (P_27, combined). Similarly, others noted that when demonstrations became more refined over time, after the second round, with one participant stating that *"the robot learned more with more trials"* (P_8, offline). Several participants associated improved robot performance with clearer, more confident demonstrations, suggesting that inefficiencies or uncertainty during demonstration could negatively affect task execution: *"if there is some uncertainty... then the robot does not have enough ability to complete the task"* (P_2, offline).

Learning as an Iterative Process Accumulating Across Demonstrations: Participants frequently described the robot's learning as incremental, improving as more demonstrations were provided and as their own performance stabilized. Many participants believed that repetition and refinement across trials were key to improving the robot's performance. One participant explained that as their own performance improved, *"the speed increased, the efficiency increased... if the robot considers the last few attempts... it would perform very well"* (P_19, real-time). Another similarly noted that with *"more iterations... it will know how it will be done easily"* (P_26, combined). This perception was shared across conditions, with participants suggesting that later demonstrations carried more useful information than early exploratory ones. However, some participants also expressed uncertainty about how demonstrations were weighted internally, particularly when early and late demonstrations differed substantially.

Partial Understanding and Uncertainty About Robot Interpretation: Despite generally positive perceptions, participants often expressed uncertainty about how accurately or consistently the robot interpreted their demonstrations. Some participants questioned whether the robot truly "learned" from them or merely recorded motions. One participant remarked, *"I'm not sure if it's learning from me... it's definitely recording what I'm doing"* (P_14, real-time), suggesting ambiguity in the learning process. Others noticed discrepancies between their demonstrations and the robot execution, observing that it does not *"mimic your movement exactly... there are some noise or some extra movement"* (P_22, real-time). Participants in the combined condition echoed this mixed understanding, noting that while the robot was generally responsive, there was *"a little*

bit of... inaccuracy towards the end" (P_34, combined). Several participants attributed these issues to inconsistent demonstrations or insufficient system understanding of task context, rather than to outright failure.

Role of Feedback in Supporting Performance on the Second Task

Internalization of Joint Limits and Range of Motion: Across all conditions, participants reported that feedback (real-time and/or offline) during the insertion task, helped them internalize the robot's joint limits and range of motion, which directly supported more effective performance in the second task. Participants in the offline condition described gaining an understanding of the robot's movement capabilities through experience and repetition, enabling smoother performance later. One participant noted that they "*knew the range of motion of the robot... how should I move it first so that it can do the second motion... effectively*" (P_6, offline), while another stated they could "*navigate it relatively easier than the first two tasks*" after understanding joint behavior (P_3, offline). This effect was especially pronounced in the real-time feedback condition, where AR made joint limits explicit. Participants frequently reported remembering these limits even after the AR was removed, stating that "*by the time I removed the AR headset... I already know the limitation of the robot*" (P_20, real-time) and that AR helped them "*try to not... reach the limit*" (P_23, real-time) during the second task. Similarly, participants in the combined condition emphasized that AR allowed them to "*see the approximate limits... so in the second task... I don't push it to its limits*" (P_36, combined).

AR Feedback as a Scaffold for Learning and Confidence Building: Participants widely perceived AR feedback as a learning scaffold that supported confidence. Visualization participants described AR as intuitive and informative, noting that it allowed them to "*visually see how much I was moving... and understand to what level I can go*" (P_15, real-time). Several participants emphasized that this scaffolding was especially valuable for the more complex second task, which involved coordinated rotations and precise manipulation: "*for the second task... it's more complicated... you have to rotate... and then pour*" (P_16, real-time). However, participants also acknowledged limitations of relying solely on AR. One participant (P_17, real-time) estimated that AR provided "*70% of the information,*" but without continuous feedback they had to rely on memory and prediction, which "*becomes a bit complicated*". This suggests that AR supported learning and confidence, but its removal increased cognitive demands during task execution.

6.5 Discussion

This work set out to explore how robots can communicate their internal constraints to novice users during kinesthetic Learning from Demonstration, and how such communication reshapes the interaction between human teachers and robot learners. Rather than treating feedback as a post-hoc diagnostic tool or a purely technical aid, our findings position real-time AR feedback as an interaction design strategy for making robot learning legible during embodied teaching. In this section, we discuss how our results contribute design knowledge by contextualizing constraint feedback as a form of dialogue, examining how actionable and escalating cues shape user behavior, and articulating transferable design principles for communicating invisible system states.

6.5.1 Effects of Constraint Visualizations

Joint Limits Visualization: The joint-limit visualization showed significant impact on user interaction, with participants consistently adjusting their demonstrations to avoid exceeding joint limits. In the real-time and combined conditions, users relied on AR cues to proactively guide motion, whereas offline participants developed this understanding retrospectively through failures and playback (see Section 5.3 for quantitative results). Participants also reported internalizing joint limits over time and transferring this knowledge to subsequent tasks (see Section 5.10.1 for qualitative results), indicating that this visualization strongly supported both immediate performance and learning.

Self-Collision Visualization: The self-collision visualization showed a more nuanced impact on user interaction, with evidence emerging across both quantitative and qualitative findings. Quantitatively, improvements in performance were less pronounced compared to joint-limit visualization, suggesting a more limited effect on proactive behavior (see Section 5.4). Qualitatively, however, participants demonstrated increased awareness of collisions through reflections on failure events (see Section 5.10.2). In the real-time condition, users explicitly linked failures to their ways of demonstrating the given tasks and subsequently adopted more cautious and controlled strategies, such as slowing down or simplifying movements. This indicates that while the visualization supported users in recognizing and responding to collision-related issues, its influence was primarily reactive, with less evidence of proactive planning or anticipation.

Manipulability Visualization: The manipulability visualization did not show a significant impact on user performance. This is likely due to the nature of the task, which inherently constrained users from entering near singular configurations (i.e., extreme cases where the end effector's movement is reduced), thereby limiting opportunities

for the visualization to influence behavior. Quantitatively, no significant differences were observed across conditions (see Section 5.5), and qualitatively, participants did not explicitly report awareness of manipulability-related concepts. Future work should consider designing experiments where manipulability is investigated independently, allowing for a more focused and rigorous evaluation of its impact on user understanding and performance.

6.5.2 Designing for Mutual Legibility in Kinesthetic Teaching

A central challenge in kinesthetic robot teaching is that users must reason about constraints (e.g. joint limits, self-collisions, singularities) that are not directly perceptible through physical interaction alone. While users can feel resistance or abrupt stopping, such sensations provide little insight into why a configuration is problematic or how it might be corrected. Our findings suggest that these breakdowns are less about user skill and more about a lack of mutual legibility between human and robot. By visualizing internal robot states in situ, the AR feedback system transformed teaching from a one-sided act of demonstration into a more reciprocal interaction. Instead of guessing what the robot could or could not accommodate, users were able to see how their actions related to the robot’s internal constraints. This aligns with previous works in HRI on intelligibility and explainable interaction: the robot does not merely execute motions but actively “shows” its limits [246], enabling users to adjust their behavior accordingly.

Importantly, this shift reframes **Learning from Demonstration as a communicative process rather than a simple data collection step**. Teaching quality improved not only because users avoided errors, but because the interaction itself supported sense-making about how the robot experiences the task.

6.5.3 From Awareness to Action: Designing Actionable Constraint Feedback

The focus group study revealed that simply making constraints visible is insufficient. Participants consistently emphasized that awareness cues, such as color changes indicating proximity to a limit, must be coupled with guidance that helps users recover from problematic configurations. This distinction between knowing that something is wrong and knowing what to do next proved critical in the lab study. Our AR system operationalized this insight by layering feedback: initial awareness was communicated through subtle, color-coded cues, while more explicit guidance (e.g., arrows or intensified highlights) emerged as users approached or crossed critical thresholds. The study results

suggest that this approach supported smoother demonstrations and reduced time spent in unproductive or constrained configurations.

From an interaction design perspective, this highlights the importance of **feedback that is not merely informative but actionable**. Effective real-time feedback must support decision-making in the moment, translating abstract constraints into concrete possibilities for movement. This principle extends beyond robotics to other interactive systems where users must navigate invisible or complex limitations in real-time [247, 248].

6.5.4 Escalation, Minimalism, and Managing Cognitive Load

A recurring concern in both the focus group and prior XR literature is the risk of overwhelming users with too much visual information. Our design addressed this tension through escalation and progressive disclosure: feedback remained invisible during safe operation and emerged gradually as constraints became relevant. This strategy proved effective in balancing awareness and cognitive load. Participants reported lower perceived workload, and observational data suggested that users were rarely distracted by the AR overlays when the robot operated within safe regions. By anchoring feedback spatially to the robot and limiting its presence to moments of risk, the system avoided becoming a constant source of visual noise. From a design standpoint, this demonstrates how **escalation can function as a general interaction pattern for XR systems** [240]. Rather than presenting all available information upfront, systems can remain visually quiet and intervene only when user action requires guidance [249]. This approach respects user attention while preserving access to critical information when it matters most.

6.5.5 Feedback as a Scaffold for Learning, Not a Crutch

One of the most revealing aspects of the study was the second task, in which all feedback was removed. Users who had previously received real-time AR feedback continued to demonstrate safer and more effective teaching behaviors, suggesting that they had internalized aspects of the robot's constraints. This finding reframes real-time feedback not as a permanent assistive layer, but as a temporary learning scaffold beyond general task practice. By receiving immediate, context-specific feedback during early interactions, users developed mental models of the robot's movements and limitations. Once these models were established, users were able to apply them even without visual support.

This distinction highlights **feedback design as a means of supporting learning and understanding rather than dependency**. This aligns with scaffolding theory, in which instructional support is provided during initial learning and gradually withdrawn as learners internalize task structure [250, 251]. In contrast to offline execution-based feedback, which supports reflection after the training, real-time feedback enabled learning through embodied action, aligning with theories of experiential and situated learning [252, 253]. By coupling perception and action in real time, feedback becomes part of the task itself, supporting the development of sensorimotor understanding that is difficult to achieve through post-hoc reflection alone.

Our proposed real-time feedback system is designed to support user understanding of the constraints instead of reducing exploration during kinesthetic demonstrations. Furthermore, rather than constraining the search space, the system provides visual feedback that helps users provide effective demonstrations considering the robot inherent constraints.

6.5.6 Design Principles for Communicating Invisible Constraints during LfD for Novice Users

Synthesizing findings from both studies, we adapt Vi et al.'s 11 guidelines for XR applications on head-mounted displays [254] to *LfD tasks that involve novice users* and propose five guidelines.

- **Design flexible spatial environments to maximize efficiency.** In XR systems for LfD, users teach robots by demonstrating task-specific actions. Efficiency in this context can be measured through number of tasks completed over time (task completion rate) and time required to complete a single demonstration (active demonstration time).

Our findings show that real-time and combined feedback conditions significantly outperform offline approaches (see Section 5.1 and 5.2). Both real-time and combined setups lead to higher task completion rates and reduced demonstration time, with no significant performance difference between them.

From a spatial design perspective, these results suggest that real-time feedback alone is sufficient to maximize efficiency. Incorporating mix elements, such as switching between XR and external displays for task execution verification, does not provide additional efficiency benefits, while introducing unnecessary spatial and cognitive overhead.

- **Use cues to help through experiences which build upon real-world knowledge.** In XR systems for LfD, cues should align with users' real-world knowledge to

reduce cognitive load and improve understanding. Our findings show that **color-based indicators**(e.g., green-to-red) are effective for communicating joint limits, **overlay visualizations** help identify self-collisions directly on the robot, and **ellipsoids at the end-effector** intuitively represent manipulability.

- **Balance Comfort and Cognitive Load Through Minimal and Contextual Feedback.** Kinesthetic teaching in LfD is physically and cognitively demanding. XR feedback should minimize strain by avoiding excessive overlays, respecting personal space, and reducing unnecessary movement. Using progressive disclosure, feedback should remain unobtrusive during safe operation and appear only when relevant, ensuring both physical comfort and manageable cognitive load.
- **Design Feedback that is Accurate, Consistent, and Reflective of Robot Capabilities.** XR feedback must faithfully represent the robot’s kinematics and limitations to build trust and correct understanding. Providing consistent, real-time responses with clear state distinctions (e.g., safe, near-limit, violation) helps users develop reliable mental models of how the robot behaves and what actions are feasible.
- **Support Learning Through Scaffolded Feedback and Exploration.** LfD is fundamentally a learning process. XR feedback should act as a scaffold by providing strong, real-time guidance during early interactions and gradually reducing support as users gain experience. This enables users to internalize robot constraints and continue effective teaching even without feedback.

While derived from a robotics context, these principles can be applied broadly to interactive systems that need to externalize hidden system states. For example, through XR interfaces that visually embed system intent and status in the user’s environment, improving situational awareness [236, 255]; auditory sonification that conveys latent states or system uncertainty via sound [136]; or motion and expressive behaviors that make internal processes interpretable through movement patterns [256].

6.5.7 Comparison with Existing Visualization Tools

Our work specifically targets the visualization of robot-inherent constraints—such as joint limits and self-collisions—during kinesthetic demonstrations, which is not the primary focus of existing tools like *MoveIt*⁵. While *MoveIt* provides valuable visualizations

⁵<https://moveit.ai/>

of kinematic structures (e.g., frames and link relationships), it does not explicitly visualize the current state of joint angles in a way that is directly interpretable during interaction.

In contrast, our approach introduces explicit, real-time visual encodings of joint angle limits, enabling users—particularly novices—to better understand how close they are to constraint boundaries and thus perform more dexterous and controlled demonstrations. Although prior tools provide baseline visualizations, they are not tailored to this interaction context or user group. Our design complements existing systems and can be integrated into *MoveIt* to enhance understanding of robot constraints.

6.5.8 Limitations and Future Work

While the results demonstrate the potential of our AR-based approach, they present several limitations related to the study population, experimental design, and technical constraints of the AR system, and highlight directions for future work.

- **Generalizability and Study Design** This study involved university students, which may limit generalizability to professional settings. Future work will evaluate industrial and healthcare workers to assess domain-specific effects. The manipulability visualization introduces an abstract representation that may be interpreted differently by novice users, potentially contributing to variability in user performance. Although notable differences in task completion times were observed, more detailed statistical analysis of this variability remains for future work. We will also incorporate established guidelines from information visualization and accessibility to further improve interface design and usability [257, 258].
- **AR System and Tracking Limitations.** We used state-of-the-art hardware; however, occasional instability in the Meta Quest 3 affected AR consistency. Our experimental setup relied on a single QR-code-based spatial anchor for AR alignment, which exhibited occasional alignment drift (< 1 cm), particularly when out of view, potentially impacting perception of spatial constraints. To mitigate this, we implemented continuous QR tracking and designed the AR-based task such that the QR code remained within the user's field of view for most of the interaction. Despite these measures, residual instability may still have influenced user performance, highlighting device reliability as a critical factor in AR-based experiments.
- **Ergonomics and User Comfort.** The headset's weight (515 g) may cause discomfort and motion-sickness symptoms such as dizziness and headaches. While

minimized in this study due to short, simple tasks (approximately 30 minutes), longer or more demanding use may increase both physical discomfort and simulator sickness.

6.6 Summary

This work examined how real-time feedback can function as a communicative and educational mechanism in embodied human-agent/robot interaction. By designing and evaluating an AR-based system that makes otherwise invisible robot constraints perceptible during kinesthetic teaching, we showed how feedback can support safer, more efficient demonstrations while helping users form more accurate mental models of the robot's capabilities.

Our contribution lies in reframing constraint feedback as an interaction design problem, empirically grounding design principles for real-time and escalating feedback, and demonstrating how such feedback can scaffold user learning rather than foster long-term dependence. While evaluated in a robotic teaching context, these insights extend to a broader class of interactive systems in which users must engage with complex, hidden system states through repeated interactions.

Several directions for future research emerge from this work. First, longitudinal studies could examine how real-time feedback shapes learning and skill retention over extended use, including whether and when such feedback should be gradually withdrawn or adapted. Second, future systems could explore adaptive or personalized feedback strategies that respond to user expertise, task complexity[242], or contextual risk[259]. Finally, extending these design principles beyond robotics, to domains such as physical rehabilitation, or human-AI/robot collaboration [260, 261], offers opportunities to further investigate how real-time, perceptual feedback can support understanding, trust, and agency in interactions between humans and autonomous systems.

Chapter 7

Conclusion and Future Work

This thesis investigated how LfD can be made more effective and more accessible for novice users by focusing on the information content of demonstrations and by supporting human teaching. Rather than treating demonstrations as interchangeable training data, we argue that what matters is not only whether the robot can complete a task, but also how well it executes the task and how reliably novices can provide useful demonstrations.

First, we showed that demonstration quality can be quantified using motion-related features and that these quality measures relate to both task success and execution quality. This enables a more informative evaluation of LfD models: beyond reporting success rates, one can diagnose which demonstrations lead to undesirable execution and identify quality thresholds where performance begins to degrade.

Second, we introduced a quality-aware LfD framework that learns from mixed demonstrations, including failed and suboptimal trials. By explicitly optimizing motion-related criteria during learning and reproduction, the framework improves execution quality while maintaining task completion, with consistent gains in simulation and on a real 7-DoF robot. These results support the broader point that failed demonstrations can be treated as informative negative evidence rather than discarded data.

Third, we demonstrated that robot kinematic redundancy is not merely a mechanical property, but an interaction factor that shapes how novices teach. Changes in redundancy influence workload, demonstration efficiency, and failure rates, and they ultimately affect downstream robot learning performance. This suggests that LfD systems should be designed with the user’s ability to explore the robot’s configuration space in mind, rather than assuming that more flexibility always translates into easier teaching.

Finally, we designed and evaluated an AR-based real-time feedback system that makes otherwise hidden robot constraints perceptible during kinesthetic teaching. By communicating constraints and impending violations in context, the system supports safer and more efficient demonstrations and helps users form more accurate mental models of the robot’s capabilities. Overall, the thesis highlights that improving LfD requires jointly addressing learning algorithms, demonstration quality, and human-centered feedback mechanisms.

Future work can extend these contributions in several ways. (1) Broaden the set of quality metrics (beyond manipulability and joint-space jerk) and test their generality across tasks, robots, and learning algorithms. (2) Develop learning methods that adaptively weight demonstrations and quality objectives during training, and study principled ways to detect cut-off regions where execution quality becomes unacceptable. (3) Investigate redundancy-aware teaching support, including interfaces and control modes that adapt the robot’s redundancy to the user’s skill level and the task context. (4) Conduct longitudinal studies to understand how real-time feedback affects learning, retention, and reliance over time, and explore adaptive feedback that personalizes timing and modality based on user expertise, task complexity [242], and contextual risk [259]. (5) Transfer the proposed feedback design principles to other domains (e.g., rehabilitation and broader human–AI/robot collaboration) to study how perceptual feedback can support understanding, trust, and agency [260, 261].

Appendix A

An Appendix

A.1 Appendix

A.1.1 Focus Group Drawings

FIGURE A1: Group-level best drawings for joint limits. Participants proposed joint-limit communication in XR using a bottom-up hierarchy that reflects increasingly sophisticated feedback layers. At the foundation, they emphasized intuitive awareness cues, especially color-based indicators that communicate safety at a glance. Building on this, they highlighted the need to quantify proximity to mechanical limits for more precise control. Participants also stressed placing these cues spatially on or near the joint. As the hierarchy progresses, alert mechanisms should escalate as users approach or exceed critical thresholds. Finally, at the highest level, participants emphasized the importance of customization.

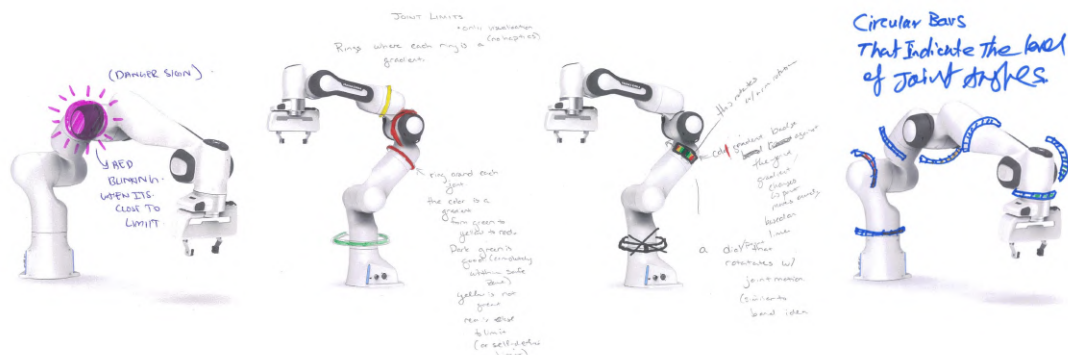


FIGURE A2: Group-level best drawings for self-collision. Participants described a hierarchy of needs for managing robot self-collisions in XR, starting with basic collision avoidance and progressing toward deeper spatial understanding. At the foundation, users need clear indicators of collision proximity to detect potential contact. Building on this, participants emphasized *actionable guidance* to help users recover safely. They also noted that the form of feedback, from simple 2D cues to richer 3D meshes, affects how well users interpret spatial relationships and anticipate collisions. Finally, they highlighted the need to manage attentional load by distinguishing normal states from genuine danger to avoid overwhelming users.

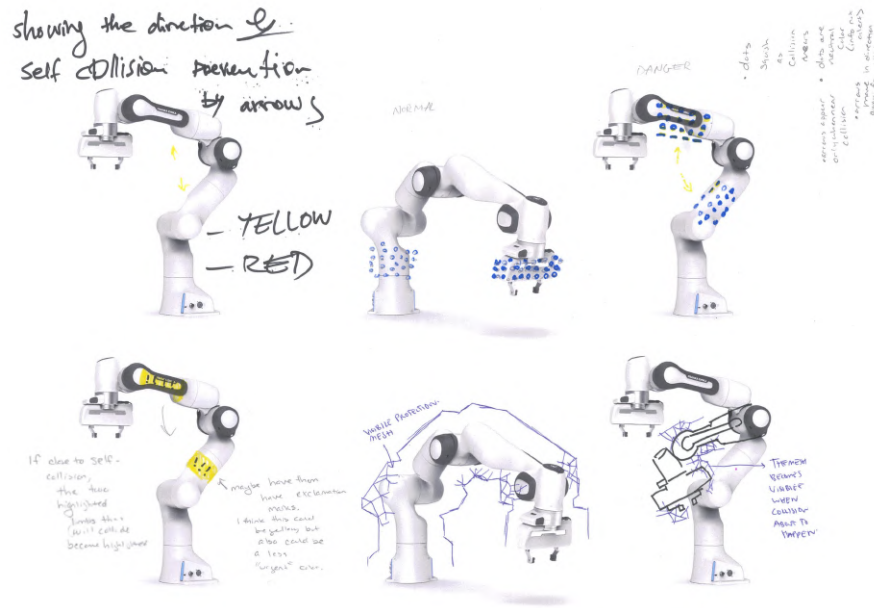
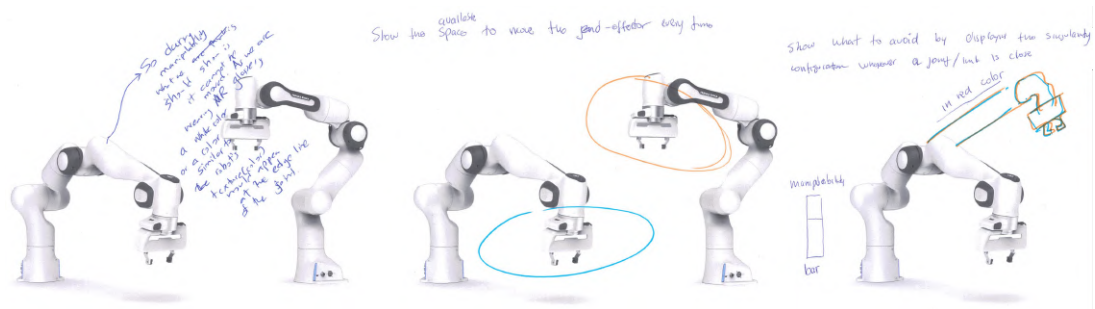


FIGURE A3: Group-level best drawings for manipulability. Participants described a layered set of needs for supporting manipulability in XR during robot learning from demonstration. First, they emphasized presenting information in the right visual form and location so users can distinguish between local joint issues and global posture effects. They then highlighted the need for clear alerts when the robot approaches poor manipulability. Building on this, participants stressed *actionable guidance*, such as directional cues or preventive mechanisms, to help users adjust demonstrations and avoid singular configurations.



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